AOS-CX 10.18.xxxx Monitoring Guide (5420, 6200 Switch Series)

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About this document

This document describes features of the AOS-CX network operating system. It is intended for administrators responsible for installing, configuring, and managing HPE Aruba Networking switches on a network.

Subtopics

Applicable products

This document applies to the following products:

- HPE Aruba Networking 5420 Switch Series (S0U67A, S0U55A, S0U63A, S0U64A, S0U65A, S0U75A, S0U72A, S0U78A, S0U58A, S0U73A, S0U74A, S0U71A, S0U76A, S0U70A, S0U77A, S0U60A, S0U61A, S0U62A, S0U66A, S0U68A)
- HPE Aruba Networking 6200 Switch Series (JL724A, JL725A, JL726A, JL727A, JL728A, R8Q67A, R8Q68A, R8Q69A, R8Q70A, R8Q71A, R8V08A, R8V09A, R8V10A, R8V11A, R8V12A, R8Q72A, JL724B, JL725B, JL726B, JL727B, JL728B, S0M81A, S0M82A, S0M83A, S0M84A, S0M85A, S0M86A, S0M87A, S0M88A, S0M89A, S0M90A, S0G13A, S0G14A, S0G15A, S0G16A, S0G17A)

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Command syntax notation conventions

Convention	Usage
example - text	Identifies commands and their options and operands, code examples, filenames, pathnames, and output displayed in a command window. Items that appear like the example text in the previous column are to be entered exactly as shown and are required unless enclosed in brackets (**[ ]**).
example - text	In code and screen examples, indicates text entered by a user.
Any of the following: • <example - text> • <example - text> • example - text • example - text	Identifies a placeholder—such as a parameter or a variable—that you must substitute with an actual value in a command or in code: • For output formats where italic text cannot be displayed, variables are enclosed in angle brackets (**< >**). Substitute the text—including the enclosing angle brackets—with an actual value. • For output formats where italic text can be displayed, variables might or might not be enclosed in angle brackets. Substitute the text including the enclosing angle brackets, if any, with an actual value.
|	Vertical bar. A logical **OR** that separates multiple items from which you can choose only one. Any spaces that are on either side of the vertical bar are included for readability and are not a required part of the command syntax.
{ }	Braces. Indicates that at least one of the enclosed items is required.
[ ]	Brackets. Indicates that the enclosed item or items are optional.
… or ...	Ellipsis: • In code and screen examples, a vertical or horizontal ellipsis indicates an omission of information. • In syntax using brackets and braces, an ellipsis indicates items that can be repeated. When an item followed by ellipses is enclosed in brackets, zero or more items can be specified.

About the examples

Examples in this document are representative and might not match your particular switch or environment.

The slot and port numbers in this document are for illustration only and might be unavailable on your switch.

Understanding the CLI prompts

When illustrating the prompts in the command line interface (CLI), this document uses the generic term **switch**, instead of the host name of the switch. For example:

```
switch>
```

The CLI prompt indicates the current command context. For example:

```
switch>
```

Indicates the operator command context.

```
switch#
```

Indicates the manager command context.

```
switch(CONTEXT-NAME)#
```

Indicates the configuration context for a feature. For example:

```
switch(config-if)#
```

Identifies the **interface** context.

Variable information in CLI prompts

In certain configuration contexts, the prompt may include variable information. For example, when in the VLAN configuration context, a VLAN number appears in the prompt:

```
switch(config-vlan-100)#
```

When referring to this context, this document uses the syntax:

```
switch(config-vlan-<VLAN-ID>)#
```

Where <VLAN-ID> is a variable representing the VLAN number.

Identifying switch ports and interfaces

Physical ports on the switch and their corresponding logical software interfaces are identified using the format: **member/slot/port.**

On the HPE Aruba Networking 6200 Switch Series

- member: Member number of the switch in a Virtual Switching Framework (VSF) stack. Range: 1 to 8. The primary switch is always member 1. If the switch is not a member of a VSF stack, then member is 1.

- slot: Always 1. This is not a modular switch, so there are no slots.
- port: Physical number of a port on the switch.

For example, the logical interface **1/1/4** in software is associated with physical port 4 in slot 1 on member 1.

On the HPE Aruba Networking 6400 and 5420 Switch Series

- member: Always 1. VSF is not supported on this switch.
- slot: Specifies physical location of a module in the switch chassis.
  - Management modules are on the front of the switch in slots 1/1 and 1/2.
  - Line modules are on the front of the switch starting in slot 1/3.
- port: Physical number of a port on a line module.

For example, the logical interface **1/3/4** in software is associated with physical port 4 in slot 3 on member 1.

Monitoring hardware through visual observation

Subtopics

Confirming normal operation of the switch by reading LEDs

This task describes using the switch LEDs to confirm that the switch is operating normally.

> **NOTE**
> For complete information on LED behaviors for your AOS-CX switch, refer to the **Installation and Getting Started Guide** for that switch series, available for download from the section of the .

Procedure

1. Quick check: Verify that the chassis has power and there are no fault conditions.
   On the front of the switch, verify that the states of the following LEDs are On Green:

2.

    - Power
    - Health

    Verify that the Health LEDs of all installed line modules are On Green.

3. Verify that the Health LEDs of all installed management modules are On Green.
4. Verify that the network ports are operating normally.
    a. On the active management module, check the Status Front section. Verify that each LED that indicates a line module is in one of the following states:
        - On Green (normal operation)
        - Off (no line module installed)
    b. On each line module, verify that each port LED is in one of the following states:
        - On Green, Half-Bright Green, or Flickering Green (normal operation)
        - Off (no cable connected or port off by default in config)
5. Verify that the power supplies are operating normally.
    a. On the active management module, check the Status Front section. Verify that each LED that indicates a power supply is in one of the following states:
        - On Green (normal operation)
        - Off (no power supply installed)
    b. On each power supply, verify that LEDs are in the following states:
        - Power LED: On Green
        - Fault LED: Off
6. Verify that the rear components are operating normally by checking the Status Rear section of the active management module:
    a. Verify that the LEDs for the fabric modules are in one of the following states:
        - On Green (normal operation)
        - Off (component not installed)
    b. Verify that the LEDs for the fan trays and fans are On Green.

7. Verify that the standby management module is ready to take over as the active management module. On the standby management module, verify the states of the following LEDs:
    - Health LED is On Green.
    - Management state standby (Stby) LED is On Green.

Detecting if the switch is not ready for a failover event

This task describes using the switch LEDs to detect if the switch is not ready for the loss of a fabric module or for a failover from the active management module to the standby management module.

> **NOTE**
>
> Although you can detect power supply failures by viewing the LEDs, you must use software commands to determine if the power supply redundancy is sufficient to power the chassis if a power supply fails. For complete information on LED behaviors for your AOS-CX switch, refer to the **Installation and Getting Started Guide** for that switch series, available for download from the section of the .

Procedure

1. Detect if the standby management module is shut down.
   If the standby management module is shut down, the LED states are as follows:
    - The standby management module health LED is Off.
    - The standby management state active (Actv) LED is Off.
    - The standby management state standby (Stby) LED is Off.
    - On the active management module in the Status Front Management Modules section, the LED for the standby management module is Off. For example, if the active management module is Management Module LED 5, Management Modules LED 6 is Off.
2. Detect if the standby management module is in a transient state. If the standby management module is booting, updating, or in another transient state, the LED states are as follows:
    - The standby management module health LED is Slow Flash Green when the service operating system is running or during an operating system update.
    - The standby management module Booting LED is Slow Flash Green when the AOS-CX operating system is booting.
    - The standby management state active (Actv) LED is Off.

- The standby management state standby (Stby) LED is Off.
- On the active management module in the Status Front Management Modules section, the LED for the standby management module is Slow Flash Green.

3. Detect if a fabric module is shut down or not present. If a fabric module is shut down or not present, the LED states are as follows:
   - On the active management module, in the Status Rear section, the LED for the fabric module is Off.
   - On the rear display module, the LED for the fabric module is Off.
   - On the fabric module, the health LED is Off. However, the fabric module is behind fan 1 and is not directly visible.

Finding faulted components using the switch LEDs

This task describes using the switch LEDs to find components that are in a fault condition.

> **NOTE**
> All green LEDs—except for chassis power LEDs and the Usr1 LED—are off when the LED mode is set to Light Faults (The Usr1 LED of the LED Mode section of the active management module is On Green and the default behavior for the Usr1 LED is being used.). For complete information on LED behaviors for your AOS-CX switch, refer to the **Installation and Getting Started Guide** for that switch series, available for download from the section of the .

Procedure

1. Find the switch that has the fault condition, which is indicated by a chassis health LED in the state of Slow Flash Orange.
   The chassis health LED is located on the front of the switch and on the rear panel of the switch.
2. If you are at the back of the switch, on the rear panel, look for LEDs that are in the Slow Flash Orange state:
   The Status Rear area has LEDs for power supplies, fabric modules, fan trays, and fans. The number on the LED represents the unit number of the component.

   If the only LED in a state of Slow Flash Orange is the Chassis health LED, go to the front of the switch.
3. At the front of the switch, on the active management module, look for LEDs that are in the Slow Flash Orange state:

- The Status Front area has LEDs for power supplies, line and fabric modules, and management modules. The number on the LED indicates the slot number of the component.
- The Status Rear area has LEDs for fabric modules and fan trays, with a single LED for all the fans in the fan tray. The number on the LED represents the slot or bay number of the component.

4. Use the number indicated by the LED that is flashing to locate the slot that contains the faulted component.
   The fabric modules are located behind the fan trays, and the fabric module number corresponds to the fan tray number.
5. At the front of the switch, on line modules, look for LEDs that are in the Slow Flash Orange state: Module LEDs and Port LEDs indicate faults if their states are Slow Flash Orange.

IP Flow Information Export

IP Flow Information Export (IPFIX) is an embedded network flow analysis tool that compiles characteristic and measured properties of flows and sends flow reports to internal or external flow collectors. IPFIX is configurable via the command-line or REST interfaces. With IPFIX, customers configure flow records with match (key) fields and collection (non-key) fields. Match fields are the set of fields that define a flow, such as IP address or UDP port. Collection fields are the set of fields that identify information to collect for a flow, such as packet and byte counters.

A flow exporter defines where and how to export flow reports. Flow exporters are created as standalone entities in the **config** context to provide flow monitors the ability to export flow reports.

Compatibility with Traffic Insight

The AOS-CX traffic insight feature allows monitoring of large amount of data that it collects from various flow exporters like IPFIX, and provides the ability to filter, aggregate, and sort the data based on user flow monitor requests. Traffic insight tracks different monitor requests simultaneously and provides monitor reports per request. For more information on configuring the Traffic Insight features, refer to the AOS-CX Security Guide.

Compatibility with Application Recognition

If the **application recognition** feature is enabled, then the application data and the flow properties collected by AR and IPFIX are exported to external or internal IPFIX collectors. For more information on configuring the Application Recognition and Traffic Insight features, refer to the AOS-CX Security Guide.

Information Elements

The IPFIX Information Elements (IE) are entities that are defined and maintained by the Internet Assigned Numbers Authority (IANA). They are characterized by a unique piece of information they can provided about a flow. Information Elements may be either private or public. Private Information Elements are exported with a Private Enterprise Number (PEN).

AOS-CX can act as an intermediate collecting process for flow reports from hardware to append certain additional IPFIX information elements to the flow reports. When configured, the software will act as an intermediate exporting process to export the augmented flow reports to any configured flow exporters.

AOS-CX supports the standard and private information elements shown in the tables below.

NOTE

Not all switches support all information elements. View a list of information elements supported by your switch using the command **show flow monitor <monitor-name> information-elements**. All Standard Information Element registration information can be found on the IANA website, at **https://www.iana.org/assignments/ipfix/**.

Standard Information Elements
octetDeltaCount
packetDeltaCount
protocolIdentifier
sourceTransportPort
sourceIPv4Address
ingressInterface
destinationTransportPort
destinationIPv4Address
sourceIPv6Address
destinationIPv6Address
sourceIPv6Address
destinationIPv6Address
classId
sourceMacAddress
vlanId

Private Information Elements
(No private information elements are currently supported by the 5420 Switch series)
tlsClientVersion (ID 1029, 4823 Aruba)
tlsServerVersion (ID 1030, 4823 Aruba)
ja3 (ID 1031, 4823 Aruba)
ja3s (ID 1032, 4823 Aruba)
supportedNextProtocol (ID 1033 , 4823 Aruba)
CertificateIssuer (ID 1034, 4823 Aruba)
CertificateIssueDate (ID 1035, 4823 Aruba)
CertificateExpiryDate (ID 1036, 4823 Aruba)
applicationType (ID 1100, 4823 Aruba)

Standard Information Elements	Private Information Elements
ipVersion	tcp3WayHsServerRespTime (ID 1101, 4823 Aruba) (6200 Switch series only)
destinationMacAddress	tcp3WayHsClientRespTime (ID 1102, 4823 Aruba) (6200 Switch series only)
applicationDescription	dropGroup1 (ID 1200, 14823 Aruba)
applicationId	dropReason1 (ID 1201, 14823 Aruba)
applicationName	dropStartMilliseconds1 (ID 1202, 14823 Aruba)
flowEndReason	dropEndMilliseconds1 (ID 1203, 14823 Aruba)
flowStartMicroseconds	dropGroup2 (ID 1210, 14823 Aruba)
flowEndMicroseconds	dropReason2 (ID 1211, 14823 Aruba)
dnsResponseCode	dropStartMilliseconds2 (ID 1212, 14823 Aruba)
ingressPhysicalInterface	dropEndMilliseconds2 (ID 1213, 14823 Aruba)
egressPhysicalInterface	dropGroup3 (ID 1220, 14823 Aruba)
applicationCategoryName	dropReason3 (ID 1221, 14823 Aruba)
(N/A)	dropStartMilliseconds3 (ID 1222, 14823 Aruba)
	dropEndMilliseconds3 (ID 1223, 14823 Aruba)
	dropGroup4 (ID 1230, 14823 Aruba)

		Private Information Elements
		dropReason4 (ID 1231, 14823 A ruba)
		dropStartMilliseconds4 (ID 1232, 14823 Aruba)
		dropEndMilliseconds4 (ID 1233, 14823 Aruba)
		cngQueueThreshold (ID 490, 14 823 Aruba)
		cngQueueObservedTimeStamp (ID 491, 14823 Aruba)

About individual Information Elements

The following IEs are used for both IPv4 and IPv6 flows:

- cngSourceIP
- cngDestinationIP

They are always 16 bytes wide. The IP version of the flow is indicated in the **ipVersion** IE. For IPv4 addresses, the first 12 octets are zero and the IPv4 address is encoded in the last 4 octets. For IPv6 addresses, the full 16 bytes encode the address.

The following IEs reflect the hardware interface ID used by the associated flow:

- ingressPhysicalInterface
- egressPhysicalInterface

The hardware interface ID does not match the front-panel port number of the interface. The hardware interface ID can be mapped to the front-panel port number by using the REST API to query for **hw_intf_info** for all interfaces and saving the **switch_intf_id** value together with the associated front-panel name to form a mapping table.

Example using curl and jq:

```
bash
$ curl -X GET
    "{SWITCH}/rest/{VERSION}/system/interfaces?
attributes=hw_intf,info,name&depth=2"
    -H "x-csrf-token: {CSRFTOKEN}"
    -b {AUTH COOKIE FILE}
```

```
    | jq 'to_entries
          | map({port: .key, switch_intf_id:
(.value.hw_intf_info.switch_intf_id | tonumber)})
          | sort_by(.switch_intf_id)
          | map({(.port): .switch_intf_id})
          | add'
{
  "1/1/6": 1,
  "1/1/2": 2,
  "1/1/1": 3,
  ...
}
```

The **IPFIX Drop Exceptions** feature is a network debugging and triaging tool supported on 5420 and 6200 Switch series. It is configured using the **collect drop ingress-exceptions** command in the **config-flow-record** context. The private information elements 1200-1233 contain possible drop reason information for why dropped packets are seen for a flow. Up to four drop reasons are supported for each flow. Data in consecutive values (for example, 1200, 1201, 1202, 1203) indicate the data in those fields are associated with each other.

For each drop reason, AOS-CX reports the drop group name, drop reason name, and drop start and end timestamps in milliseconds.

Table 1. Drop reasons for the IPFIX Drop Exceptions feature

Drop reason	Example Cause	Supported Platforms
Unknown VLAN	The packet has an unknown VLAN	5420, 6200, 6300, 6400, 8100, 8325, 8325P, 8325H, 8360, 10000 Switch series
MTU Failure	The packet size is larger than the IP Maximum Transmission Unit (MTU) configured in the interface.	5420, 6200, 6300, 6400, 8100, 8325, 8325P, 8325H, 8360 Switch series
Unroutable Unicast	The unicast packet can't be routed as destination address is not resolved.	5420, 6200, 6300, 6400, 8100, 8360 Switch series.
Unicast TTL expired	For IPv4: The unicast packet has a Time - to - Live (TTL) value of 0 or 1. For IPv6: The unicast packet has a hop limit value of 0 or 1.	5420, 6200, 6300, 6400, 8100, 8360 Switch series.
Security Ingress IP Lockdown	Packet's source IP address, VLAN, MAC or Interface does not match the IP binding entry.	5420, 6200, 6300, 6400, 8100, 8360 Switch series.

Drop reason	Example Cause	Supported Platforms
Security Ingress MAC Lockout	The packet is sourced from a locke d - out MAC address.	5420, 6200, 6300, 6400, 8100, 8 360 Switch series.
IPTCAM invalid IP address	Packet has invalid source or destin ation IP address. (For example, an invalid IPv4 add ress of **0.0.0.0** or **127.0.0.0**, or **24 0.0.0.0**, or an invalid IPv6 address of **::** or **::1**.)	5420, 6200, 6300, 6400, 8100, 8 360 Switch series.
Security learn	Packet is received on a security - enabled interface and is learned fo r the first time.	5420, 6200, 6300, 6400, 8100, 8 360 Switch series.
Security drop	Packet is received on a security - enabled interface but is not allowe d, as it as not authorized on the V LAN.	5420, 6200, 6300, 6400, 8100, 8 360 Switch series.
Meter drop	The packet is dropped because of exceeding the rate limit configured on port or via a port - access polic y.	5420, 6200, 6300, 6400, 8100, 8 360 Switch series.
ASIC drop	The packet is dropped by ASIC be cause of parity error, parser error o r invalid lookup entry.	5420, 6200, 6300, 6400, 8100, 8 360 Switch series

For 5420, 6200 6300, 6400 Switch Series:

If the **drop end** timestamp has the same value as the **drop star**t timestamp, that means only one packet of the flow was dropped. If the drop end timestamp is different from the drop start timestamp, that means more than one packet was dropped within 30 seconds.

Subtopics

- **Flow monitors**
- **Flow Records**
- **Flow Exporters**
- **Destinations**
- **Configuring IP Flow Information Export on 5420 and 6200 Switches**
- **Role-based IPFIX**
- **FAQs and Troubleshooting**
- **Flow monitoring commands**

Flow monitors

A flow monitor is applied to an interface to perform network traffic monitoring. A flow monitor consists of a flow record, a flow cache, and optional flow exporters. A flow record must be created and assigned to the flow monitor for the monitoring process to function. Flow data is compiled from the network traffic on the interface and stored in the flow cache based on the match (key) and collect (non-key) fields in the flow record. Data from the flow cache is exported by the flow exporters assigned to the flow monitor. 6200 series support a maximum of sixteen flow monitors with a limit of two flow exporters that can be applied to a single flow monitor.

Flow Records

A flow record defines match (key) fields and collection (non-key) fields. Match fields are the set of fields that define a flow, such as IP address or UDP port. Collection fields are the set of fields that identify information to collect for a flow, such as packet and byte counters. On the 5420, 6200, 6300, 6400, 8100, and 8360 switch series a maximum of sixteen flow records can be created.

There are six mandatory match fields, of which the IP match fields must be of the same type (IPv4 or IPv6).

> **NOTE**
> A flow record is invalid if it does not contain one of the supported sets of match fields.

The supported sets of match fields are:

- IPv4 version
- IPV4 source address
- IPv4 destination address
- IPv4 protocol
- Transport destination port
- Transport source port
- IPv6 version
- IPv6 source address
- IPv6 destination address
- IPv6 protocol
- Transport destination port

- Transport source port

Flow Exporters

A flow exporter defines where and how to export flow reports. Flow exporters are created as standalone entities in the **config** context to provide flow monitors the ability to export flow reports. series support a maximum of sixteen flow monitors with a limit of two flow exporters that can be applied to a single flow monitor.

Destinations

About this task

The destination specifies where flow reports are sent. There is one possible destination for a flow exporter within the 5420 and 6200 switch series:

Procedure

Hostname or IP address of a device with an optional VRF .

Results

A flow exporter can only send flow reports to one destination. The destination type specifies which destination to use. If no destination type is specified, the default destination type is the hostname or IP address of a device If no destination type is specified, the default destination type is the hostname or IP address of a device with an optional VRF . If a VRF is not specified, the default VRF will be used. A destination of each type can be configured, but only the one corresponding to the destination type is used. If there is no destination corresponding to the destination type, then the flow exporter configuration is incomplete. If a new destination of a particular type is configured, it will replace the destination of that type that was previously configured.

- Traffic Insight instance

Configuring IP Flow Information Export on 5420 and 6200 Switches

The following list describes the steps required to configure a IP flow information export (IPFIX) solution:

- Step one: Create flow records
- Step two: Configure flow exporter(s)

- Step three: Configure monitor(s)
- Step four: Apply a flow monitors to interface(s)

> **NOTE**
> IPv6 related commands are only applicable to switches that support IPv6 protocol.

Step one: Create Flow Records

Flow Records are used to define the data that will be added to the IPFIX template. This example configures one record for IPv4 and one for IPv6.

```
switch(config)# flow record flowRecordv4
switch(config-flow-record)# match ip protocol
switch(config-flow-record)# match ip source address
switch(config-flow-record)# match ip destination address
switch(config-flow-record)# match ip version
switch(config-flow-record)# match transport destination port
switch(config-flow-record)# match transport source port
switch(config-flow-record)# collect counter bytes
switch(config-flow-record)# collect counter packets
switch(config-flow-record)# collect application name
switch(config-flow-record)# collect timestamp absolute first
switch(config-flow-record)# collect timestamp absolute last
switch(config-flow-record)# collect application https url
switch(config)# flow record flowRecordv6
switch(config-flow-record)# match ipv6 protocol
switch(config-flow-record)# match ipv6 source address
switch(config-flow-record)# match ipv6 destination address
switch(config-flow-record)# match ipv6 version
switch(config-flow-record)# match transport destination port
switch(config-flow-record)# match transport source port
switch(config-flow-record)# collect counter bytes
switch(config-flow-record)# collect counter packets
switch(config-flow-record)# collect application name
switch(config-flow-record)# collect timestamp absolute first
switch(config-flow-record)# collect timestamp absolute last
switch(config-flow-record)# collect application https url
```

Next, use the **show flow record** command to verify the configuration.

Step two: Configure flow exporter(s)

In this step, you can define an exporter to send to an external destination by hostname or IP address, or to an internal destination such as Traffic Insight. The example below configures IPFIX to export data to an external address/hostname:

```
switch(config)# flow exporter flowExternal
switch(config-flow-exporter)# destination type hostname-or-ip-addr
switch(config-flow-exporter)# destination 11.1.1.1
switch(config-flow-exporter)# show flow exporter
-----------------------------------------------------------------------------
----
Flow exporter 'flowExternal
-----------------------------------------------------------------------------
----
Status                   : Accepted
Export Protocol          : ipfix
Destination Type         : Hostname or IP address
Destination              : 11.1.1.1
Transport Configuration
Protocol             : udp
Port                 : 4739
```

To configure IPFIX to export to Traffic Insight, first configure Traffic Insight.

```
switch(config)# traffic-insight TI
switch(config-ti-TI)# source ipfix
switch(config-ti-TI)# monitor topN type topN-flows
switch(config-ti-TI)# monitor appFlow type application-flows
switch(config-ti-TI)# monitor raw-mon type raw-flows
switch(config-ti-TI)# enable
```

Next, configure the flow exporter for Traffic Insight

```
switch(config)# flow exporter flowExpTI
switch(config-flow-exporter)# export-protocol ipfix
switch(config-flow-exporter)# destination type traffic-insight
switch(config-flow-exporter)# destination traffic-insight TI
```

You can use the **show flow exporter** command to verify the flow exporter configuration for Traffic Insight

```
switch(config)# show flow exporter flowExpTI
-----------------------------------------------------------------------------
----
Flow exporter 'flowExpTI'
-----------------------------------------------------------------------------
----
Status                   : Accepted
Export Protocol          : ipfix
Destination Type         : Traffic Insight
```

```
Destination              : TI
Transport Configuration
Protocol             : udp
Port                 : 4739
```

Finally, use the **show run traffic-insight** command to verify the Traffic Insight configuration:

```
switch(config)# show running-config traffic-insight
traffic-insight TI
enable
source ipfix
!
monitor topN type topN-flows entries 5
monitor appFlow type application-flows
monitor raw type raw-flows
```

Step three: Configure the monitor(s)

First, configure an IPv4 flow monitor.

```
switch(config)# flow monitor flowMonv4
switch(config-flow-monitor)# record flowRecordv4
Switch (config-flow-monitor)# exporter flowExternal
switch(config-flow-monitor)# exit
```

Next, configure an IPv6 flow monitor.

```
switch(config)# flow monitor flowMonv6
switch(config-flow-monitor)# record flowRecordv6
switch(config-flow-monitor)# exporter flowExternal
switch(config-flow-monitor)# exit
```

Once both flow monitors are created, use the **show flow monitor** command to verify the flow monitor configurations.

```
switch(config-flow-monitor)# show flow monitor
----------------------------------------------------------------------------
----
Flow monitor 'flowMonv4'
----------------------------------------------------------------------------
----
Status                     : Accepted
Flow Record                : flowRecordv4
Flow Exporter(s)           : flowExternal
Cache Configuration
Inactive Timeout       : 30
Active Timeout         : 1800
----------------------------------------------------------------------------
----
Flow monitor 'flowMonv6'
```

```
-----------------------------------------------------------------------------
----
Status                    : Accepted
Flow Record               : flowRecordv6
Flow Exporter(s)          : flowExternal
Cache Configuration
Inactive Timeout          : 30
Active Timeout  : 1800
```

```
switch(config-flow-monitor)# show flow monitor
-----------------------------------------------------------------------------
----
Flow monitor 'flowMonv4'
-----------------------------------------------------------------------------
----
Status                    : Accepted
Flow Record               : flowRecordv4
Flow Exporter(s)          : flowExternal
Cache Configuration
Inactive Timeout      : 120
Active Timeout        : 1800
-----------------------------------------------------------------------------
----
Flow monitor 'flowMonv6'
-----------------------------------------------------------------------------
----
Status                    : Accepted
Flow Record               : flowRecordv6
Flow Exporter(s)          : flowExternal
Cache Configuration
Inactive Timeout          : 120
Active Timeout  : 1800
```

Role-based IPFIX

If IPFIX monitoring is configured on a port, a network administrator who wants to monitor traffic from clients must preconfigure IPFIX monitor before onboarding the clients. Also, if a client moves from one port to another, the monitor must be reconfigured accordingly. To reduce all these complexities, 5420, 6200, 6300, and 6400 switch series allow you to configure an IPFIX monitor on a port access role.

An IPFIX deployment with monitoring configured on a port uses TCAM to get flow statistics. Since TCAM is a high-demand resource which is shared across multiple protocols including the security protocols, security policy applications could fail if IPFIX consumes most of the TCAM resources. Port access clients can use a large amount of of TCAM resources for policy applications, making it difficult support port access clients with TCAM based IPFIX.

With role-based IPFIX, when a user plugs a device into to a specific port, their user role is applied to the port. Whenever the user plugs that device into to a different port again, the role is applied to that specific port. This gives mobility and accessibility to the user and provides permissions specific to the user's role when the user's device moves across ports. Role-based IPFIX enables colorless port behavior.

There are some mutually exclusive features that will disable role-based IPFIX when they are enabled on the 6200, 6300, and 6400 platforms. The mutually exclusive features are:

- UBT
- MAC Lockout
- Extended Router MAC
- Source Port Filter
- IP Lockdown
- L3VNI

The following behaviors are observed in a deployment with role-based IPFIX:

- If a user applies the IPFIX monitor configuration on the port, it uses the existing TCAM infrastructure.
- If a port has both Role based IPFIX and Port based IPFIX, the Port based IPFIX implementation takes the priority.
- If IPFIX monitor configuration on a port moves from role based to TCAM based or vice-versa, existing IPFIX flows learned on that port are deleted.
- When the user role has an IPFIX configuration and fails to apply it, port access logs it, but it does not prevent the client from getting authorized.
- If port access is in the client mode and if there is more than one client with a conflicting IPFIX monitor configuration, traffic from all the clients will be exported through the IPFIX monitor configured for the first client.
- Role-based IPFIX and TCAM port-based IPFIX implementations can co-exist on a switch.
- Flows from unauthorized clients are not monitored.

These are some limitations to role-based IPFIX:

- Role based IPFIX can be enabled only through port access role assignment.
- Starting with 10.17, role-based IPFIX supports monitoring in both directions. If only ingress is enabled, it monitors traffic in the ingress direction. Port-based IPFIX continues to support only ingress flow monitoring.

FAQs and Troubleshooting

- The following messages are displayed to indicate an illegal argument:
  - % The flow exporter <EXPORTER-NAME> does not exist.
  - % The flow record <RECORD-NAME> does not exist.
  - % The flow monitor <MONITOR-NAME> does not exist.
  - Invalid destination IP address or hostname entered.
  - Unable to create the flow exporter. The maximum allowed number of flow exporters (<max>) has been reached.
  - Unable to create the flow record. The maximum allowed number of flow records (<max>) has been reached.
  - Unable to create the flow monitor. The maximum allowed number of flow monitors (<max>) has been reached.
  - Flow monitor cannot be applied while interface is part of LAG <LAG-NAME>.
  - Flow monitor could not be applied.
  - Flow monitor could not be unapplied.

Flow monitoring commands

Subtopics

- show flow monitor
- show flow record
- show flow-tracking
- show tech ipfix
- show ip flow-table-utilization
- show IPv4 flow endpoint statistics
- show IPv6 flow session statistics

`alert-threshold aggregate active-flows percentage`

Syntax

```
[no] alert-threshold aggregate active-flows percentage <1-100>
```

Description

Sets the alert threshold as a percentage of maximum IP flow table entries per linecard or stack member. An alert is triggered when the number of active flows reaches or exceeds this percentage. The **show ip flow-table-utilization** command displays the IP flow table size on each LC or stack member.

Parameter	Description
`<1-100>`	Percentage of the maximum allowed aggregate active flows for alert.

Examples

Configuring the aggregate active flows alert threshold per linecard or stack member:

```
switch(config-flow-tracking)# alert-threshold aggregate active-flows
percentage 50
```

Removing the aggregate active flows alert threshold:

```
switch(config-flow-tracking)# no alert-threshold aggregate active-flows
percentage
```

Command History

Release	Modification
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-flow-tracking`	Administrators or local user group members with execution righ ts for this command.

`alert-threshold endpoint active-flows`

Syntax

```
[no] alert-threshold endpoint active-flows <50-1000>
```

Description

Sets the alert threshold for the number of active flows per endpoint. An alert is triggered when an endpoint's active flow count reaches or exceeds this threshold.

Parameter	Description
`<50-1000>`	Endpoint session count alert threshold.

Examples

Configuring the endpoint active flows alert threshold:

```
switch(config-flow-tracking)# alert-threshold endpoint active-flows 50
```

Removing the endpoint active flows alert threshold:

```
switch(config-flow-tracking)#no alert-threshold endpoint active-flows
```

Command History

Release	Modification
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-flow-tracking`	Administrators or local user group members with execution rights for this command.

`alert-threshold endpoint new-connections accepted`

Syntax

```
[no] alert-threshold endpoint new-connections accepted <10-500>
```

Description

Sets the alert threshold for the number of new connections accepted by an endpoint within a 30-second monitoring window. An alert is triggered when an endpoint's accepted connection count reaches or exceeds this threshold.

Parameter	Description
`<10-500>`	Endpoint new connections accepted alert threshold.

Examples

Configuring the endpoint new connections accepted alert threshold:

```
switch(config-flow-tracking)# alert-threshold endpoint new-connections
accepted 50
```

Removing the endpoint new connections accepted alert threshold:

```
switch(config-flow-tracking)#no alert-threshold endpoint new-connections
accepted
```

Command History

Release	Modification
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-flow-tracking`	Administrators or local user group members with execution rights for this command.

`alert-threshold endpoint new-connections initiated`

Syntax

```
[no] alert-threshold endpoint new-connections initiated <10-500>
```

Description

Sets the alert threshold for the number of new connections initiated by an endpoint within a 30-second monitoring window. An alert is triggered when an endpoint's initiated connection count reaches or exceeds this threshold.

Parameter	Description
`<10-500>`	Endpoint new connections initiated alert threshold.

Examples

Configuring the endpoint new connections initiated alert threshold:

```
switch(config-flow-tracking)# alert-threshold endpoint new-connections
initiated 50
```

Removing the endpoint new connections initiated alert threshold:

```
switch(config-flow-tracking)#no alert-threshold endpoint new-connections
initiated
```

Command History

Release	Modification
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-flow-tracking`	Administrators or local user group members with execution righ ts for this command.

`diag-dump ipfix basic`

Syntax

```
diag-dump ipfix basic
```

Description

Displays diagnostic information for IPFIX.

Examples

```
diag-dump ipfix basic
==========================================================================
[Start] Feature ipfix Time : Tue Apr 11 02:23:03 2023
==========================================================================
--------------------------------------------------------------------------
[Start] Daemon ipfixd
--------------------------------------------------------------------------
```

```
- IPFIX Record Cache dump -
- IPFIX Record ipfix -
....
:- IPFIX Monitor v6ti completed -
- End of IPFIX Monitor Cache dump -
--------------------------------------------------------------------------
[End] Daemon ipfixd
--------------------------------------------------------------------------
--------------------------------------------------------------------------
[Start] Daemon ops-switchd
--------------------------------------------------------------------------
Key format: <traffic_type>_<coalescence_id>_<agent_id>_<asic_port>
Key                               TCAM Entry ID    Count
--------------------------------- ---------------- -----
1_1532781829_3_20                 0xffff7c7e7a00   1
1_3217499901_1_12                 0xffff91187580   1
1_3217499901_1_13                 0xffff91183d80   1
1_3217499901_1_14                 0xffff91186e80   1
....
--------------------------------------------------------------------------
[End] Daemon ops-switchd
--------------------------------------------------------------------------
==========================================================================
[End] Feature ipfix
==========================================================================
Diagnostic-dump captured for feature ipfix
```

Command History

Release	Modification
10.15	Command introduced on 6200 Switch series.
10.16	Command introduced on the 5420 switch series.

Command Information

Platforms	Command context	Authority
5420 6200	Manager (#)	Administrators or local user group members with execution righ ts for this command.

`description`

Syntax

```
description <DESCRIPTION>
no description <DESCRIPTION>
```

Description

Configures the description for the following telemetry options:

- flow monitor in the **config-flow-monitor** context
- flow congestion monitor in the **config-flow-congestion-monitor** context
- flow exporter in the **config-flow-exporter** context
- flow record in the **config-flow-record** context

The no form deletes a flow congestion monitor.

Parameter	Description
`<DESCRIPTION>`	Displays a string of 256 characters, maximum, including spaces.

Examples

Adding or modifying the description of flow monitor, **flow-monitor-1**:

```
switch(config)# flow monitor flow-monitor-1
switch(config-flow-monitor)# description Used for analyzing basic ipv4
traffic
```

Adding or modifying the description of flow record, **flow-record-1**:

```
switch(config)# flow record flow-record-1
switch(config-flow-record)# description Used for basic traffic analysis
```

Adding or modifying the description of flow exporter, **flow-exporter-1**:

```
switch(config)# flow exporter flow-exporter-1
switch(config-flow-exporter)# description Exports flows to 10.2.3.45:2055
```

Removing the description of flow monitor, **flow-monitor-1**:

```
switch(config)# flow exporter flow-monitor-1
switch(config-flow-exporter)# no description
```

Command History

Release	Modification
10.16	Command introduced

Command Information

Platforms	Command context	Authority

5420
6200
	`config-flow-record`
	`config-flow-monitor`
	`config-flow-congestion-monitor`
	`config-flow-exporter`
		Administrators or local user group members with execution righ ts for this command.

`exporter`

Syntax

```
exporter <EXPORTER-NAME>
no exporter <EXPORTER-NAME>
```

Description

Assigns a flow exporter to **flow monitor** in the **config-flow-monitor** context and **flow congestion monitor** in the **config-flow-congestion-monitor context**.

The **no** form of this command removes the flow exporter from the selected monitor.

Parameter	Description
`<EXPORTER-NAME>`	Specifies the name of the flow exporter assigned to the monitor.

NOTE

A maximum of two flow exporters can be applied to each flow monitor, and one flow exporter can be applied to each flow congestion monitor.

A maximum of one flow exporter can be applied to each flow monitor, and one flow exporter can be applied to each flow congestion monitor.

Examples

Assigning flow exporter, **flow-exporter-1**, to flow monitor, **flow-monitor-1**:

```
switch(config)# flow monitor flow-monitor-1
switch(config-flow-monitor)# exporter flow-exporter-1
```

Removing a flow exporter assigned to flow monitor, **flow-monitor-1**:

```
switch(config)# flow monitor flow-monitor-1
switch(config-flow-monitor)# no exporter flow-exporter-1
```

Attempting to assign non-existent flow exporter, **flow-exporter-5**, to flow monitor, **flow-monitor-1**:

```
switch(config)# flow monitor flow-monitor-1
switch(config-flow-monitor)# exporter flow-exporter-5
Flow exporter 'flow-exporter-5' does not exist.
switch(config-flow-monitor)#
```

Assigning flow exporter, **flow-exporter-1**, to flow congestion monitor, **congestion-monitor-1**:

```
switch(config)# flow congestion-monitor congestion-monitor-1
switch(config-flow-congestion-monitor)# exporter flow-exporter-1
```

Removing a flow exporter assigned to flow congestion monitor:

```
switch(config)# flow congestion-monitor congestion-monitor-1
switch(config-flow-congestion-monitor)# no exporter flow-exporter-1
```

Attempting to assign non-existent flow exporter, **flow-exporter-5**, to flow congestion monitor, **congestion-monitor-1**:

```
switch(config)# flow congestion-monitor congestion-monitor-1
switch(config-flow-congestion-monitor)# exporter flow-exporter-5
Flow exporter 'flow-exporter-5' does not exist.
switch(config-flow-congestion-monitor)#
```

Assigning more than one flow exporter to flow monitor, **flow-monitor-2**:

```
switch(config)# flow monitor flow-monitor-2
switch(config-flow-monitor)# exporter flow-exporter-2
switch(config-flow-monitor)# exporter flow-exporter-3
```

Attempting to assign flow exporter, **flow-exporter-6**, to monitor, **flow-monitor-3**, when two exporters are assigned:

```
switch(config)# flow monitor flow-monitor-3
switch(config-flow-monitor)# exporter flow-exporter 4
switch(config-flow-monitor)# exporter flow-exporter 5
switch(config-flow-monitor)# exporter flow-exporter 6

Cannot assign more than two flow exporters to a flow monitor.

switch(config-flow-monitor)#
```

Command History

Release	Modification
10.16	Command introduced

Command Information

Platforms	Command context	Authority
5420 6200	`config-flow-monitor` `config-flow-congestion-monitor`	Administrators or local user group members with execution righ ts for this command.

`flow exporter`

Syntax

```
flow exporter <name>
   description <description>
   destination
```

```
<hostname>
[vrf <vrfname>]
<ipaddr>
[vrf <vrfname>]
<ip6addr>
[vrf <vrfname>]
type
hostname-or-ip-addr
traffic-insight}
traffic-insight <instance-name>
   no ...
   template data timeout <timeout>
   transport udp <port>
```

Description

A flow exporter is the part of the IP Flow Information Export (IPFIX) feature that defines how a flow monitor exports flow reports. You can assign the same flow exporter configuration to more than one flow monitor. Each flow exporter includes a destination setting that identifies the device to which the flow reports are sent

Parameter	Description
`<name>`	Name of the flow exporter, up to 64 characters.
`description <description>`	A description of the flow exporter, up to 256 characters and spa ces.
`destination`	Configure the export destination
`<hostname>`	The exporter sends flow records to the specified hostname dest ination. The hostname can be a string of up to 64 characters.
`<IPaddr>`	The exporter sends flow records to this IPv4 address destinatio n.
`<ip6addr>`	The exporter sends flow records to this IPv6 address destinatio n.
`vrf <vrfname>`	You can optionally include the name of the destination VRF in t he destination definition on 5420, 6200, 6300, 6400, 8100, 83 60, 9300S Switch series.
`type`	Configure the type of the destination.
`hostname-or-ip-addr`	Define the destination type as a hostname or IP address.
`traffic-insight <name>`	Define the destination type as a traffic insight instance.

Parameter	Description
`no ...`	Negate any configured parameter.
`traffic-insight <INSTANCE-NAME>`	Specify the a Traffic Insight instance to be used as the destinati on.
`template data timeout <timeout>`	A flow exporter template describes the format of exported flow reports. Therefore, flow reports cannot be decoded properly wi thout the corresponding templates. This setting defines how o ften the flow exporter will resend templates to the flow monitor. The supported range is 1 - 86400 seconds, and the default is 6 00 seconds.
`transport udp <port>`	Transport protocol and port for sending flow record reports. Th e default port is port 4739.

Usage

The following table shows the maximum supported of flow monitors and flow exporters for each switch model.

Switch	Maximum Flow Monitors	Maximum Flow Exporters
5420	16	Two flow exporters can be applied to a single flow monitor.
6200	16	Two flow exporters can be applied to a single flow monitor.

Examples

The following example creates a flow exporter configuration named **exporter-1**:

```
switch(config)# flow exporter exporter-1
switch(config-flow-exporter)# dscp 34
switch(config-flow-exporter)# destination 192.0.2.1 vrf VRF1
switch(config-flow-exporter)# template data timeout 1200
switch(config-flow-exporter)# description Exports flows to 192.0.2.1
```

The following example sets a Traffic Insight instance as the destination for a flow exporter:

```
switch(config)# flow exporter exporter-3
switch(config-flow-exporter)# destination type traffic-insight
switch(config-flow-exporter)# destination traffic-insight instance-1
```

The following example adds a destination of each possible type and set **hostname-or-ip-addr** as the type to use:

```
switch(config)# flow exporter exporter-4
switch(config-flow-exporter)# destination collector-1

```

```
switch(config-flow-exporter)# destination traffic-insight instance-1
switch(config-flow-exporter)# destination type hostname-or-ip-addr
```

The following example sets an IPv4 address as the destination for a flow exporter:

```
switch(config)# flow exporter exporter-1
switch(config-flow-exporter)# destination type hostname-or-ip-addr
switch(config-flow-exporter)# destination 192.168.0.1
```

The following example sets a hostname as the destination for a flow exporter:

```
switch(config)# flow exporter exporter-1
switch(config-flow-exporter)# destination type hostname-or-ip-addr
switch(config-flow-exporter)# destination collector1
```

The following example sets an IPv6 address as the destination for a flow exporter:

```
switch(config)# flow exporter exporter-1
switch(config-flow-exporter)# destination type hostname-or-ip-addr
switch(config-flow-exporter)# destination 2001:db87::8a2e:370a:7334
```

The following example sets an IPv4 address as the destination for a flow exporter with the VRF to which the IPv4 address belongs:

```
switch(config)# flow exporter exporter-2
switch(config-flow-exporter)# destination type hostname-or-ip-addr
switch(config-flow-exporter)# destination 192.0.2.1 vrf VRF1
```

The following example sets a Traffic Insight instance as the destination for a flow exporter:

```
switch(config)# flow exporter exporter-3
switch(config-flow-exporter)# destination type traffic-insight
switch(config-flow-exporter)# destination traffic-insight instance-1
```

The following example adds a destination of each possible type and set **hostname-or-ip-addr** as the type to use:

```
switch(config)# flow exporter exporter-4
switch(config-flow-exporter)# destination collector-1
switch(config-flow-exporter)# destination traffic-insight instance-1
switch(config-flow-exporter)# destination type hostname-or-ip-addr
```

The following example removes the destination type of **traffic-insight** from a flow exporter:

```
switch(config)# flow exporter exporter-3
switch(config-flow-exporter)# no destination traffic-insight
```

The following example removes the destination type of **hostname-or-ip-addr** from a flow exporter:

```
switch(config)# flow exporter exporter-1
switch(config-flow-exporter)# no destination
```

Command History

Release	Modification
10.16	Command introduced on the 5420 switch series.
10.15	Command introduced on 6200 Switch series.

Command Information

Platforms	Command context	Authority
5420 6200	`config` `config-flow-exporter`	Administrators or local user group members with execution righ ts for this command.

`flow monitor`

Syntax

```
flow monitor <name>
    exporter <name>
cache timeout {inactive <timeout}|{active <timeout}
    description <description>
    record <name>
```

Description

On HPE Aruba Networking 5420, 6200, 6300, 6400, 8325, 8325H, 8325P, 8100, 8360, 9300, 9300S, and 10040 Switch series, a flow monitor is the part of the IP Flow Information Export (IPFIX) feature that performs network monitoring for the selected interface. A flow monitor configuration consists of a flow record, a flow cache, and one or more associated flow exporters. A flow monitor compiles data from the network traffic on the interface and stores it in the flow cache in a format defined by the flow record. The flow exporters associated with the monitor then export data from the flow cache to the flow exporter destination.

> **NOTE**
> HPE Aruba Networking 5420, 6200, 6300, 6400, 8100, 8360 Switch series support a maximum of sixteen flow monitors with a limit of two flow exporters that can be applied to a single flow monitor. If no software augmentation of flows is required, there is no need to configure a flow collector or flow monitor.

Parameter	Description
`<name>`	Name of the flow monitor, up to 64 characters.
`cache timeout active <timeout>`	Use the cache timeout parameter to define an active or inactive timeout for the flow monitor. A flow monitor closes a flow sessio n that is active for longer than the active timeout or inactive for longer than the inactive timeout. The active timeout range is 30 - 604800. The default active ti me out value is 1800 and inactive timeout value is 30.
`cache timeout inactive <timeout>`	Use the cache timeout parameter to define an inactive timeout for the flow monitor. A flow monitor closes a flow session that is active for longer than the active timeout or inactive for longer t han the inactive timeout. For 5420, 6200, 6300, 6400, 8100, 8360, 9300S, 10040 Switc h Series, the inactive timeout range is 30 - 604800. The defaul t active time out value is 1800 and inactive timeout value is 30.
`description`	A description up to 256 characters long, including spaces.
`exporter <name>`	Assign a flow exporter to a flow monitor.
`record <name>`	(For HPE Aruba Networking 5420, 6200, 6300, 6400, 8100, 8 325, 8325P, 8325H, 8360, 9300, 9300S, 10040 Switch series) Assigns a flow record to a flow monitor.

Examples

The following example creates a flow monitor configuration named **monitor-1**.

```
switch(config)# flow monitor monitor-1
switch(config-flow-monitor)# description Monitor for analyzing basic ipv4
traffic
switch(config-flow-monitor)# exporter flow-exporter-1

switch(config-flow-monitor)# record flow-record-1
switch(config-flow-monitor)# cache timeout inactive 120
switch(config-flow-monitor)# cache timeout active 1500
```

The following workflow changes the flow record assigned to a flow monitor.

```
switch(config)# flow monitor flow-monitor-1
switch(config-flow-monitor)# record flow-record-2
```

Create more than the maximum number of allowed flow monitors

```
switch(config)# flow monitor monitor-1
switch(config)# flow monitor monitor-2
<--OUTPUT OMITTED FOR BREVITY-->
switch(config)# flow monitor monitor-16
switch(config)# flow monitor monitor-17
No more than 16 flow monitors can be configured. Another flow monitor
must be removed first.
```

Assign more than one flow exporter to flow monitor **flow-monitor-2**

```
switch(config)# flow monitor flow-monitor-2
switch(config-flow-monitor)# exporter flow-exporter-2
switch(config-flow-monitor)# exporter flow-exporter-3
```

Add or modify the description of flow record **flow-record-1**

```
switch(config)# flow record flow-record-1
switch(config-flow-record)# description Used for basic traffic analysis
```

Add or modify the description of flow exporter **flow-exporter-1**

```
switch(config)# flow exporter flow-exporter-1
switch(config-flow-exporter)# description Exports flows to 10.2.3.45:2055
switch(config-flow-exporter)# description Exports flows to Traffic Insight
```

Command History

Release	Modification
10.16	Command introduced on the 5420 switch series.
10.15	Command introduced on 6200 Switch series.

Command Information

Platforms	Command context	Authority
5420 6200	`config` `config-flow-monitor`	Administrators or local user group members with execution righ ts for this command.

`flow record`

Syntax

```
flow record <record-name>
   match
ip {source|destination} address
ip protocol|version
ipv6 {source|destination} address
ipv6 protocol|version
transport {source | destination} port
   collect
application name
application https url
application dns response-code
forwarding-status
drop ingress-exceptions
egress vlan
egress interface
egress queue
counter {packets|bytes}
tcp establishment-time
transport {source | destination} port
timestamp absolute first
timestamp absolute last
   descriptioN <description>
```

Description

Creates or modifies a flow record and switches to the **config-flow-record** context for the flow record. Define data to be included in a flow record by configuring flow record match and collect fields.

A flow record defines match (key) fields and collection (non-key) fields. Customers configure flow records with **match** (key) fields and **collect** (non-key) fields. Match fields are the set of fields that define a flow, such as IP address or UDP port. Collect fields are the set of fields that identify information to collect for a flow, such as packet and byte counters.

Traffic with matching attributes (for example, traffic coming from the same interface, sent to the same destination with the same protocol) are classified as a single flow. Information for some or all of the matched settings can be collected and exported to a destination defined by the flow exporter assigned to the flow monitor.

> **NOTE**
>
> Traffic must match a match rule definition before it can be collected and sent. You cannot collect and send data that is not matched.

> **NOTE**
>
> For 5420, 8325, 8325H, 8325P, 9300 and 9300S Switch series, a maximum of one flow record can be created.
>
> For 6200, 6300, 6400, 8100, and 8360 Switch series, a maximum of 16 flow records can be created.

> **NOTE**
>
> It is advised to configure collect egress interface and queue also when threshold is being configured.

Parameter	Description
`<record-name>`	Name of the flow monitor, up to 64 characters.
`match`	match traffic according to one or more of the following key attri butes: • **ip source**: Match traffic from the same IPv4 source. • **ip destination**: Match traffic to the same IPv4 destination. • **ip protocol**: Match traffic using the same IP version • **ip version**: Match traffic using the same IP protocol • **ipv6 source**: Match traffic from an IPv6 source. • **ipv6 destination**: Match traffic to an IPv6 destination. • **ipv6 protocol**: Match traffic using the same IPv6 version • **ipv6 version**: Match traffic using the same IPv6 protocol • **transport {source | destination} port**: Match traffic by sou rce or destination transport port
`description`	A description for the flow record up to 256 characters long, incl uding spaces.
`collect`	Configures data fields to be included a flow record.

Parameter	Description
	• **application name:** Specify the application name as a non - key field in a flow record. • **application https url**: Specify the HTTP/HTTPS applicatio n URL as a non - key field in a flow record. • **application dns response - code**: Specify the DNS param eters and DNS response code as a non - key field in the flo w record. • **application tcp establishment - time**: Specifies TCP conn ection establishment time as a non - key field in a flow reco rd • **drop ingress - exceptions**: On 5420. 6200. 6300, 6400, 8 100, 8325, 8325P, 8325H, 9300 and 10000 Switch series, this enables the IPFIX drop - reason feature, which specifie s drop ingress - exceptions as a non - key field in a flow re cord. • **counter bytes**: Collect counter data for bytes in the flow. By te count represents the number of incoming bytes since the previous report. • **counter packets**: Collect counter data for packets in the flo w. Packet count represents the number of incoming packets since the previous report. • **egress interface**: Specifies an egress interface as a non - key field in a flow record. • **egress queue**: Specifies an egress queue as a non - key fiel d in a flow record. • **egress vlan**: Specifies an egress VLAN ID as a non - key fie ld in a flow record. • **fowarding status**: Specifies forwarding status as a non - key field in a flow record • **timestamp absolute first**: Collect absolute timestamp of th e first packet observed. • **timestamp absolute last**: Collect absolute timestamp of th e last packet observed. • **tcp establishment - time**: Specifies TCP connection establi shment time as a non - key field in a flow record.

Examples

Adding IPv4 and transport match fields to **flow-record-1:**

```
switch(config)# flow record flow-record-1
switch(config-flow-record)# match ip source address
switch(config-flow-record)# match ip destination address
switch(config-flow-record)# match ip protocol
switch(config-flow-record)# match ip version
switch(config-flow-record)# match transport source port
switch(config-flow-record)# match transport destination port
switch(config-flow-record)# description Record used for basic ipv4 traffic
analysis
```

The following example adds a TCP connection establishment time to flow record **flow-record-1** as a collect field. During this process, ARC inspects the first few packets of each flow and only those packets are copied to the switch. Time measurement for established TCP flows is possible during the TCP connection establishment phase, and all three packets are received in the order.

```
switch(config-flow-record)# collect application tcp establishment-time
```

Adding counter and timestamp collect fields to **flow-record-1**:

```
switch(config)# flow record flow-record-1

switch(config-flow-record)# collect counter packets
switch(config-flow-record)# collect counter bytes
switch(config-flow-record)# collect timestamp absolute first
switch(config-flow-record)# collect timestamp absolute last
```

Adding IPv4 match fields to flow record **flow-record-1** using the **ip** keyword:

```
switch(config)# flow record flow-record-1
switch(config-flow-record)# match ip source address
switch(config-flow-record)# match ip destination address
switch(config-flow-record)# match ip protocol
switch(config-flow-record)# match ip version
```

Adding IPv6 match fields to flow record **flow-record-2**:

```
switch(config)# flow record flow-record-2
switch(config-flow-record)# match ipv6 source address
switch(config-flow-record)# match ipv6 destination address
switch(config-flow-record)# match ipv6 protocol
switch(config-flow-record)# match ipv6 version
```

Adding IPv6 collect fields to flow record **flow-record-2**:

```
switch(config)# flow record flow-record-2
switch(config-flow-record)# collect application tcp establishment-time
switch(config-flow-record)# collect egress-vlan
switch(config-flow-record)# collect egress interface
switch(config-flow-record)# collect forwarding-status
switch(config-flow-record)# collect egress queue
```

Adding IPv4 match fields to flow record **flow-record1**:

```
switch(config)# flow record flowrecord1
switch(config-flow-record)# match ip source address
switch(config-flow-record)# match ip destination address
switch(config-flow-record)# match ip protocol
switch(config-flow-record)# match ip version
switch(config-flow-record)# match transport source port
switch(config-flow-record)# match transport destination port
switch(config-flow-record)# collect application name
switch(config-flow-record)# collect counter bytes
switch(config-flow-record)# collect counter packets
switch(config-flow-record)# collect egress interface
switch(config-flow-record)# collect egress queue
switch(config-flow-record)# collect egress vlan
switch(config-flow-record)# collect forwarding-status
switch(config-flow-record)# collect application https url
switch(config-flow-record)# collect application tcp establishment-time
switch(config-flow-record)# collect timestamp absolute first
switch(config-flow-record)# collect timestamp absolute last
```

Removing the IPv4 destination address match field from flow record **flow-record-1** using the **ip** keyword:

```
switch(config)# flow record flow-record-1
switch(config-flow-record)# no match ip destination address
```

Removing the IPv4 destination address match field from flow record **flow-record-1** using the **ipv4** keyword:

```
switch(config)# flow record flow-record-1
switch(config-flow-record)# no match ipv4 destination address
```

Removing the transport destination port match field from flow record **flow-record-1:**

```
switch(config)# flow record flow-record-1
switch(config-flow-record)# no match transport destination port
```

Adding queue congestion threshold to **flow-record-1** as a collect field:

```
switch(config)# flow record flow-record-1
switch(config-flow-record)# collect egress queue threshold
```

NOTE

To export queue threshold attribute, capture-flows threshold profile should be configured on the egress interfaces. If threshold profile is not configured, congested flows can't be detected and the IEs values are exported as 0. If threshold profile is configured, but collect field is not configured, these IEs are not exported. So in ether case, congested flows are not known. For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.17	The **collect drop ingress - exceptions** parameter is introduced on 5420, 6200, 6200, 8325H, 9300 and 10000 Switch series. Support for **egress vlan** and **egress interface** parameters is added on 8100 an d 8360 Switch series.
10.16	Command introduced on the 5420 switch series.
10.15.1000	The **collect tcp establishment - time** parameter is introduced on 6200 Switch series.
10.15	Command introduced on the 6200 and 9300S Switch series
10.13	Added **application https url** and **dns response - code** parameters.

Command Information

Platforms	Command context	Authority
5420 6200	`config` `config-flow-record`	Administrators or local user group members with execution righ ts for this command.

`flow statistics`

Syntax

```
flow statistics [enable] | ingress-only
no flow statistics [enable] | ingress-only
```

Description

Configures collection of statistics for flows that are tracked.

Parameter	Description
`enable`	Enables collection of statistics for flows that are tracked.

Parameter	Description
`ingress-only`	Enables collection of statistics for flows that are tracked on ing ress interface - only.

CAUTION

If any of the mutually exclusive configurations are present on the switch, flow statistics will not be enabled and an error message is displayed.

NOTE

Flow statistics gets operational only after you run the flow-tracking enable command; systems with flow-tracking already enabled must disable it before configuring the statistics and then re-enable, or toggle it afterwards.

Examples

Enable flow statistics collection:

```
switch(config-flow-tracking)# flow statistics enable
```

Disable flow statistics collection:

```
switch(config-flow-tracking)#no flow-statistics enable
```

Enabling flow collection on ingress interface only:

```
switch(config-flow-tracking)# flow statistics ingress-only
```

Disabling flow collection on ingress interface only:

```
switch(config-flow-tracking)#no flow statistics ingress-only
```

Command History

Release	Modification
10.16	Command introduced on the 5420 switch series.
10.15	Command introduced on 6200 Switch Series

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

flow-tracking

Syntax

```
flow-tracking
   enable
   icmp-ageout
   interface-flow-limit
   flow
statistics
enable
ingress-only
   tcp-ageout
   track
icmp
   udp-ageout
   no ...
```

Description

Configures flow tracking for TCP and UDP flows, and optionally, ICMP flows. The **no** form of this command deletes the flow tracking configuration context.

In order to optimize the flow removal process, flows that have aged-out are flushed in batches. A flow that has aged out is flushed only when the next batch processes. This can cause some flows to stay inactive for a slightly longer time than the value configured here.

Parameter	Description
`enable`	Enables flow tracking.
`icmp-ageout`	Configures an age - out time for ICMP flows, in seconds. Range: 10 to 86400. Default: 15.
`interface-flow-limit`	Configures global concurrent flow limit for flow tracking enable d interfaces. Range: For 6200 Switch series, 64 to 5000. For 54 20 Switch series, 64 to 10000. Default: none.
`flow`	Configures flow parameters.
`  statistics`	Configures flow statistics.
`enable`	Enables flow statistics collection.

Parameter	Description
`ingress-only`	Restricts flow tracking to the ingress direction. **NOTE** The **application visibility** feature requires traffi c flows in both the ingress and egress direction s. Configuring this option prevents features that depend on bi - directional flows from functionin g.
`tcp-ageout`	Configures age - out time for established TCP flows in seconds. Range: 120 to 86400. Default: 600.
`track icmp`	Enables tracking of ICMP flows, in addition to the TCP/UDP flow s tracked by default.
`udp-ageout`	Configures age - out time for established UDP flows in seconds . Range: 30 to 86400. Default: 30.

Examples

Configuring flow tracking:

```
switch(config)# flow-tracking
switch(config-flow-tracking)#
```

Deleting flow tracking:

```
switch(config)# no flow-tracking
switch(config)#
```

Enabling flow tracking:

```
switch(config)# flow-tracking
switch(config-flow-tracking)# enable
```

Disabling flow tracking:

```
switch(config)# flow-tracking
switch(config-flow-tracking)# no enable
```

Configuring an established ICMP flow age-out to 600 seconds:

```
switch(config)# flow-tracking
switch(config-flow-tracking)# icmp-ageout 600
```

Removing an established ICMP flow age-out of 600 seconds:

```
switch(config)# flow-tracking
switch(config-flow-tracking)# no icmp-ageout 600
```

Configuring an established TCP flow age-out to 1000 seconds:

```
switch(config)# flow-tracking
switch(config-flow-tracking)# tcp-ageout 1000
```

Removing an established TCP flow age-out of 1000 seconds:

```
switch(config)# flow-tracking
switch(config-flow-tracking)# no tcp-ageout 1000
```

Configuring an established UDP flow age-out to 1000 seconds:

```
switch(config)# flow-tracking
switch(config-flow-tracking)# udp-ageout 1000
```

Removing an established UDP flow age-out of 1000 seconds:

```
switch(config)# flow-tracking
switch(config-flow-tracking)# no udp-ageout 1000
```

Configuring global level interface flow limit to 256 interfaces:

```
switch(config)# flow-tracking
switch(config-flow-tracking)# interface-flow-limit 256
```

Removing global level interface flow limit to 256 interfaces:

```
switch(config)# flow-tracking
switch(config-flow-tracking)# no interface-flow-limit 256
```

Command History

Release	Modification
10.17	The **flow statistics ingress - only** parameter was introduced.
10.16	Command introduced on the 5420 switch series.
10.15	Command introduced on 6200 Switch Series

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`ip|ipv6 flow monitor (interface)`

Syntax

```
[no] ip|ipv6 flow monitor (interface)
```

Description

Enable flow monitoring on inbound and outbound interfaces by assigning a flow monitor to that interface. Only physical interfaces and LAG interfaces can be monitored. A flow monitor cannot be applied to an interface that is part of a LAG. If an unsupported application is attempted, an error message will be displayed. If the flow monitor is associated with a flow record that contains application fields as collect fields, then Application Recognition should be enabled on the same interface.

The **[no]** form of command disables the flow monitoring.

Examples

Enable a flow monitor configuration named **flow-monitor-1** for IPv4 traffic on a physical interface.

```
switch(config)# interface 1/1/1
switch(config-if)# ip flow monitor flow-monitor-1 in
```

Associate a flow monitor configuration named **flow-monitor-2** for IPv4 traffic on a LAG interface.

```
switch(config)# interface lag 1
switch(config-if)# ip flow monitor flow-monitor-2 in
```

Associate a flow monitor configuration named **flow-monitor-3** for IPv6 traffic on a physical interface.

```
switch(config)# interface 1/1/1
switch(config-if)# ip flow monitor flow-monitor-3 in
```

Associate a flow monitor configuration named **flow-monitor-4** for IPv6 traffic on a physical interface.

```
switch(config)# interface lag 1
switch(config-if)# ipv6 flow monitor flow-monitor-1 in
```

Command History

Release	Modification
10.16	Command introduced on the 5420 switch series.
10.15	Command introduced on the 6200 Switch Series

Command Information

Platforms	Command context	Authority
5420 6200	`config` `config-flow-monitor`	Administrators or local user group members with execution righ ts for this command.

`show flow exporter`

Syntax

```
show flow exporter [<name>]
[statistics]

[vsx-peer]
```

Description

Displays flow exporter configuration and status. When no exporter name is specified, the output of this command displays information for all flow exporters.

The output of this command can indicate the following status types:

- Accepted
- Rejected (Internal error: exporter does not exist)
- Rejected (Internal error: destination type does not exist)
- Rejected (Destination type is hostname or IP address, but no destination is specified)

- Rejected (Destination type is hostname or IP address, but the specified hostname or IP address is invalid)
- Rejected (Destination type is Traffic Insight, but no destination is specified)
- Rejected (Destination type is Traffic Insight, but the specified Traffic Insight instance does not exist)
- Rejected (Destination type is Traffic Insight, but the specified Traffic Insight instance is not enabled)
- Rejected (Destination type is Traffic Insight, but the specified Traffic Insight instance source is not IPFIX)
- Rejected (Internal error: destination type is Traffic Insight, but the specified Traffic Insight instance is invalid)
- Rejected (Internal error: the specified destination VRF is invalid)
- Rejected (Internal error: the specified destination VRF is not ready)

Parameter	Description
`<name>`	Name of the flow exporter.
`statistics`	Adds statistical information about the flow exporter to the outp ut.
`vsx-peer`	Displays flow collector configuration for the VSX peer.

Examples

Display the configuration of all flow exporters:

```
switch# show flow exporter
-----------------------------------------------------------------------------
----
Flow exporter 'exporter-1'

-----------------------------------------------------------------------------
----
Reports sent             : 0
```

Display information with no flow exporters configured

```
switch# show flow exporter
No flow exporters configured.
switch# show flow exporter statistics
No flow exporters configured.
```

Display a flow exporter information with TI as a destination

```
switch# show flow exporter exporter-5
--------------------------------------------------------------------------------
----
Exporter Name           : exporter-5
--------------------------------------------------------------------------------
----
Description             : Exporter configured with TI as the destination
Status                  : Rejected (Destination type is Traffic Insight,
but the specified Traffic Insight instance does not exist)
Export Protocol         : ipfix
Destination Type        : Traffic Insight
Destination             : instance-1
Transport Configuration
Protocol            : UDP
Port                : 2055
```

Display information for a flow exporter that has multiple destinations of different types configured and **hostname-or-ip-addr** is specified as the destination type to use.

```
switch# show flow exporter exporter-6
--------------------------------------------------------------------------------
----
Flow exporter 'exporter-6'
--------------------------------------------------------------------------------
----
Description             : TI and hostname configured, but type is hostname-
or-ip-addr
Status                  : Accepted
Export Protocol         : ipfix
Destination Type        : Hostname or IP address
Destination             : collector-1
Destination VRF         : mgmt
Transport Configuration
Protocol            : UDP
Port                : 4821
```

Command History

Release	Modification
10.16	Command introduced on the 5420 Switch series.
10.15	Command introduced on the 6200 Switch series

Command Information

Platforms	Command context	Authority
5420 6200	Manager (`#`)	Administrators or local user group members with execution righ ts for this command.

`show flow monitor`

Syntax

```
show flow monitor
[statistics | information-elements]

show flow monitor <MONITOR-NAME>
[statistics]
```

Description

Displays flow monitor configuration and status. When no monitor name is specified, the output of this command displays information for all flow monitors.

Parameter	Description
`<name>`	Name of the flow monitor.
`statistics`	Includes the **statistics** parameter to display additional flow and cache statistics.
`information-elements`	Displays the configured information elements for the flow moni tor.

Usage

The output of this command can indicate the following status types:

- Possible status types for all switches
    - Accepted

- Rejected (Internal error: monitor does not exist)
- Rejected (The state of one or more of the assigned flow exporters is rejected)

- Possible Status types for 4100i, 5420 ,6000, 6100, 6200, 6300, 6400, 8100, 8325 ,8325P, 8325H, 8360, 9300, 9300S, and 10040 Switch series:
  - Rejected (A record must be assigned to the monitor, but no record is assigned)
  - Rejected (The state of the assigned record is rejected)
  - Rejected (Internal error: failure in processing the record configuration)

The possible statistics for a flow monitor are:

Statistics name	Meaning	Switches that support this statistic
Current Entries	Current number of flows in the flow cache for this flow monitor	5420 6200
Flows Added	Total number of flows added to the flow cache for this flow monitor since it was created	5420 6200
Total Flows Terminated	Total number of flows removed from the flow cache for this flow monitor since it was created due to any flow end reason	5420 6200
Flows Aged	Number of flows removed from the flow cache for this flow monitor since it was created due to active or inactive timeout	5420 6200
Active Timeout	Number of flows removed from the flow cache for this flow monitor since it was created due to active cache timeout.	
Inactive Timeout	Number of flows removed from the flow cache for this flow monitor since it was created due to inactive cache timeout.	5420 6200
End of Flow Detected	Number of flows removed from the flow cache for this flow monitor since it was created due to the detection of signals indicating the end of the flow. For example, a TCP FIN/RST flag.	
Forced End	Number of flows removed from the flow cache for this flow monitor since it was created due to some external event. For	5420 6200

Statistics name	Meaning	Switches that support this statistic
	example, the shutdown of a flow monitor.	
Current Dropped Entries	Current number of dropped flows in the flow cache for this flow monitor	
Current Forwarded Entries	Current number of forwarded flows in the flow cache for this flow monitor	

Examples

Display information for a flow monitor on 6200, 6300, 6400, 8100 or 8360 Switch series:

```
switch# show flow monitor 'monitor-1
--------------------------------------------------------------------------
----
Flow monitor 'monitor-1'
--------------------------------------------------------------------------
----
Description                : Used for IPv4 traffic analysis
Status                     : Accepted
Flow Record                : record-1
Flow Exporter(s)           : exporter-1, exporter-2
Cache Configuration
Inactive Timeout       : 1800
Active Timeout         : 300
--------------------------------------------------------------------------
----
```

Display information for a flow monitor on a 5420, 9300, 9300S, 10040 Switch series:

```
switch# show flow monitor monitor-1
--------------------------------------------------------------------------
----
Flow monitor 'monitor-1'
--------------------------------------------------------------------------
----
Description                : Used for IPv4 traffic analysis
Status                     : Accepted
Flow Record                : record-1
Flow Exporter(s)           : exporter-1
Cache Configuration
Inactive Timeout       : 1800
Active Timeout         : 300
```

Display information and statistics for a flow monitor on 5420, 6200, 6300, 6400, 8100 or 8360 Switch series:

```
show flow monitor statistics
----------------------------------------------------------------------------
----
Flow monitor 'monitor-1'
----------------------------------------------------------------------------
----
Current Entries          : 2
Flows Added              : 6
Total Flows Terminated   : 4
  Flows Aged             : 2
    Active Timeout       : 1
    Inactive Timeout     : 1
  End of Flow Detected   : 2
  Forced End             : 0
  Flows Aged             : 4
```

Command History

Release	Modification
10.16	Command introduced on the 5420 Switch series.
10.15	Command introduced on the 6200 Switch series.

Command Information

Platforms	Command context	Authority
5420 6200	`config` `config-flow-monitor`	Administrators or local user group members with execution righ ts for this command.

`show flow record`

Syntax

```
show flow record [<name>]
```

Description

Display flow record configuration and status. When no record name is specified, the output of this command displays information for all flow records.

The output of this command can indicate the following status types:

- Accepted
- Rejected (Internal error: failed to process record)
- Rejected (Mix of IPv4 and IPv6 match fields is not allowed. Specify match fields of the same IP version (IPv4 or IPv6))

Parameter	Description
`<name>`	Name of the flow record.

Examples

> **NOTE**
>
> IPv6 related commands are only applicable to switches that support IPv6 protocol.

Display the configuration of a flow record named **flow-record-1**.

```
switch# show flow record record-1
-----------------------------------------------------------------------------
----
Flow record  'record-1'
-----------------------------------------------------------------------------
----
Description             : Used for IPv4 traffic analysis
Status                  : Accepted
Match Fields
    ipv4 destination address
    ipv4 protocol
    ipv4 source addressipv4 version
    transport destination port
    transport source port
Collect Fields
    application name
counter bytes
    counter packets
    application https URL
```

```
    drop ingress-exceptions
```

Display the information of a specific flow record.

```
switch# show flow record record-1
----------------------------------------------------------------------------
----
Flow record  'record-1'
----------------------------------------------------------------------------
----
Description             : Used for IPv4 traffic analysis
Status                  : Accepted
Match Fields
    ipv4 destination address
    ipv4 protocol
    ipv4 source address
    ipv4 version
    transport destination port
    transport source port
Collect Fields
    application https url
    counter bytes
    counter packets
```

Display information for all flow records

```
switch# show flow record
----------------------------------------------------------------------------
----
Flow record  'record-1'
----------------------------------------------------------------------------
----
Description             : Used for IPv4 traffic analysis
Status                  : Accepted
Match Fields
    ipv4 destination address
    ipv4 protocol
    ipv4 source address
    ipv4 version
    transport destination port
    transport source port
Collect Fields
    counter bytes
    counter packets
```

```
--------------------------------------------------------------------------------
----
Flow record  'record-2'
--------------------------------------------------------------------------------
----
Description              : Used for IPv6 traffic analysis
Status                   : Accepted
Match Fields
    ipv6 destination address
    ipv6 protocol
    ipv6 source address
    ipv6 version
    transport destination port
    transport source port
Collect Fields
    application name
    counter bytes
    counter packets
```
```

Display information for a specific flow record

```
switch# show flow record record-3
--------------------------------------------------------------------------------
----
Flow record  'record-3'
--------------------------------------------------------------------------------
----
Description              : Used for IPv4 traffic analysis
destination address, protocol, transport destination port, and transport
source port.)
Match Fields
ipv4 destination address
ipv4 protocol
ipv4 source address
Collect Fields
counter bytes
counter packets
```

Display information with no flow records configured

```
switch# show flow record
No flow records configured
```
```

Command History

Release	Modification
10.17	The output of this command includes **drop ingress - exceptions** information for the 5420 and 6200 Switch series.
10.16	Command introduced on the 5420 switch series.
10.15	Command introduced on the 6200 Switch series

Command Information

Platforms	Command context	Authority
5420 6200	Manager (`#`)	Administrators or local user group members with execution rights for this command.

`show flow-tracking`

Syntax

```
show flow-tracking
```

Description

Displays flow-tracking and statistics collection configurations and status.

Examples

Display the configuration of role based flow tracking on a 5420 or 6200 Switch series.

```
switch (config)# show flow-tracking
Flow Tracking Global Configuration
Configuration status                : Enabled
Operational status                  : Enabled
Failure Reason                      : NA
UDP Ageout                          : 60   (Seconds)
TCP Ageout                          : 600  (Seconds)
ICMP Ageout                         : 15   (Seconds)
Interface Flow limit                : None
Tracked Protocols                   : TCP, UDP
```

```
Alert Thresholds:
     Aggregate active flows percentage : 50
     Endpoint active flows             : 250
     Endpoint new connection initiated : 50
     Endpoint new connection accepted  : 60
Statistics Collection
  Configuration Status          : Enabled
  Operational Status            : Enabled
  Ingress-only                  : Enabled
  Failure Reason                : NA
Flow Tracking Port Configuration
Interface          App Recognition      IPFIX         Operation Status
-----------          -------------      ----------        ----------
1/1/1                Disabled           Disabled          Disabled
1/1/2                Disabled           Disabled          Disabled
1/1/3                Disabled           Disabled          Disabled
1/1/4                Disabled           Disabled          Disabled
1/1/5                Disabled           Disabled          Disabled
1/1/6                Disabled           Disabled          Disabled
1/1/7                Disabled           Disabled          Disabled
1/1/8                Disabled           Disabled          Disabled
1/1/9                Disabled           Disabled          Disabled
1/1/10               Disabled           Disabled          Disabled
1/1/11               Disabled           Disabled          Disabled
1/1/12               Disabled           Disabled          Disabled
1/1/13               Disabled           Disabled          Disabled
1/1/14               Disabled           Disabled          Disabled
1/1/15               Enabled            Disabled          Enabled
1/1/16               Disabled           Disabled          Disabled
1/1/17               Disabled           Disabled          Disabled
1/1/18               Disabled           Disabled          Disabled
1/1/19               Disabled           Disabled          Disabled
1/1/20               Disabled           Disabled          Disabled
1/1/21               Disabled           Disabled          Disabled
1/1/22               Disabled           Disabled          Disabled
1/1/23               Disabled           Disabled          Disabled
1/1/24               Disabled           Disabled          Disabled
1/1/25               Disabled           Disabled          Disabled
1/1/26               Disabled           Disabled          Disabled
1/1/27               Disabled           Disabled          Disabled
1/1/28               Disabled           Disabled          Disabled
```

```
switch(config)# show flow-tracking
Flow Tracking Global Configuration
```

```
Configuration status            : Enabled
Operational status              : Enabled
Failure Reason                  : NA
UDP Ageout                      : 30  (Seconds)
TCP Ageout                      : 600 (Seconds)
ICMP Ageout                     : 15  (Seconds)
Interface Flow limit            : None
Tracked Protocols               : TCP, UDP
Statistics Collection
   Configuration Status         : Enabled
   Operational Status           : Enabled
   Ingress-only                 : Enabled
   Failure Reason               : NA
Flow Tracking Port Configuration
Interface       App Recognition    Reflexive ACL         IPFIX
Operation Status
-----------     -----------       ----------------      ----------
----------
1/1/1              Enabled             Disabled            Enabled
Enabled
1/1/2              Enabled             Disabled            Disabled
Enabled
1/1/3              Enabled             Disabled            Disabled
Enabled
1/1/4              Enabled             Disabled            Disabled
Enabled
1/1/5              Enabled             Disabled            Disabled
Enabled
1/1/6              Enabled             Disabled            Disabled
Enabled
1/1/7              Enabled             Disabled            Enabled
Enabled
1/1/8              Enabled             Disabled            Disabled
Enabled
1/1/9              Enabled             Disabled            Disabled
Enabled
1/1/10             Disabled            Disabled            Disabled
Disabled
1/1/13             Enabled             Disabled            Enabled
Enabled
1/1/14             Enabled             Disabled            Disabled
Enabled
1/1/15             Enabled             Disabled            Disabled
```

```
Enabled
1/1/16              Enabled             Disabled            Disabled
Enabled
1/1/17              Enabled             Disabled            Disabled
Enabled
1/1/18              Enabled             Disabled            Disabled
Enabled
1/1/19              Enabled             Disabled            Disabled
Enabled
1/1/20              Enabled             Disabled            Disabled
Enabled
1/1/21              Enabled             Disabled            Disabled
Enabled
1/1/23              Enabled             Disabled            Disabled
Enabled
1/1/24              Enabled             Disabled            Enabled
Enabled
1/1/28              Disabled            Disabled            Disabled
Disabled
```

Command History

Release	Modification
10.17	Added **Ingress - only** information for Statistics Collection.
10.16	Command introduced on the 5420 switch series.
10.15	Command introduced on 6200 Switch Series.

Command Information

Platforms	Command context	Authority
5420 6200	Manager (`#`)	Administrators or local user group members with execution righ ts for this command.

`show tech ipfix`

Syntax

```
show tech ipfix
```

Description

Shows the IPFIX configuration settings.

If applicable source IP address or source interface is configured for the IPFIX protocol, that configuration is used.

For 6200, 6300,6400,8360,8100 and 9300S Switch Series, If a valid source is configured, the exporter sends flows to an external collector using the effective configured source IP address as the source IP address of the flow packets. In the context of this application, a valid source IP address is any IP address configured in the exporter's VRF namespace.

Examples

The example shows the IPFIX configuration settings.

```
switch#show tech ipfix
====================================================
Show Tech executed on Tue Apr 11 02:43:06 2023
====================================================
====================================================
[Begin] Feature ipfix
====================================================
*********************************
Command : show flow exporter
*********************************
----------------------------------------------------------------------------
----
Flow exporter 'ipfix'
----------------------------------------------------------------------------
----
Status                   : Accepted
Export Protocol          : ipfix
Destination Type         : Traffic Insight
Destination              : t1
Transport Configuration
Protocol             : udp
Port                 : 4739
----------------------------------------------------------------------------
----
Flow exporter 'V6E1'
----------------------------------------------------------------------------
----
....
```

```
=========================================================
[End] Feature ipfix
=========================================================
```

Command History

Release	Modification
10.16	Command introduced on the 5420 switch series.
10.15	Command introduced. on 9300S and 6200 Switch Series

Command Information

Platforms	Command context	Authority
5420 6200	Manager (`#`)	Administrators or local user group members with execution righ ts for this command.

`show ip flow-table-utilization`

Syntax

```
show ip flow-table-utilization
```

Description

Displays the current IP flow table utilization per linecard (LC), including both the maximum capacity and the current number of active flows.

Examples

Configuring the endpoint new connections accepted alert threshold:

```
switch# show ip flow-table-utilization
Line Modules
=================
Slot name | flow-table capacity  | flow-count
----------|----------------------|------------
1/1       | 28672                | 15000
1/3       | 28672                |  5000
```

Command History

Release	Modification
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Manager (`#`)	Administrators or local user group members with execution righ ts for this command.

`show IPv4 flow endpoint statistics`

Syntax

```
show ip flows endpoint <A.B.C.D> [vrf <vrf_name>]
```

Description

Displays the number of active flows of VRF with name **vrf_name**, the flows observed in the last 30 seconds, and details about new connections initiated and accepted by this endpoint during the same period. If the **vrf_name** is not specified, default VRF active flows are displayed.

Parameter	Description
`IPv4 address`	Show endpoint statistics for the IPv4 address. Required
`vrf`	Display the routes in the VRF. Optional
`vrf_name`	Specify the VRF name.

Parameter	Description
	Required

NOTE

The output displays the following information:

- **VRF:** The specified vrf name.
- **Endpoint address:** The specified IPv4 address.
- **Active flows:** Number of active flows for the endpoint.
- **Flows seen last 30sec:** Number of flows observed in the last 30 seconds.
- **No of new connections (last-30sec):**
  - **Initiated:** Number of new connections initiated by the endpoint in the last 30 seconds.
  - **Accepted:** Number of new connections accepted for the endpoint in the last 30 seconds.

Examples

This example shows the number of active flows of the default VRF:

```
switch# show ip flows endpoint 192.168.1.1
VRF                                  : default
Endpoint address                     : 192.168.1.1
Active flows                         : 10
Flows seen last 30sec                : 20
Number of new connections (last-30sec):
   Initiated                         : 2
   Accepted                          : 0
```

Command History

Release	Modification
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Manager (`#`)	Administrators or local user group members with execution righ ts for this command.

`show IPv6 flow session statistics`

Syntax

```
show ipv6 flows endpoint &lt;A.B.C.D&gt; [vrf &lt;vrf_name&gt;]
```

Description

Displays the number of active flows of VRF with name **vrf_name**, the flows observed in the last 30 seconds, and details about new connections initiated and accepted by this endpoint during the same period. If the **vrf_name** is not specified, default VRF active flows are displayed.

Parameter	Description
`IPv6 address`	Show endpoint statistics for the IPv6 address. Required
`vrf`	Display the routes in the VRF. Optional
`vrf_name`	Specify the VRF name.

Parameter	Description
	Required

NOTE

The output displays the following information:

- **VRF:** The specified vrf name.
- **Endpoint address:** The specified IPv6 address.
- **Active flows:** Number of active flows for the endpoint.
- **Flows seen last 30sec:** Number of flows observed in the last 30 seconds.
- **No of new connections (last-30sec):**
  - **Initiated:** Number of new connections initiated by the endpoint in the last 30 seconds.
  - **Accepted:** Number of new connections accepted for the endpoint in the last 30 seconds.

Examples

This example shows the number of active flows of the specified VRF:

```
switch# show ipv6 flows endpoint 2001::1 vrf red
VRF                                  : red
Endpoint address                     : 2001::1
Active flows                         : 10
Flows seen last 30sec                : 20
Number of new connections (last-30sec):
   Initiated                         : 2
   Accepted                          : 0
```

Command History

Release	Modification
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Manager (`#`)	Administrators or local user group members with execution righ ts for this command.

Queue Monitoring

AOS-CX provides multiple features that assist with monitoring queuing behavior such as Queue Monitoring.

The Queue Monitoring feature allows the switch to collect queue statistics at specific time intervals, including queue depth, and display this information as a queue statistics history. This data provides a history of activity for each monitored queue and can be viewed directly on the switch. These aggregate statistics can be used to monitor network health, troubleshoot network issues, and identify normal network behavior patterns and performance anomalies. The feature is disabled by default, but can be enabled on a maximum of 52 interfaces per switch. For more information on enabling this feature, see queue-monitor.

Queue statistics history data can be displayed in the command-line interface as a list or a histogram reflecting the last eight hours of queue statistics history, or as a table showing the previous five minutes of queue statistics history.

As performance anomalies are identified, the interface queue experiencing the anomaly can further analyzed to identify the cause of the behavior. For example, traffic patterns for flows competing for network bandwidth can be examined to determine if corrective action is needed to address potential problems.

If no queue depth is specified, the output of the **show interface queue-monitor** command displays statistics for queues with a minimum depth of one kbyte, which includes all queues which have reached a depth greater than zero.

Subtopics

Queue statistics history

Queue monitoring stores collected queue depth data in a time-series database within the switch. This data, accessible on the switch for reviewing the queue statistics history, helps with identifying network issues or behavior patterns.

Queue depth

Queue depth is the memory size required to buffer packets in a queue during congestion. The value changes depending on the arrival rate of the packets destined to egress a given queue, and the rate at which that queue is able to transmit packets.

Data retention limits

Data retention limits are enforced within the switch by reducing time precision of the stored data within specific windows of time. The retention duration for queue statistics data is derived from the frequency of the polling interval.

Table 1. Queue monitoring data retention schedule

Polling interval	Maximum statistic age
1 seconds	8 hours
5 seconds	8 hours
10 seconds	8 hours
30 seconds	8 hours
60 seconds	8 hours

NOTE

The switch has limited storage for retaining information. For longer monitoring durations, use an external monitor with the queue monitoring feature to collect and store data.

Queue monitoring commands

Subtopics

show interface queue-monitor
show interface queue-monitor status
show queue-monitor status

`clear queue-monitor interface`

Syntax

```
clear queue-monitor interface [<IFNAME>|<IFRANGE>]
```

Description

Clear the data collected on an interface since enabling queue monitoring. If no interfaces are specified, perform a clear of data collected on all interfaces with queue monitoring enabled.

Parameter	Description
`<IFNAME>`	Name of the interface. Format: **<MEMBER>/<SLOT>/<PORT>**
`<IFRANGE>`	Range of interfaces. Format: **<MEMBER>/<SLOT>/<PORT> - <MEMBER>/<SLOT>/<PORT>**

Examples

Clearing queue monitor data collected on interface 1/1/1:

```
config)# clear queue-monitor interface 1/1/1
Warning: clearing collected queue monitor statistics will be reflected
in all CLI sessions, any agents running in the analytics engine, and any
external systems monitoring switch statistics.
Continue (y/n)? y
```

Clearing queue monitor data collected on interfaces 1/1/1 and 1/1/2:

```
config)# clear queue-monitor interface 1/1/1-1/1/2
Warning: clearing collected queue monitor statistics will be reflected
in all CLI sessions, any agents running in the analytics engine, and any
external systems monitoring switch statistics.
Continue (y/n)? y
```

Clearing all queue monitor data collected on all interfaces:

```
config)# clear queue-monitor interface
Warning: clearing collected queue monitor statistics will be reflected
in all CLI sessions, any agents running in the analytics engine, and any
external systems monitoring switch statistics.
Continue (y/n)? y
```

Related Commands

Command	Description
queue - monitor	This command enables the queue monitoring feature, allowing the switch to coll ect queue statistics at 10 - second time intervals. These aggregate statistics ca n be used to monitor network health, troubleshoot network issues, and identify normal network behavior patterns and performance anomalies. This feature is di sabled by default, but can be enabled on a maximum of 52 interfaces per switch .

Command History

Release	Modification
10.17	Command introduced on 5420, 6200, 6300 and 6400 Switch series.

Command Information

Platforms	Command context	Authority
5420 6200 (except f or JL724A, JL 725A, JL726A , JL727A, JL7 28A)	`config`	Administrators or local user group members with execution righ ts for this command.

`queue-monitor`

Syntax

```
queue-monitor
no queue-monitor
```

Description

This command enables the queue monitoring feature, allowing the switch to collect queue statistics. then display this information as a queue statistics history. These aggregate statistics can be used to monitor network health, troubleshoot network issues, and identify normal network behavior patterns and performance anomalies. This feature is disabled by default, but can be enabled on a maximum of 52 interfaces per switch.

Examples

Enabling the queue monitoring feature on interface 1/1/1:

```
(config)# interface 1/1/1
(config-if)# queue-monitor
```

Disabling the queue monitoring feature on interface 1/1/1:

```
(config)# interface 1/1/1
(config-if)# no queue-monitor
```

Related Commands

Command	Description
show queue - monitor status	This command is used to view the global state of the queue monitor feature. Ou tput includes the memory consumed by the feature, the statistics being monito red for collection, and the interfaces currently enabled for monitoring.

Command History

Release	Modification
10.17	Command introduced on 5420, 6200, 6300 and 6400 Switch series.

Command Information

Platforms	Command context	Authority
5420	`config-if`	Administrators or local user group members with execution righ ts for this command.

Platforms	Command context	Authority
6200 (except for JL724A, JL725A, JL726A, JL727A, JL728A)		

`show interface queue-monitor`

Syntax

```
show interface {<IFNAME>} queue-monitor
histogram [filter  [since <1-10000> days|hours|minutes|seconds][type {queue-
depth}
            ]
list [filter [since <1-10000> days|hours|minutes|seconds][threshold {kbps
<1-4294967295>}|{kbytes <1-4294967295>}|{packets <1-4294967295>}] [type
{queue-depth [

            ]] [slot <slot-id>]
table [filter [threshold {kbps <1-4294967295>}|{kbytes <1-4294967295>}|
{packets <1-4294967295>}] [type {queue-depth}
            ]]
```

Description

Display queue statistics history in the CLI as either a list or a histogram, or as a table, showing the previous five minutes of queue statistics history.

Parameter	Description
`interface [<IFNAME>]`	Specifies the name of an Ethernet port or LAG on the switch. If the optional **<IFNAME>** parameter is omitted, the output of this command displays all queue statistic data for all interfaces. Format: **<MEMBER>/<SLOT>/<PORT>** or **lag <ID>**
`histogram|list|table`	Specifies the output format for queue monitor output. • histogram - Queue statistics history is presented in histogram format for the last 8 hours. • list - Queue statistics history is presented in the list format for the last 8 hours

Parameter	Description
	• table - Queue statistics history is presented in tabular for mat for the last 5 minutes.
`filter`	The **list** format is able to display all of the collected queue stati stics history data when no filters are applied. To limit the inform ation displayed in the output of this command, use the filter pa rameters to filter the output to the format and data you require.
`since <1-10000> days|hours|minutes|seconds`	Show events no older than a specified age, measured in second s, days, hours or minutes.
`type {queue-depth}`	Filter queue monitoring data by the specified filter type. If you do not specify a filter type, the type defaults to **queue - depth** . To see filtered output for other statistics types, you must speci fy the type. • **queue - depth**: Show queue depth data.
`threshold {kbps|kbytes|packets <1-4294967295>}`	Show data that exceeds a specified threshold in kilobytes, num ber of packets, or kilobits per second.

Examples

Displaying the queue statistics history data for all monitored interfaces in list format:

```
switch# show interface queue-monitor list
Time Range: 8 hours (2023-08-07 09:11:30 to 2023-08-08 17:11:30 (UTC+00:00))
Time                 Interface  Queue  Statistic               Value
----------------------------------------------------------------------------
----
2023-08-08 17:11:10  1/1/8      4      Queue Depth             12345 kbytes
2023-08-08 17:11:10  1/1/9      4      Queue Depth             5432 kbytes
2023-08-08 17:11:20  1/1/8      4      Queue Depth             12345 kbytes
2023-08-08 17:11:20  1/1/9      4      Queue Depth             5432 kbytes
2023-08-08 17:11:30  1/1/8      4      Queue Depth             12345 kbytes
2023-08-08 17:11:30  1/1/9      4      Queue Depth             5432 kbytes
```

The following example displays a portion of output of the **show interface 1/1/8 queue-monitor** command, showing queue statistics history data in tabular format for interface 1/1/8 with a min-depth of 2 kbytes.

```
switch# show interface 1/1/8 queue-monitor table min-depth kbytes 2
1/1/8 Statistics History
Collection Interval: 10 seconds
Time Range: 2023-08-08 17:07:00 to 2023-08-08 17:12:00 (UTC)
Min-Depth Filter: 2 KB
                          Queue Depth - bytes
```

```
Time (s)        Q0        Q1        Q2        Q3        Q4        Q5
Q6        Q7
-------------------------------------------------------------------------------
----
-300             -         -         -         -         -         -
-         -
-295             -         -         -         -         -         -
-         -
-290             -         -         -         -         -         -
-         -
-285             -         -         -         -         -         -
-         -
-280             -         -         -         -         -         -
-         -
-275             -         -         -         -         -         -
-         -
-270             -         -         -         -         -         -
-         -
-265             -         -         -         -         -         -
-         -
-260             -         -         -         -         -         -
-         -
-255             -         -         -         -         -         -
-         -
-250             -      2078         -         -         -         -
-         -
-245             -      9914         -         -         -         -
-         -
-240             -    171102         -         -         -         -
-         -
-235             -    996262         -         -         -         -
-         -
-230             -      5444         -         -         -         -
-         -
-225             -         -         -         -         -         -
-         -
-220             -         -         -         -         -         -
-         -
-215             -         -         -         -         -         -
-         -
-210             -         -         -         -         -         -
-         -
-205             -         -         -         -         -         -
```

```
-        -
-200             -        -        -        -        -        -
-        -
...
```

Displaying the queue statistics history data for interface 1/1/8 in histogram format.

```
switch(config)# show interface 1/1/8 queue-monitor histogram
1/1/8 Statistics History
Time Range: 2023-08-08 09:12:15 to 2023-08-08 17:12:15 (UTC)
Samples per Queue: 13
                         Count at Depth
Depth (KB)           Q0       Q1       Q2       Q3       Q4       Q5       Q6       Q7
----------------------------------------------------------------------------
0                    13       13       13       13        7       13       13       10
1-64                  0        0        0        0        6        0        0        3
65-128                0        0        0        0        0        0        0        0
129-256               0        0        0        0        0        0        0        0
257-512               0        0        0        0        0        0        0        0
513-1024              0        0        0        0        0        0        0        0
1025-2048             0        0        0        0        0        0        0        0
2049-4096             0        0        0        0        0        0        0        0
4097-8192             0        0        0        0        0        0        0        0
8193-16384            0        0        0        0        0        0        0        0
16385-32768           0        0        0        0        0        0        0        0
32769-65536           0        0        0        0        0        0        0        0
```

Output of the global **show** command when queue monitoring is not enabled on any interfaces.

```
switch(config)# show interface queue-monitor list
Queue monitoring is not enabled on any interfaces.
```

Output of the **show** command when the interface requested does not have queue monitoring enabled.

```
switch(config)# show interface 1/1/8 queue-monitor table
Interface 1/1/8 does not have queue monitoring enabled.
```

Output of the global **show** command with the list presentation speficied when there is no relevant data to show. Note: the table and histogram presentation formats output all enabled interface data.

```
switch(config)# show interface queue-monitor list
Enabled interfaces have not collected any data.
```

Output of the **show** command for an interface with the list presentation specified but there is no relevant data to show.

```
switch(config)# show interface 1/1/8 queue-monitor list
Interface 1/1/8 has not collected any data.
```

Related Commands

Command	Description
queue - monitor	This command enables the queue monitoring feature, allowing the switch to coll ect queue statistics at 10 - second time intervals. These aggregate statistics ca n be used to monitor network health, troubleshoot network issues, and identify normal network behavior patterns and performance anomalies. This feature is di sabled by default, but can be enabled on a maximum of 52 interfaces per switch .

Command History

Command Information

Platforms	Command context	Authority
5420 6200 (except f or JL724A, JL 725A, JL726A , JL727A, JL7 28A)	Operator (`>`) or Manage r (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show interface queue-monitor status`

Syntax

```
show interface {<IFNAME> | <IFRANGE>} queue-monitor status
```

Description

Displays the queue monitoring status for a specific interface or range of interfaces. If no interface is specified, the command will display the queue monitoring status for all interfaces configured.

Parameter	Description
`{<IFNAME> | <IFRANGE>}`	Use the **queue - monitor status** parameters to show whether t he feature is enabled or disabled, and information about memor y consumption and monitored interfaces. Include the optional **<I FNAME>** parameter to filter the output of this command to sho

Parameter	Description
	w data for just the specified interface, or include both the <**IFNAME >** parameter and the additional **<IFRANGE>** parameter also to view data for a range of interfaces. If no interface is specified, the command will display the queue monitor status for all interfaces that have queue monitoring configured.

Examples

Showing queue monitoring status for interface **1/1/1** when queue monitoring is not configured on that interface.

```
switch# show interface 1/1/1 queue-monitor status
Interface 1/1/1 does not have queue monitoring enabled.
```

Showing the queue monitoring status for interface **1/1/1** when queue monitoring is enabled on that interface.

```
switch# show interface 1/1/1 queue-monitor status
Interface 1/1/1
Status               : Enabled
Monitored Statistics : queue-depth
```

Showing the queue monitoring status for a range of interfaces when queue monitoring is enabled on those interfaces.

```
switch# show interface 1/1/1-1/1/2 queue-monitor status
Interface 1/1/1
Status               : Enabled
Monitored Statistics : queue-depth


Interface 1/1/2
Status               : Enabled
Monitored Statistics : queue-depth
```

Showing the queue monitoring status for all interfaces that have queue monitoring enabled, but the interface does not support any statistics.

```
switch# show interface queue-monitor status
Interface 1/1/1
Status               : Blocked (No queue statistics supported for
monitoring)
```

```
Interface 1/1/2
Status                 : Blocked (No queue statistics supported for
monitoring)
Interface 1/1/3 does not have queue monitoring enabled.
```

Showing the queue monitoring status for all interfaces where queue monitoring is enabled on interface **1/1/1**.

```
switch# show interface queue-monitor status
Interface 1/1/1
Status                 : Enabled
Monitored Statistics : queue-depth



Interface 1/1/2 does not have queue monitoring enabled.
Interface 1/1/3 does not have queue monitoring enabled.
```

Showing the queue monitoring status for all interfaces when the feature pack is invalid and queue monitoring is configured only on interface **1/1/1**.

```
switch# show interface queue-monitor status
Interface 1/1/1
Status                 : Blocked (Invalid feature pack)
Interface 1/1/2 does not have queue monitoring enabled.
Interface 1/1/3 does not have queue monitoring enabled.
```

Showing the queue monitoring status for interface **1/1/1** when the feature pack is invalid and queue monitoring is configured on that interface.

```
switch# show interface 1/1/1 queue-monitor status
Interface 1/1/1
Status                 : Blocked (Invalid feature pack)
```

Command History

Release	Modification
10.17	Command introduced

Command Information

Platforms	Command context	Authority
5420 6200 (except for JL724A, JL725A, JL726A	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Platforms	Command context	Authority
, JL727A, JL728A)		

`show queue-monitor status`

Syntax

```
show queue-monitor status
```

Description

Displays the global state of the queue monitoring feature. The output includes the global enable/disable status, the memory consumed by the feature, the statistics being collected, and the interfaces currently enabled for monitoring.

Examples

Showing queue monitoring status when queue monitoring is not enabled on any interfaces.

```
switch# show queue-monitor status
Feature Status             : Running
Memory Consumption         : 0 kbytes
Monitored Interfaces/Max : 0/52
```

Showing the queue monitoring status when queue monitoring is enabled on interface **1/1/1**.

```
switch# show queue-monitor status
Feature Status             : Running
Memory Consumption         : 184 kbytes
Polling Interval           : 10 seconds
Data Retention Duration    : 8 hours
Monitored Interfaces/Max : 1/52
Monitored Interfaces       : 1/1/1
```

Related Commands

Command	Description
queue - monitor	This command enables the queue monitoring feature, allowing the switch to collect queue statistics at 10 - second time intervals. These aggregate statistics can be used to monitor network health, troubleshoot network issues, and identify normal network behavior patterns and performance anomalies. This feature is di

Command	Description
	sabled by default, but can be enabled on a maximum of 52 interfaces per switch .

Command History

Command Information

Platforms	Command context	Authority
5420 6200 (except for JL724A, JL725A, JL726A, JL727A, JL728A)	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Congestion Event Detection

The Congestion Event Detection feature is an extension of Queue Monitoring that allows users to configure alert thresholds for queue depth, queue transmit rate, and queue drops. Alerts can be viewed on-switch and can also be sent to the switch event log on a selective basis.

Congestion Event Detection is supported on 8325, 8325H, 8325P, 9300, 9300S, 10000, and 10040 switch series. This feature enables users to configure queue statistics thresholds which are actively checked by the switch on interfaces where monitoring and thresholds are applied. This mechanism leverages queue monitoring as the source of periodic queue statistics samples for comparison against the configured thresholds. If a statistic has crossed a threshold, information about the occurrence is saved within the switch and an optional action can also be taken at the time of the occurrence. Accumulated occurrences (events) can be viewed on the switch through the CLI or requested through the REST API interface.

Subtopics

[Using collected data]
[Congestion event detection commands]

Using collected data

Queue monitoring provides the ability to look into switch congestion-related performance issues and identify when problems have occurred. The methods available for reviewing performance issues are:

- Congestion events
- Per-interface congestion histograms
- Per-interface congestion-detailed view

As anomalies are identified, the interface queue that experienced the anomaly is the starting point for identifying the cause of the behavior. Examine traffic patterns for flows competing for network bandwidth to determine if corrective action is needed to address potential problems.

Congestion event detection commands

Subtopics

apply congestion-event profile

Syntax

```
apply congestion-event profile [<NAME>]
no apply congestion-event profile [<NAME>]
```

Description

Applies a congestion event profile to an interface.

The **no** form removes the congestion event profile from the interface.

Parameter	Description
`<NAME>`	Specifies the name of a congestion event profile. Range: 1 to 64 alphanumeric characters, including period (.), un derscore (_), and hyphen (-).

Usage

Apply the congestion event profile to the interface to begin the process of actively checking the monitored queue statistics for congestion events based on the thresholds configured in the applied congestion event profile. User-specified congestion event profiles and profile entries may be modified and deleted while applied on an interface. If a congestion event profile or entry is modified while applied on an interface, any ongoing event occurrences will end immediately.

Examples

Applying congestion event profile, **prof1**, to an interface where the queues on the interface have a monitoring source:

```
switch(config-if)# apply congestion-event profile prof1
```

Applying non-existent congestion event profile, **prof22**, to an interface:

```
switch(config-if)# apply congestion-event profile prof22
Profile prof22 does not exist.
```

Applying congestion event profile, **prof1**, to an interface where the queues on the interface do not have a queue data monitoring source. Without a source for queue statistics, no congestion event threshold checking will be performed.

```
switch(config-if)# apply congestion-event profile prof1
Warning: Queue monitoring must also be enabled using the 'queue-monitor'
command for congestion events to be detected.
```

Removing any congestion event profile from an interface.

```
switch(config-if)# no apply congestion-event profile
```

Removing congestion event profile, **prof1**, from an interface:

```
switch(config-if)# no apply congestion-event profile prof1
```

Removing congestion event profile, **prof2**, from an interface where congestion event profile, **prof1**, is currently applied.

```
switch(config-if)# no apply congestion-event profile prof2
The profile to remove does not match the currently configured profile.
```

Command History

Release	Modification
10.16	Command introduced

Command Information

Platforms	Command context	Authority
	`config-if`	Administrators or local user group members with execution righ ts for this command.

`congestion-event profile`

Syntax

```
congestion-event profile <NAME>
no congestion-event profile <NAME>
```

Description

Creates a new congestion event profile and switches to the **config-cng-prof** context for the profile.

The no form deletes the profile and removes it from all applied interfaces.

> **NOTE**
> If the specified profile already exists, this command switches to the **config-cng-prof** context for the named profile.

Parameter	Description
`<name>`	Name of the congestion event profile, up to 64 characters.

Usage

A congestion event profile contains zero or more congestion event entries that specify the thresholds to use for identifying conditions of interest that should be saved as an event occurrence. Use **show congestion-event profile [NAME]** to view the status of all congestion event profiles or their settings.

Examples

Creating a congestion-event profile named **prof1**:

```
switch(config)# congestion-event profile prof1
```

Deleting a congestion event profile named **prof1**:

```
switch(config)# no congestion-event profile prof1
```

Creating a congestion event profile with an invalid name:

```
switch(config)# congestion-event profile prof##
Invalid name. Please enter a string of up to 64 alphanumeric,
underscore, hyphen, and period characters.
```

Command History

Release	Modification
10.18	Command introduced on the 10040 Switch series
10.16	Command introduced

Command Information

Platforms	Command context	Authority
	`config`	Administrators or local user group members with execution righ ts for this command.

IP Flow Path Trace

IP Flow Path Trace (flowtraced) feature helps to trace the complete path taken by an Application Flow while traversing from Source device to Destination device. There is a new dedicated daemon (flowtraced) created to support the feature. A new proprietary UDP protocol is designed to handle the feature requirements. These CPU generated packets traverse through the network and collect all the necessary telemetry information from each of the HPE ANW OS-CX devices traversed where flow path trace is supported. All the information collected from each of the device traversed is stored in

the payload as TLVs.

> **NOTE**
> IP Flow Path Trace is supported on 5420, 6200, 6300, 6400, 8360 and 8100 Switch Series.

Subtopics

[IP Flow Path Trace commands]

Limitations, Conflicts or Exclusions

- Platform should support Forwarding Info infra to fetch the nexthop and interface details.
- Traffic Insight feature needs to be running on all devices in the path to match on the Policy/ACL drop matches. If Traffic Insight is not supported on the devices, then the Flow drop will not be detected.
- If Traffic Insight feature is not supported or enabled along with Application recognition,Flow Tracking,IPFix, TI with AppFlow monitor on the first-hop device, then the flow trace query should have the complete 5 tuple information. Partial query having only the Client IP and destination (AppName, DomainName) will result in lookup failure.
- Client-Insight ARPtoGW needs to be enabled on the first Access switch if acting as an L2 device to make sure that the MAC-IP association of the GW are available.
- Client-IP-Tracker needs to be enabled on the first Access switch if acting as an L2 device to make sure that the MAC-IP association of the Client/Source are available.
- If the Clients have static IPs, the **gateway-ip** needs to be provided in the flow trace query.
- The feature sends out proprietary packets using UDP Source L4 Port set to 55000 and Destination L4 Port set to 55001, which are private ports as per IANA.
- In case of Flow Trace capability unaware devices present on the path, the pathtrace post that device could result in an incorrect path.
- If the last-hop device IP is not provided along with the 5-tuple query, then packet would get leaked to the destination device if the last hop switch is not Flow Trace capable.
- The MAC and Routing tables may be constantly changing. The egress port received is valid for that snapshot in time. It may not remain the same in the future.
- This feature is not applicable for broadcast, multicast or unknown unicast packets.
- IPv6 support is not provided.
- Below drops are not captured in current release:
    - GBP Policy drops
    - QoS drops
    - Per-Hop latency
    - Path Latency

    - GBP relay issues
    - PBR drops
- Support for UBT, IPSEC or GRE tunnels is not present.
- Periodic probes for the 5-tuple query is not supported.
- Number of non-flowtrace capable devices in the path (applicable for L3 only).
- Automatic Reverse Flow Trace is not supported.
- If Application provided is not active in TI for the provided source IP, then the flow trace query will fail.
- If destination details are not provided, then the flow trace query will fail.
- Uses IP exception CoPP class which can be updated while executing the **flowtrace** query.
- If the ARP table in case of L3 or the CIPT/CIARPGW table in case of L2 does not have the MAC-IP binding on the first-hop device, then the query is not initiated and failed.
- LLDP neighbor information is used to figure out if the adjacent device is a Flow Trace capable or not. So all the devices in the network need to have LLDP enabled to make sure that the device classification is correct.
- If the TLV packet payload increases above the MTU of the egress interface, the flow trace will be stopped on that device and a partial trace information will be sent back to the initiating device. Each node will add multiple TLV information depending on the attributes getting collected.
- It is advisable to have the MTU in the network to be jumbo capable to make sure the flow trace is complete especially when there are more than 7-8 nodes between the source and destination device.

IP Flow Path Trace commands

Subtopics

IP Flow Path Trace

`IP Flow Path Trace`

Syntax

```
flowtrace {source-ip-address <IP-ADDR>} {destination-type <ipv4|domainname|
appname> destination <string>}
```

```
[source-l4-port <L4-PORT> destination-l4-port <L4-PORT>] [transport-
protocol <PROTOCOL>]
[vrf <VRF-NAME>] [last-hop-device-ip <IP-ADDR>] [gateway-ip <IP-ADDR>]
```

Description

This command is used to get the complete flow path trace of the 5 tuple query which is initiated from the CLI. The Source IP and the Destination details are mandatory for the command to initiate the query. The other fields are optional parameters. If **last-hop-device-ip** is not provided, then the proprietary packets will traverse until the last hop, HPE ANW OS-CX, where flow path trace is supported, or until the destination device, in case the device before destination does not support flow path trace. The **gateway-ip** is also optional. This is useful especially when the first-hop device is an L2 device to get the next-hop MAC details needed for forwarding info lookup.

> **NOTE**
> The flow path trace command does not validate the transport protocol of the flows and provides output using the user-specified protocol number instead of verifying the actual transport protocol.

Parameter	Description
`IP-ADDR`	Specify the IPv4 address (A.B.C.D). Required
`L4-PORT`	Specify the L4 Port (1 - 65535). Optional
`PROTOCOL`	Specify the transport protocol number (1 - 255) Optional
`VRF-NAME`	Specify the VRF. Optional
`DOMAIN-NAME`	Specify the Domain Name. Optional
`APP-NAME`	Specify the Application Name. Optional

Examples

This example shows the complete Flow Path trace successful until the last-hop-device:

```
switch# flowtrace source-ip-address 50.0.0.5 destination-type ipv4
destination 80.0.0.2 transport-protocol 6 source-l4-port 23456
destination-l4-port 34567 vrf default last-hop-device-ip 60.0.0.1
IP Flow Trace Summary:
----------------------
src_ip      : 50.0.0.5                    dst_ip      : 80.0.0.2
src_port    : 23456                       dst_port    : 34567
protocol    : 6
----------------------------------------------------------------------------
----
PathNode  1 :
Device Type        : FlowTrace_Capable          Device ID           :
SGXXXXXXAL
System MAC         : 90:aa:aa:aa:aa:80          Egress Subnet       : NA
Ingress Interface  : 1/1/43                     Egress Interface    : 1/1/45
Ingress VLAN       : 10                         Egress VLAN         : 10
Flow Status        : Forwarded
PathNode  2 :
Device Type        : FlowTrace_Capable          Device ID           :
SGXXXXXXBL
System MAC         : f8:bb:bb:bb:bb:00          Egress Subnet       : ECMP
70.0.0.0/24
Ingress Interface  : 1/3/47                     Egress Interface    : 1/5/47
Ingress VLAN       : 10                         Egress VLAN         : 20
Flow Status        : Forwarded
PathNode  3 :
Device Type        : FlowTrace_Capable          Device ID           :
SGXXXXXXCL
System MAC         : f8:cc:cc:cc:cc:00          Egress Subnet       :
60.0.0.0/24
Ingress Interface  : 1/6/3                      Egress Interface    : lag20-
>1/3/2
Ingress VLAN       : 20                         Egress VLAN         : 30
Flow Status        : Forwarded
PathNode  4 :
Device Type        : Last_Hop_Device            Device ID           :
SGXXXXXXDL
System MAC         : 90:dd:dd:dd:dd:00          Egress Subnet       :
80.0.0.0/24
Ingress Interface  : 1/2/2                      Egress Interface    : 1/1/1
Ingress VLAN       : 30                         Egress VLAN         : 40
Flow Status        : Forwarded
```

In this example the flow Path trace failed because of a Policy drop in an intermediate device:

```
switch# flowtrace source-ip-address 50.0.0.5 destination-type ipv4
destination 80.0.0.2 transport-protocol 6 source-l4-port 23456
destination-l4-port 34567 vrf default last-hop-device-ip 60.0.0.1
IP Flow Trace Summary:
----------------------
src_ip      : 50.0.0.5                    dst_ip      : 80.0.0.2
src_port    : 23456                       dst_port    : 34567
protocol    : 6
----------------------------------------------------------------------------
----
PathNode  1 :
Device Type        : FlowTrace_Capable         Device ID          :
SGXXXXXXAL
System MAC         : 90:aa:aa:aa:aa:80         Egress Subnet      : NA
Ingress Interface  : 1/1/43                    Egress Interface   : 1/1/45
Ingress VLAN       : 10                        Egress VLAN        : 10
Flow Status        : Forwarded
PathNode  2 :
Device Type        : FlowTrace_Capable         Device ID          :
SGXXXXXXBL
System MAC         : f8:bb:bb:bb:bb:00         Egress Subnet      : ECMP
70.0.0.0/24
Ingress Interface  : 1/3/47                    Egress Interface   : 1/5/47
Ingress VLAN       : 10                        Egress VLAN        : 20
Flow Status        : Forwarded
PathNode  3 :
Device Type        : FlowTrace_Capable         Device ID          :
SGXXXXXXCL
System MAC         : f8:cc:cc:cc:cc:00         Egress Subnet      : ECMP
70.0.0.0/24
Ingress Interface  : 1/3/47                    Egress Interface   : 1/5/47
Ingress VLAN       : 10                        Egress VLAN        : 20
Flow Status        : Dropped
Drop Reason:
Policy : testPolicy1
Class  : testClass1
RuleID : 30
```

This example shows Flow Path trace via an intermediate Non-Flow trace capable device.

```
switch# flowtrace source-ip-address 50.0.0.5 destination-type ipv4
destination 80.0.0.2 transport-protocol 6 source-l4-port 23456
destination-l4-port 34567 vrf default last-hop-device-ip 60.0.0.1
IP Flow Trace Summary:
----------------------
```

```
src_ip      : 50.0.0.5                  dst_ip      : 80.0.0.2
src_port    : 23456                     dst_port    : 34567
protocol    : 6
----------------------------------------------------------------------------
----
PathNode  1 :
Device Type         : FlowTrace_Capable         Device ID           :
SGXXXXXXAL
System MAC          : 90:aa:aa:aa:aa:80         Egress Subnet       : NA
Ingress Interface   : 1/1/43                    Egress Interface    : 1/1/45
Ingress VLAN        : 10                        Egress VLAN         : 10
Flow Status         : Forwarded
PathNode  2 :
Device Type         : FlowTrace_Capable         Device ID           :
SGXXXXXXBL
System MAC          : f8:bb:bb:bb:bb:00         Egress Subnet       : ECMP
70.0.0.0/24
Ingress Interface   : 1/3/47                    Egress Interface    : 1/5/47
Ingress VLAN        : 10                        Egress VLAN         : 20
Flow Status         : Forwarded
PathNode  3 :
Device Type         : NonFlowTrace_Capable      Device ID           :
SGXXXXXXCL
System MAC          : f8:cc:cc:cc:cc:00
Flow Status         : Forwarded
PathNode  4 :
Device Type         : Last_Hop_Device           Device ID           :
SGXXXXXXDL
System MAC          : 90:dd:dd:dd:dd:00         Egress Subnet       :
80.0.0.0/24
Ingress Interface   : 1/2/2                     Egress Interface    : 1/1/1
Ingress VLAN        : 30                        Egress VLAN         : 40
Flow Status         : Forwarded
```

This example shows a successful complete Flow Path trace for an Application Based query:

```
switch# flowtrace source-ip-address 10.0.0.10 destination-type app_name
destination ssh
IP Flow Trace Summary:
----------------------
src_ip      : 10.0.0.10                 dst_ip      : 60.0.0.2
src_port    : 23456                     dst_port    : 22
protocol    : 6            app_name     : ssh
----------------------------------------------------------------------------
----
```

```
PathNode  1 :
Device Type         : FlowTrace_Capable         Device ID           :
SGXXXXXXAL
System MAC          : 90:ea:ea:ee:aa:50         Egress Subnet       : NA
Ingress Interface   : 1/1/1                     Egress Interface    : 1/1/48
Ingress VLAN        : 10                        Egress VLAN         : 10
Flow Status         : Forwarded
PathNode  2 :
Device Type         : FlowTrace_Capable         Device ID           :
SGXXXXXXBL
System MAC          : f8:bb:bb:bb:bb:00         Egress Subnet       :
70.0.0.0/24
Ingress Interface   : 1/1/47                    Egress Interface    : 1/5/47
Ingress VLAN        : 10                        Egress VLAN         : 20
Flow Status         : Forwarded
PathNode  3 :
Device Type         : Last_Hop_Device           Device ID           :
SGXXXXXXCL
System MAC          : f8:cc:cc:cc:cc:00         Egress Subnet       :
60.0.0.0/24
Ingress Interface   : 1/6/3                     Egress Interface    : 1/6/6
Ingress VLAN        : 20                        Egress VLAN         : 30
Flow Status         : Forwarded
```

This example shows a successful complete Flow Path trace for a VxLAN based topology:

```
switch# flowtrace source-ip-address 10.0.0.10 destination-type app_name
destination ssh
IP Flow Trace summary:
----------------------
src_ip      : 10.0.0.10                 dst_ip      : 60.0.0.2
src_port    : 23456                     dst_port    : 22
protocol    : 6
----------------------------------------------------------------------------
----
PathNode  1 :
Device Type         : FlowTrace_Capable          Device ID           :
SGXXXXXXAL
System MAC          : 90:aa:aa:aa:aa:80          Egress Subnet       : NA
Ingress Interface   : 1/1/43                     Egress Interface    : 1/1/45
Ingress VLAN        : 10                         Egress VLAN         : 10
Flow Status         : Forwarded
PathNode 2:
Device Type     : Tunnel_Entry_Device    Device ID       : SGXXXXXXBL
System MAC          : BB:A4:7D:29:C4:50          Egress IP           :
```

```
30.0.0.1
Ingress Interface  : 1/1/10                   Egress Interface   : 1/1/12
Egress VLAN        : 30                       Flow Status        :
Forwarded
PathNode 3:
Device Type        : Flow_Trace Capable       Device ID          :
SGXXXXXXCL
System MAC         : CC:A4:7D:29:C4:50        Egress IP          :
40.0.0.1
Ingress Interface  : 1/1/13                   Egress Interface   : 1/1/15
Egress VLAN        : 40                       Flow Status        :
Forwarded
PathNode 4:
Device Type        : Tunnel_End_Device        Device ID          :
SGXXXXXXDL
System MAC         : DD:A4:7D:29:C4:50        Egress IP          :
50.0.0.1
Ingress Interface  : 1/1/4                    Egress Interface   : 1/1/5
Egress VLAN        : 30                       Flow Status        :
Forwarded
PathNode 5:
Device Type        : Last_Hop_Device          Device ID          :
SGXXXXXXEL
System MAC         : EE:A4:7D:29:C4:60        Egress IP          :
60.0.0.1
Ingress Interface: 1/1/20                     Egress Interface   : 1/1/21
Egress VLAN: 50                               Flow Status        :
Forwarded
```

This example shows an error due to a missing Source IP:

```
switch# flowtrace destination-type ipv4 destination 50.0.0.50
transport-protocol 6 source-l4-port 23456 destination-l4-port 34567
vrf default last-hop-device-ip 50.0.0.1
Missing Required Parameter Source IP
```

This example shows an error since the there is no active Application:

```
switch# flowtrace source-ip-address 100.0.0.10 destination-type appname
destination ssh
No active session to the 'ssh' application from source 100.0.0.10
```

This example shows an error since Forwarding Info infra is not successful

```
switch# flowtrace source-ip-address 10.0.0.10 destination-type ipv4
destination 50.0.0.50 transport-protocol 6 source-l4-port 23456
destination-l4-port 34567 vrf default last-hop-device-ip 50.0.0.1
Forwarding-info lookup was not successful
```

Command History

Release	Modification
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

gRPC network management interface

gRPC is a RPC framework developed by Google to create distributed systems. This protocol allows a client running on one system to call the services defined in a remote server as a local server.

The gRPC Network Management Interface (gNMI) feature provides secure, high-performance, standards-based network management and gRPC-based access to network device configuration and operational data. Using industry-standard OpenConfig models, gNMI enables the building of modern network automation and monitoring solutions.

gNMI is implemented as an HTTP/2 server supporting encrypted communication and role-based access control. It allows network management systems to interact with the device for configuration and telemetry using gRPC, supporting VRF-aware access and role-based control.

gNMI supports access over SVIs, loopback, and data ports.

gNMI allows for the following:

- Streaming of real-time telemetry data from switches
- Monitoring of interface statistics, system performance, and hardware health
- Integration with modern network management platforms
- Implementation of event-driven network automation

Refer to gNMI commands for information on enabling and configuring gNMI.

gNMI commands

`crypto pki application gnmi certificate`

Syntax

```
crypto pki application gnmi certificate <CERT-NAME>
```

Description

Configures a certificate for the gNMI server. **local-cert** is used by default. For more details, refer to the [Public Key Infrastructure guide] (./Functionality_Guide_PKI.md)

Parameter	Description
`<CERT-NAME>`	Specifies the certificate name.

Examples

Configure sign-cert for the gNMI server:

```
switch(config)# crypto pki application gnmi certificate sign-cert
```

Command History

Release	Modification
10.17	Command introduced

Command Information

Platforms	Command context	Authority
5420	`config`	Administrators or local user group members with execution righ ts for this command.

`gnmi vrf`

Syntax

```
gnmi vrf <VRF NAME>
no gnmi vrf <VRF NAME>
```

Description

Enables the gNMI server on a given VRF. gNMI can be enabled on multiple VRFs simultaneously.

> **NOTE**
>
> There is a maximum of 30 gNMI subscriptions per request and a maximum of 9 concurrent streams per switch. These are system-wide limits shared among all authenticated users regardless of connection method (local or remote).

The **no** form of this command removes the configuration.

Disabling the gNMI server on a VRF immediately closes any active streams.

Parameter	Description
`<VRF NAME>`	Specifies the VRF name.

Examples

Enable the gNMI server on VRF mgmt, this allows access to the gNMI server from the OOBM port in the "management VRF":

```
switch(config)# gnmi vrf mgmt
```

Removing the configuration of gNMI server on mgmt:

```
switch(config)# no gnmi vrf mgmt
```

Command History

Release	Modification
10.17	Command introduced

Command Information

Platforms	Command context	Authority
5420	`config`	Administrators or local user group members with execution righ ts for this command.

`show gnmi`

Syntax

```
show gnmi
```

Description

Displays the current gNMI configuration.

Examples

Display the current gNMI configuration:

```
switch(config)# show gmni
gNMI Configuration
----------------------------------------------------
VRF                  : mgmt, default
Access mode          : read-only
Global stream limit  : 9
```

Command History

Release	Modification
10.17	Command introduced

Command Information

Platforms	Command context	Authority
5420	`config`	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`diag-dump`

Syntax

```
diag-dump gnmi basic
```

Description

Provides detailed service status information for gNMI and nginx-configurator daemon.

Examples

Showing feature information:

```
==========================================================================
[Start] Feature gnmi Time : Thu Jun 26 14:30:13 2025
==========================================================================
--------------------------------------------------------------------------
[Start] Daemon nginx-configurator
--------------------------------------------------------------------------
gNMI nginx VRF service status dump:
-----------------------------------------------------------------------------
------
| VRF                                | LoadState    | ActiveState |
SubState        |
-----------------------------------------------------------------------------
------
| VRF_2                              | loaded       | active      |
running         |
-----------------------------------------------------------------------------
------

--------------------------------------------------------------------------
[End] Daemon nginx-configurator
--------------------------------------------------------------------------
==========================================================================
[End] Feature gnmi
==========================================================================
Diagnostic-dump captured for feature gnmi
```

Command History

Release	Modification
10.17.1000	Command introduced

Command Information

Platforms	Command context	Authority
5420	`config`	Administrators or local user group members with execution righ ts for this command.

`debug nginxconfigurator`

Syntax

```
debug nginxconfigurator <all|gnmi> all severity <level>
```

Description

Enables nginx configurator and YANG resolver debug logs.

Examples

Enabling nginx configurator and YANG resolver debug logs:

```
switch# debug nginxconfigurator all severity debug
switch# debug yang all severity debug
```

Reviewing debug logs:

```
switch(config)# show debug buffer module nginxconfigurator
switch(config)# show debug buffer module yang-resolver
```

Command History

Release	Modification
10.17.1000	Command introduced

Command Information

Platforms	Command context	Authority
5420	`config`	Administrators or local user group members with execution righ ts for this command.

`show tech`

Syntax

```
show tech gnmi
```

Description

Provides configuration, logs, and status information relevant to the gNMI HTTP/2 server.

Examples

Showing gNMI configuration information:

```
switch# show tech gnmi
```

```
====================================================
Show Tech executed on Thu Jun 26 14:30:33
2025========================================================================
================================
Begin] Feature gnmi
 ====================================================
 *********************************
Command : show gnmi
*********************************
gNMI Configuration
----------------------------
VRF                    : mgmt, default
Access mode            : read-only
Max streams globally   : 9

*********************************
Command : diag-dump gnmi basic
*********************************
=========================================================================
[Start] Feature gnmi Time : Thu Jun 26 14:30:33 2025
=========================================================================
-------------------------------------------------------------------------
[Start] Daemon nginx-configurator
-------------------------------------------------------------------------
gNMI nginx VRF service status dump:
----------------------------------------------------------------------------
------
| VRF                              | LoadState    | ActiveState |
SubState        |
----------------------------------------------------------------------------
------
| VRF_2                            | loaded       | active      |
running         |
----------------------------------------------------------------------------
------


-------------------------------------------------------------------------
[End] Daemon nginx-configurator
-------------------------------------------------------------------------
=========================================================================
```

```
[End] Feature gnmi
================================================================================
```

Command History

Release	Modification
10.17.1000	Command introduced

Command Information

Platforms	Command context	Authority
5420	`config`	Administrators or local user group members with execution righ ts for this command.

Boot commands

Subtopics

- Boot commands

Boot commands

Subtopics

- boot set-default
- boot system
- show boot-history

`boot set-default`

Syntax

```
boot set-default {primary | secondary}
```

Description

Sets the default operating system image to use when the system is booted.

Parameter	Description
`primary`	Selects the primary network operating system image.
`secondary`	Selects the secondary network operating system image.

Example

Selecting the primary image as the default boot image:

```
switch# boot set-default primary
Default boot image set to primary.
```

> **NOTE**
> For more information on features that use this command, refer to the Fundamentals Guide or the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (`#`)	Administrators or local user group members with execution rights for this command.

`boot system`

Syntax

```
boot system [primary | secondary | serviceos]
```

Description

Reboots all modules on the switch. By default, the configured default operating system image is used. Optional parameters enable you to specify which system image to use for the reboot operation and for future reboot operations.

Parameter	Description
`primary`	Selects the primary operating system image for this reboot and sets the configured default operating system image to **primary** for future reboots.
`secondary`	Selects the secondary operating system image for this reboot a nd sets the configured default operating system image to **secon dary** for future reboots.
`serviceos`	Selects the service operating system for this reboot. Does not c hange the configured default operating system image. The ser vice operating system acts as a standalone bootloader and rec overy OS for switches running the AOS-CX operating system an d is used in rare cases when troubleshooting a switch.

Usage

This command reboots the entire system. If you do not select one of the optional parameters, the system reboots from the configured default boot image.

You can use the **show images** command to show information about the primary and secondary system images.

Choosing one of the optional parameters affects the setting for the default boot image:

- If you select the **primary** or **secondary** optional parameter, that image becomes the configured default boot image for future system reboots. The command fails if the switch is not able to set the operating system image to the image you selected.
  You can use the **boot set-default** command to change the configured default operating system image.
- If you select **serviceos** as the optional parameter, the configured default boot image remains the same, and the system reboots all management modules with the service operating system.

If the configuration of the switch has changed since the last reboot, when you execute the **boot system** command you are prompted to save the configuration and you are prompted to confirm the reboot operation.

Saving the configuration is not required. However, if you attempt to save the configuration and there is an error during the save operation, the **boot system** command is aborted.

Examples

Rebooting the system from the configured default operating system image:

```
switch# boot system
Do you want to save the current configuration (y/n)? y
The running configuration was saved to the startup configuration.
This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? y
The system is going down for reboot.
The system is going down for reboot.
```

Rebooting the system from the secondary operating system image, setting the secondary operating system image as the configured default boot image:

```
switch# boot system secondary
Default boot image set to secondary.
Do you want to save the current configuration (y/n)? n
This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? y
The system is going down for reboot.
```

Canceling a system reboot:

```
switch# boot system
Do you want to save the current configuration (y/n)? n
This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? n
Reboot aborted.
switch#
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution righ ts for this command.

show boot-history

Syntax

```
show boot-history [all|{vsf member <1-10>}]
```

Description

Shows boot history information. When no parameters are specified, shows the most recent information about the current boot operation, and the three previous boot operations for the switch. When the **all** parameter is specified, the output of this command shows the boot information for the active management module.

.

> **NOTE**
>
> To view boot-history on a standby, the command must be sent on the conductor console.

Parameter	Description
`all`	Optional. Shows boot information for the active management m odule.
`vsf member <1-10>`	Optional. Display boot history for the specified VSF member

Usage

This command displays the boot-index, boot-ID, and up time in seconds for the current boot. If there is a previous boot, it displays boot-index, boot-ID, reboot time (based on the time zone configured in the system) and reboot reasons. Previous boot information is displayed in reverse chronological order.

The output of this command includes the following information:

Parameter	Description
`Index`	The position of the boot in the history file. Range: 0 to 3.
`Boot ID`	A unique ID for the boot . A system - generated 128 - bit strin g.
`Current Boot, up for <time>`	For the current boot, the **show boot - history** command shows the number of seconds the module has been running on the cur rent software.

Parameter	Description
`<Timestamp>: boot reason`	For previous boot operations, the **show boot - history** comma nd shows the time at which the operation occurred and the rea son for the boot. The reason for the boot is one of the following values: • <DAEMON - NAME> **crash**: The daemon identified by <DA EMON - NAME> caused the module to boot. • **Kernel crash**: The operating system software associated wit h the module caused the module to boot. • **Uncontrolled reboot**: The reason for the reboot is not kno wn. • **Reboot requested through database**: The reboot occurre d because of a request made through the CLI or other API.

Table 1. Description of reboots handled through the database

Boot History String	Description
Reboot requested by user	A user requested a switch reboot through the CLI or web UI.
Reset button pressed	The switch detected a short - press of the reset but ton
Backplane fault	A backplane fault occurred.
Configuration change	A configuration change resulted in a reboot.
Configuration version migration	A configuration version migration occurred which re quired a reboot.
Console error	The console failed to start.
Fabric fault	A fabric fault occurred.
All line modules faulted	A zero line card condition occurred.
Redundancy switchover requested	A user requested a redundancy switchover.
Redundant Management communication timeout	The standby management module has taken over fro m an unresponsive active management module.

Boot History String	Description
Redundant Management election timeout	A failure to elect a standby management module in t he allotted time.
Critical service fault (error)	A daemon critical to switch operation has stopped fu nctioning. An extra error string may be present to d escribe the error in detail.
VSF autojoin renumber	Reset triggered by VSF autojoin.
VSF member renumbered	A user requested a renumber of a VSF member.
VSF switchover requested	A user requested a VSF switchover.
VSX software update	Reset triggered by a VSX software update.
Chassis critical temperature	Chassis operating temperature exceeded.
Chassis low critical temperature	Chassis temperature below the minimum operating t hreshold.
Chassis insufficient fans	Insufficient fans to cool the chassis.
Chassis unsupported PSUs/fans	Unsupported or misconfigured PSUs or system fans.
Management module critical temperature	Management module operating temperature exceed ed.
ISSU SMM update	Standby management module reboot triggered by a n In - Service Software Upgrade (ISSU).
ISSU switchover	Redundancy switchover triggered by an In - Service Software Upgrade.
ISSU aborted	Standby management module reset triggered by fail ure during an In - Service Software Upgrade.
Rollback timer expired	Reset triggered by the ISSU rollback timer expiring.

Examples

Showing the boot history of the active management module:

```
switch# show boot-history
Management module
=================
Index : 2
Boot ID : c34a2c2499004a02bbeeff4992e1fdbd
```

```
Current Boot, up for 1 days 13 hrs 13 mins 27 secs
Index : 1
Boot ID : bfba9bc486304e57904ac717a0ccbdcd
02 Sep 23 02:55:33 : CPU request reset with 0x20201, Version:
FL.10.14.0000-1619-ga9ec1805bd442~dirty
02 Sep 23 02:55:33 : Switch boot count is 2
Index : 0
Boot ID : a88a71b7ca9a4574af7e3b811ddfdc7e
02 Sep 23 02:49:26 : Reboot requested by user, Version: FL.10.14.0000-1619-
ga9ec1805bd442~dirty
02 Sep 23 02:50:02 : Switch boot count is 1
Index : 3
Boot ID : f00ba10c8c44457f83fee303d014a89a
25 Aug 23 10:27:42 :  Power on reset with 0x1, Version: FL.10.14.0000-1465-
g9df95249d06b0~dirty
25 Aug 23 10:28:18 :  Switch boot count is 3
25 Aug 23 10:29:02 :  Primary overtemperature fault detected with 0x2 in
PSU 1/1
```

Showing the boot history of the active management module and all line modules:

```
switch#
Management module
=================
Index : 3
Boot ID : f1bf071bdd04492bbf8439c6e479d612
Current Boot, up for 22 hrs 12 mins 22 secs
Index : 2
Boot ID : edfa2d6598d24e989668306c4a56a06d
07 Aug 18 16:28:01 : Reboot requested through database
Index : 1
Boot ID : 0bda8d0361df4a7e8e3acdc1dba5caad
07 Aug 18 14:08:46 : Reboot requested through database
Index : 0
Boot ID : 23da2b0e26d048d7b3f4b6721b69c110
07 Aug 18 13:00:46 : Reboot requested through database
Line module 1/1
=================
Index : 3
10 Aug 17 12:45:46 : dune_agent crashed
...
Management module
=================
Index : 3
Boot ID : f1bf071bdd04492bbf8439c6e479d612
```

```
Current Boot, up for 22 hrs 12 mins 22 secs
Index : 2
Boot ID : edfa2d6598d24e989668306c4a56a06d
07 Aug 18 16:28:01 : Reboot requested through database
Index : 1
Boot ID : 0bda8d0361df4a7e8e3acdc1dba5caad
07 Aug 18 14:08:46 : Reboot requested through database
Index : 0
Boot ID : 23da2b0e26d048d7b3f4b6721b69c110
07 Aug 18 13:00:46 : Reboot requested through database
Line module 1/1
=================
Index : 3
10 Aug 17 12:45:46 : dune_agent crashed
...
```

```
switch# show boot-history
Management module
=================
Index : 2
Boot ID : a61ad00d10864c748bc7893a5d4af2e4
15 Dec 23 19:02:02 : Power on reset with 0x1, Version: FL.10.13.1000AF
15 Dec 23 19:02:02 : Switch boot count is 0
15 Dec 23 19:02:17 : PSU 1/1: Fault detected
Index : 1
Boot ID : 30d831bbfdfa425baf50a629ee01b185
15 Dec 23 19:01:58 : Power on reset with 0x1, Version: FL.10.13.1000AF
15 Dec 23 19:01:58 : Switch boot count is 0
```

The following example displays the boot history for the VSF member **2**.

```
switch# show boot-history vsf member 2
Member-2
=========
Index : 0
Boot ID : df99026c194a44f1944a3e7685fb4d90
Current Boot, up for 3 hrs 31 mins 39 secs
Index : 3
Boot ID : 7bf4104903fe4ad1ba4bce40e8099c76
10 Aug 17 10:02:24 : Reboot requested through database
10 Aug 17 10:02:13 : Switch boot count is 2
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution righ ts for this command.

External storage

The switch has limited capacity to store data, collected by switch features and protocols. You can provide virtually unlimited storage capacity by adding user-supplied external storage volumes. Supported volume types and storage protocols include: NFSv3, NFSv4, and SCP (sshfs).

One application of external storage is the saving and restoring of DHCP lease files over SCP or NFS network attached storage systems. SCP file system protocol uses a user mode process to emulate a network file system. The key advantage is packet level encryption and simple configuration. The key disadvantage is slow performance.

You can set up external storage volume credentials and then enable it. A storage management process acts on your requests by enabling the storage volume using the requested storage protocol. You can disable the external storage volume or set it up but leave it disable.

The feature maintains storage volume state. The states are: *disabled* (down), *connecting* (establishing connection), *operational* (up), and *unaccessible* (unavailable).

If a storage volume is unavailable, the system attempts to reconnect periodically. Multiple volumes could connect concurrently. If one connection times out the others can connect immediately.

The system supports server connection through data and management ports.

Data port support requires server IP address on a default VRF.

Once a storage volume is enabled, applications can use the volume to store retrieve and delete files and directories.

Subtopics

External storage commands

Subtopics

- **address**
- **directory**
- **disable**
- **enable**
- **external-storage**
- **password (external-storage)**
- **show external-storage**
- **show running-config external-storage**
- **type**
- **username**
- **vrf**

`address`

Syntax

```
address {<IPV4-ADDR> | <IPV6-ADDR> | hostname <HOSTNAME>}
no address {<IPV4-ADDR> | <IPV6-ADDR> | hostname <HOSTNAME>}
```

Description

Specifies the NAS IP address or hostname.

The **no** form of this command deletes an IP address or hostname.

Parameter	Description
`<IPV4-ADDR>`	Specifies the NAS server IPv4 address, Global.
`<IPV6-ADDR>`	Specifies the IPv6 address of the NAS server.
	Specifies the hostname of the NAS server. String.

Parameter	Description
<HOSTNAME>	

Examples

Creating the logfiles storage volume with IP address 10.1.1.1:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# address 10.1.1.1
```

Deleting an external storage volume named logfiles:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# no address 10.1.1.1
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-external-storage-<VOLUME-NAME>`	Administrators or local user group members with execution righ ts for this command.

directory

Syntax

```
directory <DIRECTORY-NAME>
no directory <DIRECTORY-NAME>
```

Description

Selects an existing directory on the external storage volume.

The **no** form of this command clears a directory of an external storage volume.

Parameter	Description
<DIRECTORY-NAME>	Specifies the external storage directory for mapping the volume .

Examples

Creating a volume named logfiles that is mapped under /home on the server:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# directory /home
```

Clearing the directory /home:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# no directory /home
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	config-external-storage-*<VOLUME-NAME>*	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`disable`

Syntax

```
disable
no disable
```

Description

Disables the external storage volume.

The **no** form of this command enables the external storage volume. This is identical to the `enable` command.

Examples

Disabling a volume named logfiles:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# disable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-external-storage-<VOLUME-NAME>`	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`enable`

Syntax

```
enable
no enable
```

Description

Enables the external storage volume.

The **no** form of this command disables the external storage volume. This is identical to the `disable` command.

Examples

Creating and then enabling a volume named logfiles:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# enable
```

Disables the external storage volume:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# disable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-external-storage-<VOLUME-NAME>`	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

external-storage

Syntax

```
external-storage <VOLUME-NAME>
no external-storage <VOLUME-NAME>
```

Description

Creates or updates an external storage volume.

The **no** form of this command deletes an external storage volume.

Examples

Creating the logfiles storage volume:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)#
```

Deleting the logfiles storage volume:

```
switch(config)# no external-storage logfiles
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

password (external-storage)

Syntax

```
password [{plaintext | ciphertext} <PASSWORD>]
no password {plaintext | ciphertext} <PASSWORD>
```

Description

Sets the password for network attached storage server login.

The **no** form of this command clears the password for network attached storage server login.

Parameter	Description
`{ciphertext | plaintext}`	Selects the password format.
`<PASSWORD>`	Specifies the password. **NOTE** When the password is not provided on the com mand line, plaintext password prompting occurs upon pressing Enter. The entered password cha racters are masked with asterisks.

Examples

Creating a volume named logfiles with password Xj#9:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# password plaintext Xj#9
```

Creating a volume named bak1 with a prompted plaintext password:

```
switch(config)# external-storage bak1
switch(config-external-storage-bak1)# password
Enter the NAS server password: **********
Re-Enter the NAS server password: **********
```

Clearing the password for volume logfiles:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# no password plaintext Xj#9
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config-external-storage-<VOLUME-NAME>`	Administrators or local user group members with execution rights for this command.

`show external-storage`

Syntax

```
show external-storage [<VOLUME-NAME>]
```

Description

Shows external storage configuration and state for all volumes or for a specified volume.

Parameter	Description
`<VOLUME-NAME>`	Specifies the external storage volume name that the show command will use.

Examples

```
switch# show external-storage
--------------------------------------------------------------------------------------------
              Address          VRF         Username          Type          Directory
State
```

```
--------------------------------------------------------------------------------
--------
nfsvol     10.1.1.1     nas      ---          NFSv3      /home
operational
nfsfiles   20.1.1.1     nas      netstorage   NFSv4      /netstor
disabled
scpdev     nasserver    nas      scpstor      SCP        /scp
unaccessible
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

show running-config external-storage

Syntax

```
show running-config external-storage
```

Description

Shows the running configuration of the external storage.

Examples

```
switch# show running-config external-storage
external-storage nfsvol
      address   10.1.1.1
      vrf       nas
      type      nfsv4
      directoty /home
      enable
```

```
external-storage scpdev
      address   30.1.1.1
      vrf       nas
      username  switchuser
      password  ciphertext xxx
      type      scp
      directoty /home
      enable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	Operator (>) or Manage r (#)	Administrators or local user group members with execution righ ts for this command.

type

Syntax

```
type {nfsv3 | nfsv4 | scp}
no type {nfsv3 | nfsv4 | scp}
```

Description

Sets the network attached storage access type for reaching the external storage volume.

The **no** form of this command deletes an external storage volume.

Parameter	Description
`nfsv3`	Specifies the NFSv3 network access protocol.
`nfsv4`	Specifies the NFSv4 network access protocol.

Parameter	Description
`scp`	Specifies the SCP network access protocol.

Examples

Creating the logfiles volume using NFSV4:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# type nfsv4
```

Clearing the external storage access type:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# no type nfsv4
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-external-storage-<VOLUME-NAME>`	Administrators or local user group members with execution righ ts for this command.

`username`

Syntax

```
username <USER-NAME>
no username <USER-NAME>
```

Description

Sets the username for logging in to a network attached storage server.

The **no** form of this command clears a username.

Parameter	Description
<USER-NAME>	Specifies the username.

Examples

Creating a volume named logfiles with the user name nassuser:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# username nasuser
```

Clearing the user name nasuser from accessing the logfiles volume:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# no username nasuser
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	config-external-storage-*<VOLUME-NAME>*	Administrators or local user group members with execution righ ts for this command.

`vrf`

Syntax

```
vrf <VRF-NAME>
no vrf <VRF-NAME>
```

Description

Setting a VRF to reach network attached storage.

The **no** form of this command clears access of a VRF to network attached storage.

Parameter	Description
`<VRF-NAME>`	Specifies the VRF name.

Examples

Creating the logfiles volume and setting a VRF named nas to access the network attached storage:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# vrf nas
```

Clearing access of a VRF named nas to the network attached storage:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# no vrf nas
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-external-storage-<VOLUME-NAME>`	Administrators or local user group members with execution righ ts for this command.

IP-SLA

The IP Service Level Agreement (IP-SLA) is a feature that enables the measuring of network performance between two nodes in a network for different service level agreement parameters such as round-trip time (RTT), one-way delay, jitter, reachability, packet loss, and voice quality scores. These two nodes can span across area in access, distribution or core inside a LAN as well as across WAN between core to core or core to Data Centre switches. This feature helps you measure the SLA for different protocols or applications such as UDP echo, UDP jitter (for voice and video), TCP connect, HTTP, and ICMP echo. This guide provides details for managing and monitoring different types of IP-SLAs.

Subtopics

IP-SLA guidelines

- AOS-CX supports only SLA configuration through CLI and thresholds can be configured using NAE agents using WebUI/REST.
- AOS-CX supports only forever tests. On-demand tests are not supported.
- Maximum sessions: IP-SLA source 50, IP-SLA responder 8.
- NAE can effectively monitor a maximum of 300 parameters, reducing the maximum supported session by 300.
- NAE supports only syslog.

- NAE agents must be triggered for each IP-SLA test on every switch.
- If multiple IP addresses are received for a DNS query, DNS works with the first resolved IP.
- When the DNS server IP is not explicitly configured, the system automatically uses the first DNS server available in its default configuration.
- The source interface/IP option is not applicable for SLAs configured on 'mgmt' VRF, as it has only one interface.
- A system time change because of NTP or a manual change causes an incorrect calculation.
- There is no interoperability of UDP echo SLA between AOS-CX and FlexFabric switches.
- Source IP and source port combination must be unique across SLA sessions in a same switch.
- Do not use the same source port across the source and responder sessions in a switch.
- The configuration of history results is limited to a maximum of 8 IP-SLA sessions. This means that you can enable and store historical performance data, such as response times and availability, for up to 8 individual IP SLA sessions at any given time.
- NTP synchronization is a must for SLA types involving one-way delay such as UDP jitter VoIP.
- It is mandatory to set default CoPP to the maximum value when UDP jitter SLA is enabled. Otherwise, 100% packet loss can be seen and UDP jitter SLA probes will result in failure:

```
copp-policy default
     class hypertext priority 6 rate 50000 burst 64
     default-class priority 6 rate 99999 burst 9999
```

Limitations with VoIP SLAs

- A maximum of 80 concurrent VoIP SLAs can be scheduled in a 20 second slot.
- A single VoIP probe takes 20 seconds to complete.
- The default and minimum probe interval for VoIP SLA is 120 seconds.
- SLAs scheduled in the same slot, periodically sends 1000 probe packets for 120 seconds in 20 second intervals.
- Default 120 second probe interval is divided in to 6 slots of 20 seconds to avoid synchronization of all configured VoIP SLAs sending probes at the same time.

- SLAs started at the same time exceeding the concurrent limit of 80 must wait for the next 20 second VoIP slot to open before moving to ‘running’ state.
- The maximum number of VoIP SLAs supported is 80 X 6 slots = 480 SLAs.
- SLAs exceeding 480 will continue to remain in the 'waiting for VoIP slot' until any slot is freed by stopping the running SLA.
- To avoid high RTT, a single switch with more than 20 SLAs should not have single responder SLA.
- When IP is received dynamically (e.g. using DHCP) for interfaces other than management interface, IPSLA source or responder has to be configured only using interface name.

IP-SLA commands

Subtopics

`https`

Syntax

```
https {get | raw} <URL> [history-interval <HISTORY-INTERVAL>] [cache
disable] [name-server {<IPV4-ADDR-DNS-SERVER>|<IPV6-ADDR-DNS-SERVER>}]
[probe-interval <PROBE-INTERVAL>] [proxy <PROXY-URL>] [source {<IPV4-ADDR>|
<IPV6-ADDR>|IFNAME>}] [source-port <SOURCE-PORT-NUM>] [version <VERSION-
NUMBER>] [http-raw-request <RAW-PAYLOAD>]
no https {get | raw} <URL> [history-interval <HISTORY-INTERVAL>] [cache
disable] [name-server {<IPV4-ADDR-DNS-SERVER>|<IPV6-ADDR-DNS-SERVER>}]
[probe-interval <PROBE-INTERVAL>] [proxy <PROXY-URL>] [source {<IPV4-ADDR>|
<IPV6-ADDR>|IFNAME>}] [source-port <SOURCE-PORT-NUM>] [version <VERSION-
NUMBER>] [http-raw-request <RAW-PAYLOAD>]
```

Description

Configures HTTPS as the IP-SLA test mechanism. Requires destination URL and type of HTTPS request (get/raw).

The **no** form of this command removes the configuration.

> **NOTE**
> For HTTPS IP-SLA sessions, it is not required to install a certificate on the switch.

Parameter	Description
`{get	raw}`	Selects HTTPS request type as get or raw where the system will generate or provide HTTPS payload.
`<URL>`	Specifies HTTPS URL address of syntax. https://<HOST NAME/ IP - ADDRESS>:<PORT>/<PATH>.
`history-interval <HISTORY-INTERVAL>`	Configures the history interval for the IP - SLA. Set the history i nterval to minimum of two times the probe - interval for the SL A. Range: 60 to 7200.
`cache disable`	Selects cache option for the HTTPS server. By default the optio n is enabled.
`name-server {<IPV4-ADDR-DNS-SERVER>|<IPV6-ADDR-DNS-SERVER>}`	Specifies the IPv4 address of DNS server.
`probe-interval <PROBE-INTERVAL>`	Specifies the probe interval in seconds. Range: 30 to 604800.
`proxy <PROXY-URL>`	Specifies the probe interval in seconds. Range: 30 to 604800.
`source {<IPV4-ADDR>|<IPV6-ADDR>|IFNAME>}`	Selects the source, either an IPv4 address, an IPv6 address, or h ostname for SLA probes.
`source-port <SOURCE-PORT-NUM>`	Specifies the value of the source port for the IP - SLA probes.
`version <VERSION-NUMBER>`	Specifies the source interface to use for sending IP - SLA pro bes.
`https-raw-request <RAW-PAYLOAD>`	Specifies the HTTPS raw request. String.

Examples

Configuring HTTPS get with parameters:

```
switch(config-ipsla-1)# https get https://device.arubanetworks.com/root/
home.html
```

Configuring HTTPS raw with parameters:

```
switch(config-ipsla-1)# https raw https://device.arubanetworks.com/root/
home.html raw-request "GET /en/US/hmpgs/index.html"
```

Removing the HTTPS raw:

```
switch(config-ipsla-1)# no https raw https://device.arubanetworks.com/root/
home.html raw-request "GET /en/US/hmpgs/index.html"
```

Command History

Release	Modification
10.16.1000	Added new parameters **history - interval**. Also, IPv6 addresses can be used.
10.12.1000	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-ip-sla-<IP-SLA-NAME>`	Administrators or local user group members with execution righ ts for this command.

`ip-sla responder`

Syntax

```
ip-sla responder <SLA-NAME> (udp-echo | tcp-connect | udp-jitter-voip)
[<PORT-NUM>] [source {<SOURCE-IPV4-ADDR>|<SOURCE-IPV6-ADDR>|<IFNAME>}] [vrf
<VRF-NAME>] [ipv6]
no ip-sla responder <SLA-NAME> (udp-echo | tcp-connect | udp-jitter-voip)
[<PORT-NUM>] [source {<SOURCE-IPV4-ADDR>|<SOURCE-IPV6-ADDR>|<IFNAME>}] [vrf
<VRF-NAME>] [ipv6]
```

Description

Selects the IP-SLA responder. The responder can be configured for udp-echo, tcp-connect, udp-jitter-voip type. It requires the SLA name, SLA type, and port number as arguments.

The **no** form of this command removes the IP-SLA responder.

Parameter	Description
<SLA-NAME>	Specifies the IP - SLA responder name. Length: 1 to 64 chara cters.
`udp-echo`	Enables responder for udp - echo probes.
`tcp-connect`	Selects TCP connect as the IP - SLA test mechanism.
`udp-jitter-voip`	Selects VOIP jitter as the IP - SLA test mechanism.
<PORT-NUM>	Specifies the port number to listen for IP - SLA probes. Range: 1 to 65535.
`source {`*<SOURCE-IPV4-ADDR>*`|`*<SOURCE-IPV6-ADDR>*`|`*<IFNAME>*`}`	Selects the source IPv4 or IPv6 address for SLA probes or the s ource interface to use for sending IP - SLA probes.
`vrf` *<VRF-NAME>*	Specifies the name of the VRF to use.
`ipv6`	Configures IPv6 responder. This keyword is required if an IPv6 address is being used by the source interface or VRF. By defaul t, it will be considered as an IPv4 address.

Usage

The IPv6 keyword is required if an IPv6 address is being used by the source interface or VRF. Otherwise, by default, it will be considered as an IPv4 address.

Examples

Configuring IP-SLA responder for udp-echo:

```
switch(config)# ip-sla responder SLA1 udp-echo 8000 source 2.2.2.2
```

Configuring IP-SLA responder with IPv6:

```
switch(config)#ip-sla responder SLA1 udp-echo 8000 source 1/1/1 ipv6
```

Configuring IP-SLA responder for udp-jitter-voip:

```
switch(config)#ip-sla responder SLA1 udp-jitter-voip 1025 vrf <VRF>
```

Disabling IP-SLA responder:

```
switch(config)# no ip-sla responder SLA1 udp-echo 8000 source 2.2.2.2
```

Command History

Release	Modification
10.15	Added **ipv6** parameter.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`show ip-sla all`

Syntax

```
show ip-sla all
```

Description

Shows all ip-sla source configuration and status.

Examples

Showing results for ip-sla all:

```
switch# show ip-sla all
SLA Name                      : 2 (non-persistent)
Status                        : running
```

```
SLA Type                    : udp-echo
VRF                         : default
Source IP                   :
Source Interface            :
Domain Name Server          :
Payload Size                : 28
TOS                         : 0
Probe Interval(seconds)     : 60
History Interval(seconds)   : 0
Timeout Interval(seconds)   : 45
IP-SLA session status
IP-SLA Name                            : 2 (non-persistent)
IP-SLA Type                            : udp-echo
Destination Host Name/IP Address       : 10.1.1.2
Destination Port                       : 8888
Source IP Address/IFName               :
Source Port                            :
Status                                 : running
IP-SLA Session Cumulative Counters
Total Probes Transmitted               : 10
Probes Timed-out                       : 10
Bind Error                             : 0
Destination Address Unreachable        : 0
DNS Resolution Failures                : 0
Reception Error                        : 0
Transmission Error                     : 0
Operational Status                     : down
IP-SLA Latest Probe Results
Last Probe Time                        :
Packets Sent                           : 1
Packets Received                       : 0
Packet Loss in Test                    : 100%
Minimum RTT(ms)                        :
Maximum RTT(ms)                        :
Average RTT(ms)                        :
DNS RTT(ms)                            :
----------------------------------------------------------------------------
--
SLA Name                    : echo-udp-sess2 (non-persistent)
Status                      : running
SLA Type                    : udp-echo
VRF                         : default
Source IP                   :
```

```
Source Interface          :
Domain Name Server        :
Payload Size              : 28
TOS                       : 0
Probe Interval(seconds)   : 60
History Interval(seconds) : 0
Timeout Interval(seconds) : 45
IP-SLA session status
IP-SLA Name                          : echo-udp-sess2 (non-persistent)
IP-SLA Type                          : udp-echo
Destination Host Name/IP Address     : 100.1.1.2
Destination Port                     : 8888
Source IP Address/IFName             :
Source Port                          :
Status                               : running
IP-SLA Session Cumulative Counters
Total Probes Transmitted             : 10
Probes Timed-out                     : 0
Bind Error                           : 0
Destination Address Unreachable      : 0
DNS Resolution Failures              : 4
Reception Error                      : 0
Transmission Error                   : 0
Operational Status                   : Up
IP-SLA Latest Probe Results
Last Probe Time                      :
Packets Sent                         : 1
Packets Received                     : 1
Packet Loss in Test                  : 0.0000%
Minimum RTT(ms)                      :
Maximum RTT(ms)                      :
Average RTT(ms)                      :
DNS RTT(ms)                          :
----------------------------------------------------------------------------
--
```

Showing results for non-configured ip-sla all:

```
switch# show ip-sla all
IPSLA source is not configured
```

Command History

Release	Modification
10.16.1000	Updated to display **history interval**.
10.12.1000	Updated to display **https** as an IP - SLA type.
10.07 or earlier	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manager (`#`)	Administrators or local user group members with execution rights for this command.

`show ip-sla responder`

Syntax

```
show ip-sla responder <SLA-NAME> [initiator {<SOURCE-IPV4-ADDR>|<SOURCE-IPV6-ADDR>}] [<SOURCE-PORT-NUM>] [results]
```

Description

Shows the given IP-SLA responder configuration and operation status.

Parameter	Description
`<SLA-NAME>`	Specifies the SLA name.
`initiator {<SOURCE-IPV4-ADDR>|<SOURCE-IPV6-ADDR>}`	Selects the source IPv4 or IPv6 address for SLA probes to use.
`<SOURCE-PORT-NUM>`	Configures the source port for the IP - SLA test. Range: 1 to 65 535.

Parameter	Description
`results`	Displays the statistics for a given source IP and port.

Examples

Showing IP-SLA responder configuration:

```
switch# show ip-sla responder SLA3
    SLA Name              : SLA3
    IP-SLA Type           : Udp-echo
    VRF                   : Default
    Responder Port        : 8000
    Responder IP          : 2.2.2.3
    Responder Interface : 1/1/1
    Responder Status      : Running
```

Showing IP-SLA responder with initiator and results parameters:

```
switch# show ip-sla responder SLA1 initiator 2.2.2.1 8000 results
IP-SLA Type           : Udp-echo
VRF Name              : Default
Source IP             : 2.2.2.1
Source Port           : 8000
Responder Port        : 8888
Responder IP          : 2.2.2.3
Responder Interface :
Responder Status      : Running
Packets Received      : 2
Packets Sent          : 2
```

Command History

Release	Modification
10.16.1000	Added new parameters: **initiator** and **results**.
10.07 or earlier	Command introduced.

Command Information

Platforms	Command context	Authority
5420	Operator (`>`) or Manager (`#`)	Administrators or local user group members with execution rights for this command.

Platforms	Command context	Authority
6200		

show ip-sla

Syntax

```
show ip-sla <SLA-NAME> [{results |  history-results}]
```

Description

Shows the given IP-SLA source configuration and status.

Parameter	Description
<SLA-NAME>	Specifies the SLA name.
`results`	Displays the statistics calculated for an SLA type.
`history-results`	Displays the history statistics calculated for the SLA ID.

Examples

Showing results for ip-sla:

```
switch# show ip-sla xyz results
IP-SLA session status
IP-SLA Name                     : xyz
IP-SLA Type                     : tcp-connect
Destination Host Name/IP Address: 2.2.2.1
Destination Port                : 8888
Source IP Address/IFName        : 2.2.2.2
Source Port                     : 5555
Status                          : running
IP-SLA session cumulative counters
Total Probes Transmitted        : 1
Probes Timed-out                : 0
Bind Error                      : 0
Destination Address Unreachable : 0
```

```
DNS Resolution Failures          : 0
Reception Error                  : 0
Transmission Error               : 0
IP-SLA Latest Probe Results
Last Probe Time                  : 2018 Jul 13 02:00:35
Packets Sent                     : 1
Packets Received                 : 1
Packet Loss in Test              : 0.0000%
Minimum RTT(ms)                  : 12
Maximum RTT(ms)                  : 12
Average RTT(ms)                  : 12
DNS RTT(ms)                      : 0
TCP RTT(ms)                      : 12
```

Showing history results for ip-sla:

```
switch# sh ip-sla abcd history-results
```

IP-SLA Name: abcd Session Details =============== IP-SLA Type : tcp-connect Status : running Probe Interval(seconds) : 30 History Interval(seconds) : 600 Source Port : 3000 Destination Port : 4000 Source IP Address : 10.0.0.1 Dest Host Name/IP Address : 10.0.0.2 History Probe Results ===================== Packet Stats ------------ Probes Transmitted : 4 Packets Sent : 4 Packets Received : 4 Loss Percentage : 0.0000% Error Stats ----------- Transmission Errors : 0 Reception Errors : 0 Bind Errors : 0 Dest. Unreachable : 0 Probes Timed-Out : 0 DNS Resolution Failures : 0 Probe RTT Stats --------------- Min RTT(ms) : 0 Max RTT(ms) : 0 Avg RTT(ms) : 0 DNS RTT Stats ------------ Min RTT(ms) : Max RTT(ms) : Avg RTT(ms) :

Command History

Release	Modification
10.16.1000	Added new parameter **history - results**. Updated to display **history interval**.
10.12.1000	Updated to display **https** as an IP - SLA type.
10.07 or earlier	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

L1-100Mbps downshift

The speed downshift feature allows the user to link-up at sub-optimal speeds when failing to link-up at the highest advertised speed. There are fixed number of link attempts made to establish link at highest advertised speed and when all of them fail and attempt is made to link-up at a lower possible speed.

This feature requires underlying PHY to have support for the same and hence capability is only added to select set of ports. If a link cannot be established at the highest common denominator within a set number of link attempts, the PHY advertises the next highest speed using auto-negotiation.

Subtopics

Limitations with speed downshift

- Link up may be delayed as certain number of retries are done to establish the link at highest advertise speeds by both link partners before downshifting.
- Link may be established at sub-optimal speed.

L1-100Mbps downshift commands

Subtopics

`downshift enable`

Syntax

```
downshift-enable
no downshift-enable
```

Description

Enables/disables automatic speed downshift on an interface that supports downshift, generally 1GBASE-T ports. When enabled, downshift allows an interface to link at a lower advertised speed when unable to establish a stable link at the maximum speed. Downshifting only applies to physical interfaces that are not members of a LAG and is only available when auto-negotiation is enabled. When only one speed is advertised, downshift will not be triggered.

Examples

```
switch(config-if)# interface 1/1/1
switch(config-if)# downshift-enable
Warning: this is a non-standard mode for use only when standards-based
auto-negotiation is not able to establish a stable link. Enabling this
may cause the port to link at a lower than expected speed and should
not be used on ports that are members of a LAG. Support calls may require
this feature to be disabled
Continue (y/n)?
switch(config-if)#
```

When automatic downshift is enabled:

```
switch(config-if)# show running-config interface
interface 1/1/1
    downshift-enable
```

Disabling automatic speed downshift:

```
switch(config-if)# interface 1/1/1
switch(config-if)# no downshift-enable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

`show interface`

Syntax

```
show interface [<IFNNAME>|<IFRANGE>] [brief | physical]
show interface [<IFNNAME>|<IFRANGE>] [extended [non-zero] | [human-
readable]]
show interface [<IFNNAME>] monitor [human-readable]
show interface [lag | loopback | tunnel | vlan ] [<ID>] [brief]
show interface lag [<LAG-ID>] [extended [non-zero] | [human-readable]]
show interface lag [<LAG-ID>] monitor [human-readable]
```

Description

Shows active configurations and operational status information for interfaces.

Parameter	Description
`<IFNAME>`	Specifies a interface name.
`<IFRANGE>`	Specifies the port identifier range.
`brief`	Shows brief info in tabular format.
`physical`	Shows the physical connection info in tabular format.
`extended`	Shows additional statistics, including the **tx filtered** and **rx filte red** counters. • Rx filter packets are protocol packets received when the p rotocol is disabled on the switch and there is only one port i n the VLAN. Protocols include OSPF, PIM, RIP, LACP, and LL DP. • An example of a Tx filtered packet would be a multicast pac ket being filtered from going out of the ingress port.

Parameter	Description
`human-readable`	Shows statistics rounded to the nearest power of 1000, for exa mple, 1K, 345M, 2G. This is available only in the CLI interface o utput.
`non-zero`	Shows only non zero statistics.
`LAG`	Shows LAG interface information.
`monitor`	Continuously monitor interface statistics.
`LOOPBACK`	Shows loopback interface information.
`TUNNEL`	Shows tunnel interface information.
`VLAN`	Shows VLAN interface information.
`<LAG-ID>`	Specifies the LAG number. Range: 1 - 256
`<LOOPBACK-ID>`	Specifies the LOOPBACK number. Range: 0 - 255
`<TUNNEL-ID>`	Specifies the tunnel ID. Range: 1 - 255
`<VLAN-ID>`	Specifies the VLAN ID. Range: 1 - 4094
`VXLAN`	Shows the VXLAN interface information.
`<VXLAN-ID>`	Specifies the VXLAN interface identifier. Default: 1

Examples

```
switch# show interface 1/1/1
Interface 1/1/1 is up
```

```
Admin state is up
Link state: up for 2 days (since Sun Jun 21 05:30:22 UTC 2020)
Link transitions: 1
Description: backup data center link
Hardware: Ethernet, MAC Address: 70:72:cf:fd:e7:b4
MTU 1500
Type 1GbT
Full-duplex
qos trust none
Speed 1000 Mb/s
Auto-negotiation is on
Flow-control: off
Error-control: off
                Energy-Efficient Ethernet is enabled
                MDI mode: MDIX
L3 Counters: Rx Enabled, Tx Enabled
Rate collection interval: 300 seconds
Rates                           RX                   TX        Total (RX+TX)
------------- -------------------- -------------------- --------------------
Mbits / sec                   0.00                 0.00                 0.00
KPkts / sec                   0.00                 0.00                 0.00
Unicast                       0.00                 0.00                 0.00
Multicast                     0.00                 0.00                 0.00
Broadcast                     0.00                 0.00                 0.00
Utilization %                 0.00                 0.00                 0.00
Statistics                      RX                   TX                Total
------------- -------------------- -------------------- --------------------
Packets                          0                    0                    0
Unicast                          0                    0                    0
Multicast                        0                    0                    0
Broadcast                        0                    0                    0
Bytes                            0                    0                    0
Jumbos                           0                    0                    0
Dropped                          0                    0                    0
Filtered                         0                    0                    0
Pause Frames                     0                    0                    0
L3 Packets                       0                    0                    0
L3 Bytes                         0                    0                    0
Errors                           0                    0                    0
CRC/FCS                          0                  n/a                    0
Collision                      n/a                    0                    0
Runts                            0                  n/a                    0
```

```
Giants                           0                    n/a                    0
Other                            0                      0                    0
```

Showing information when interface 1/1/1 is configured:

```
MDI mode: MDIX
VLAN Mode: native-untagged
Native VLAN: 1
Allowed VLAN List: all
Rate collection interval: 300 seconds
Rates                           RX                     TX        Total (RX+TX)
------------- -------------------- -------------------- --------------------
Mbits / sec                   0.00                   0.00                 0.00
KPkts / sec                   0.00                   0.00                 0.00
Unicast                       0.00                   0.00                 0.00
Multicast                     0.00                   0.00                 0.00
Broadcast                     0.00                   0.00                 0.00
Utilization %                 0.00                   0.00                 0.00
Statistics                      RX                     TX                Total
------------- -------------------- -------------------- --------------------
Packets                          0                      0                    0
Unicast                          0                      0                    0
Multicast                        0                      0                    0
Broadcast                        0                      0                    0
Bytes                            0                      0                    0
Jumbos                           0                      0                    0
Dropped                          0                      0                    0
Filtered                         0                      0                    0
Pause Frames                     0                      0                    0
Errors                           0                      0                    0
CRC/FCS                          0                    n/a                    0
Collision                      n/a                      0                    0
Runts                            0                    n/a                    0
Giants                           0                    n/a                    0
```

Showing information when the interface is currently linked at a downshifted speed:

```
switch(config-if)# show interface 1/1/1
Interface 1/1/1 is up
...
Auto-negotiation is on with downshift active
```

Showing information when the interface is currently linked with energy-efficient-ethernet negotiated:

```
switch(config-if)# show interface 1/1/1
Interface 1/1/1 is up
...
Energy-Efficient Ethernet is enabled and active
```

```
switch(config-if)# show interface 1/1/1
Interface 1/1/1 is down
Admin state is up
State information: Disabled by VSX
Link state: down for 3 days (since Tue Mar 16 05:20:47 UTC 2021)
Link transitions: 0
Description:
Hardware: Ethernet, MAC Address: 04:09:73:62:90:e7
MTU 1500
Type SFP+DAC3
Full-duplex
qos trust none
Speed 0 Mb/s
Auto-negotiation is off
Flow-control: off
Error-control: off
VLAN Mode: native-untagged
Native VLAN: 1
Allowed VLAN List: 1502-1505
Rate collection interval: 300 seconds
Rate                                  RX                    TX         Total
(RX+TX)
---------------- -------------------- --------------------
--------------------
Mbits / sec                         0.00                  0.00
0.00
KPkts / sec                         0.00                  0.00
0.00
Unicast                             0.00                  0.00
0.00
Multicast                           0.00                  0.00
0.00
Broadcast                           0.00                  0.00
0.00
Utilization                         0.00                  0.00
0.00
Statistic                             RX                    TX
Total
---------------- -------------------- --------------------
--------------------
Packets                                  0
0                    0
Unicast                                  0
```

```
0                    0
Multicast                            0
0                    0
Broadcast                            0
0                    0
Bytes                                0
0                    0
Jumbos                               0
0                    0
Dropped                              0
0                    0
Pause Frames                         0
0                    0
Errors                               0
0                    0
CRC/FCS                              0
n/a                    0
Collision                          n/a
0                    0
Runts                                0
n/a                    0
Giants                               0
n/a                    0
```

Showing information when the interface is configured with EEE and the EEE has auto-negotiated:

```
switch(config-if)# show interface 1/1/1 physical
----------------------------------------------------------------------------
----------------------------------------------------------------
                            Link    Admin         Speed           Flow-
Control          EEE        PoE Power                       Port
Port        Type            Status    Config    Status | Config    Status |
Config    Status | Config  (Watts)     State Information  Description
----------------------------------------------------------------------------
----------------------------------------------------------------
1/1/1       1GbT            up        up        1G        auto      off
off       on        on      --          10M/100M/1G        --
```

Showing the monitor information:

> **NOTE**
>
> In monitor mode, the CLI refreshes data automatically until it is exited by entering **q**. Pressing **?** opens the help menu to display which options are available in this context.

```
Interface 1/1/1 is up
Rate                                 RX                   TX        Total
(RX+TX)
---------------- -------------------- --------------------
--------------------
MBits / sec                    30196.43             30196.43
60392.85
MPkts / sec                    58977.39             58977.40
117954.79
Unicast                            0.00                 0.00
0.00
Multicast                      58977.39             58977.40
117954.79
Broadcast                          0.00                 0.00
0.00
Utilization %                     75.49                75.49
150.98
Statistic                            RX                   TX        Total
(RX+TX)
---------------- -------------------- --------------------
--------------------
Packets                      4756527649           4756527865
9513055514
Unicast                               0
0                    0
Multicast                    4756527649           4756527865
9513055514
Broadcast                             2
0                    2
Bytes                      304417778668         304417795428
608835574096
Jumbos                                0
0                    0
Dropped                               0          19028847730
19028847730
Pause Frames                          0
0                    0
Errors                                0
0                    0
CRC/FCS                               0
n/a                    0
help: ?, quit: q
Help for Interface Monitor
```

```
h  Toggle human-readable mode
c  Clear interface statistics
Does not apply to rates
Arrows, PgUp, PgDn, Home, End
Navigate interface statistics
Delay: 2
help: ?, quit: q
```

Showing the output for interface 1/1/1 in human-readable format:

> **NOTE**
>
> In human-readable format, the **< 1** symbol for **Utilization** indicates that the amount of packets is between zero and one. This is true in cases where the number of bytes increases but the number of packets and the **Utilization** value is not displayed even in the normal output, where the human-readable parameter is not included in the command.

```
switch(config-if)# show interface 1/1/1 human-readable
Interface 1/1/1 is up
Rate                               RX                   TX         Total
(RX+TX)
---------------- -------------------- --------------------
--------------------
Bits / sec                         3M
3M                   6M
Pkts / sec                        316                  316
633
Unicast                           319                  319
638
Multicast                           0
0                    0
Broadcast                           0
0                    0
Utilization %                     < 1                  < 1
< 1
Statistic                          RX                   TX
Total
---------------- -------------------- --------------------
--------------------
Packets                          577K
577K                   1M
Unicast                          577K
577K                   1M
Multicast                           0
51                   51
```

```
Broadcast                                    0
15                    15
Bytes                                     744M
745M                    1G
Jumbos                                       0
0                     0
Dropped                                      0
0                     0
Filtered                                     0
0                     0
Pause Frames                                 0
0                     0
Errors                                       0
0                     0
CRC/FCS                                      0
n/a                     0
Collision                                  n/a
0                     0
Runts                                        0
n/a                     0
Giants                                       0
n/a                     0
```

Showing information about extended counters:

> **NOTE**
>
> The output of the `show interface extended` command varies depending on the switch model and configuration.

```
switch(config-if)# show interface 1/1/17 extended
-------------------------------------------------------------------
Interface 1/1/17
-------------------------------------------------------------------
Statistics                                              Value
-------------------------------------------------------------------
Dot1d Tp Port In Frames                                 547
Dot1d Tp Port Out Frames                                608
Dot3 In Pause Frames                                    0
Dot3 Out Pause Frames                                   0
Ethernet Stats Broadcast Packets                        19
Ethernet Stats Bytes                                    40162
Ethernet Stats Packets                                  342
...
-------------------------------------------------------------------
```

```
Error-Statistics                                      Value
----------------------------------------------------------------------
Dot1d Base Port MTU Exceeded Discards                 0
Dot3 Control In Unknown Opcodes                       0
Dot3 Stats Alignment Errors                           0
Dot3 Stats FCS Errors                                 0
Dot3 Stats Frame Too Longs                            0
Dot3 Stats Internal Mac Transmit Errors               0
Ethernet RX Oversize Packets                          0
...
```

Showing interface link-status:

```
switch# show interface link-status
-------------------------------------------------------------

Port           Type           Physical     Link         Last
                              Link State   Transitions  Change
-------------------------------------------------------------
1/1/1          1G-BT          down         0            --
1/1/2          1G-BT          up           1            1 minute ago (Fri
Mar 09 12:36:56 UTC 2018)
1/1/3          1G-BT          up           1            1 minute ago (Fri
Mar 09 12:36:56 UTC 2018)
1/1/4          --             down         0            --
1/1/5          --             down         0            --
```

Showing interface loopback 1 link-status:

```
-------------------------------------------------------------
                              Physical     Link         Last
Port           Type           Link State   Transitions  Change
-------------------------------------------------------------
loopback1      --             up           --           --
```

Showing interface 1/1/2-1/1/3 link-status:

```
-------------------------------------------------------------
                              Physical     Link         Last
Port           Type           Link State   Transitions  Change
-------------------------------------------------------------
1/1/2          1G-BT          up           1            1 minute ago (Fri
Mar 09 12:36:56 UTC 2018)
1/1/3          1G-BT          up           1            1 minute ago (Fri
Mar 09 12:36:56 UTC 2018)
```

Showing interface link-status:

```
switch# show interface link-status
-------------------------------------------------------------------------
```

```
Port            Type             Physical     Link         Link Flaps   Last
                                 Link State   Transitions Ignored      Change
---------------------------------------------------------------------------
1/1/1           1G-BT            down         0            0            --
1/1/2           1G-BT            up           1            0            1 minute
ago (Fri Mar 09 12:36:56 UTC 2018)
1/1/3           1G-BT            up           1            0            1 minute
ago (Fri Mar 09 12:36:56 UTC 2018)
1/1/4           --               down         0            0            --
1/1/5           --               down         0            0            --
```

Showing state information when interface is blocked:

```
8360(config-if)# show interface 1/1/1
Interface 1/1/1 is up (Blocked)
Admin state is up
State information: Blocked by UDLD
Link state: up for 1 minute (since Mon Jun 10 09:25:27 UTC 2024)
Link transitions: 1
Description:
Persona:
Hardware: Ethernet, MAC Address: 00:fd:45:67:85:91
MTU 1500
Type 10G-LR / 10G SFP+ LR
Full-duplex
qos trust none
Speed 10000 Mb/s
Auto-negotiation is off
Flow-control: off
Error-control: off
VLAN Mode: access
Access VLAN: 1
Rate collection interval: 300 seconds
Rate                                   RX                   TX        Total
(RX+TX)
---------------- -------------------- --------------------
--------------------
Mbits / sec                          0.00                 0.00
0.00
KPkts / sec                          0.00                 0.00
0.00
Unicast                              0.00                 0.00
0.00
Multicast                            0.00                 0.00
0.00
```

```
Broadcast                         0.00                 0.00
0.00
Utilization %                     0.00                 0.00
0.00
Statistic                           RX                   TX
Total
---------------- -------------------- --------------------
--------------------
Packets                             15
15                   30
Unicast                             12
12                   24
Multicast                            3
3                    6
Broadcast                            0
0                    0
Bytes                             1350                 1350
2700
Jumbos                               0
0                    0
Dropped                              0
0                    0
Pause Frames                         0
0                    0
Errors                               0
0                    0
CRC/FCS                              0
n/a                    0
Collision                          n/a
0                    0
Runts                                0
n/a                    0
Giants                               0
n/a                    0
```

Command History

Release	Modification
10.15	Added state information when port goes into down state.
10.11	Added `monitor` parameter.

Release	Modification
10.10	Added `human-readable` parameter.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (`>`) or Manage r (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show interface downshift-enable`

Syntax

```
show interface [<IFNNAME>|<IFRANGE>] downshift-enable
```

Description

Displays speed downshift information, including the interface speed status and configuration.

Parameter	Description
<IFNAME>	Specifies a interface name.
<IFRANGE>	Specifies the port identifier range.

Examples

Showing automatic downshift information:

```
switch(config-if)# show interface downshift-enable
-------------------------------------------------
```

```
                Downshift                Speed
Port       Enabled | Active     Status    | Config
--------------------------------------------------
1/1/1      yes       yes        100M-FDx   auto
1/1/2      yes       no         1G         auto
1/1/3      yes       no         100M-FDx   100M-FDx
1/1/4      no        no         --         auto
```

Showing automatic downshift information on per interface:

```
switch(config-if)# show interface 1/1/2 downshift-enable
--------------------------------------------------
                Downshift                Speed
Port       Enabled | Active     Status    | Config
--------------------------------------------------
1/1/2      yes       no         1G         auto
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show running-config interface`

Syntax

```
show running-config interface [<IFNNAME>|<IFRANGE>]
show running-config interface [lag | loopback | tunnel | vlan ] [<ID>]
```

Description

Displays active configurations of various switch interfaces.

Parameter	Description
<IFNAME>	Specifies a interface name.
<IFRANGE>	Specifies the port identifier range.
`LAG`	Specifies LAG interface information
`LOOPBACK`	Specifies loopback interface information.
`TUNNEL`	Specifies tunnel interface information.
`VLAN`	Specifies VLAN interface information.
<LAG-ID>	Specifies the LAG number. Range: 1 - 256.
<LOOPBACK-ID>	Specifies the LOOPBACK number. Range: 0 - 255.
<TUNNEL-ID>	Specifies the tunnel ID. Range: 1 - 255.
<VLAN-ID>	Specifies the VLAN ID. Range: 1 - 4094.
`VXLAN`	Specifies the VXLAN interface information.
<VXLAN-ID>	Specifies the VXLAN interface identifier. Default: 1.

Examples

Showing 1/1/2 interface configuration:

```
switch(config-if)# show running-config interface 1/1/2
interface 1/1/2
   no shutdown
   description DC-23
   exit
```

Showing loopback interfaces configured:

```
switch(config-if)# show running-config interface loopback
interface loopback 1
    description lb interface 1
    exit
interface loopback 2
    description lb interface 2
    exit
```

Showing loopback interfaces not configured:

```
switch(config-if)# show running-config interface loopback
No loopback interfaces configured.
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

Mirroring

Mirroring allows you to replicate all traffic arriving and/or leaving the selected system interfaces. This data can be used for collection or analysis.

The traffic replicated using mirroring can be sent to a separate interface on the same switch as the traffic source for analysis or inspection. Such a collection of interfaces and settings is called a mirror session.

A mirror session can be configured with many traffic sources but only a single output, or destination. In the initial configuration, the mirror session is disabled. You have enable the feature to start the replication.

CAUTION

Care must be taken in choosing the number and rates of sources to avoid over-saturating a session destination. A mirror session with multiple 10G sources can overwhelm a single 10G destination and important data may be lost.

This version of AOS-CX support the following mirror capabilities:

- Support for a VLAN as source for a mirror session
- Ability for a given Mirror source in one session to act as source in another Mirror Session
- Support for a Layer 2/bridged Link Aggregation Group (LAG) as Session destination
- Support for a Layer 3/Route-only Link Aggregation Group (LAG) as Session destination

NOTE

The following interface types are not supported as mirror source or destination:

- Management
- Persona
- Loopback
- VxLAN
- Subinterfaces

Hit Count Behavior for Policy Mirroring Actions

When using classifier policies with mirroring actions, it is important to note that packets mirrored via these policy actions will have their hit counts recorded within the Policy hit counts. These packets will not be recorded as output packets in the Mirror Session statistics, except in the case of Mirror-to-CPU sessions. This distinction means that traffic metrics for packets mirrored via Policy Actions should be obtained from Policy hit counts, as these packets will not appear in the Mirror Session output statistics.

Mirroring and CoPP

When configuring a Mirror Session with the destination set to the CPU, it is important to understand how mirrored traffic interacts with Control Plane Policing (CoPP).

Control plane packets that are already destined for the switch (for example, routing protocol packets and management traffic) will be processed by their respective CoPP classes. These packets will still be mirrored to the CPU as part of the Mirror-to-CPU session, but they will not be counted under the Mirror-to-CPU CoPP class. The Mirror-to-CPU CoPP class is specifically designed to manage dataplane traffic that is mirrored to the CPU. Any control plane traffic mirrored to the CPU will bypass this class and be handled by the CoPP class corresponding to its original purpose (e.g., OSPF, BGP, or ARP). Administrators should account for this distinction when analyzing traffic mirrored to the CPU.

For example, if a mirror session is configured to mirror all traffic from a source interface to the CPU, and that interface receives OSPF packets, the OSPF packets will be mirrored to the CPU. these packets will be processed by the OSPF CoPP class, not the Mirror-to-CPU CoPP class.

Subtopics

Mirror statistics

Mirror statistics are reset for a Mirror-to-CPU session when an interface is added or removed from a LAG that is a source interface in the Mirror session and during a failover.

Classifier policies and mirroring sessions

About this task

Network traffic can be mirrored to a destination interface in two ways:

- Using a mirroring session alone.
- Using Classifier Policies with mirror actions in conjunction with a mirroring session.

Basic mirroring sessions provide coarse control over the type of traffic mirrored from a source: all received, all transmitted, or both. However, a traffic class within a Classifier Policy applied to a source can provide much finer grained control of mirrored traffic. For example, a policy can match on many different aspects of the Ethernet or IPv4 or IPv6 header information in each frame or packet received or transmitted on an interface.

The steps to configure a policy and class with a mirror action are the following:

Procedure

1. Configuring a mirroring session with a destination interface.
2. Enabling the mirroring session.

3. Configuring the Classifier Policy, specifying the mirroring session ID in the mirror action.

Results

If the packets being mirrored are received from a VLAN that is not allowed on the mirror destination, the mirrored packets would be dropped at the mirror destination interface. When the mirrored packets are dropped at the destination, the mirror output packet and byte count will increment, however the packets will not be received at the mirror destination.

The mirror destination port among the active mirror sessions must be unique. That is, if an interface is configured as a source or destination in an active mirror session, the same port cannot be used as a destination in another active mirror session.

Scenario 1

Scenario 2

VLAN as a source

About this task

AOS-CX allows configuration of VLAN as a mirroring source. When a VLAN source is configured in the 'rx' direction, all packets are mirrored as they are received in the switch. When a VLAN source is configured in 'tx' direction, all packets are mirrored as they are transmitted out of the switch.

More than one source VLAN can be configured in a mirror session. Each such VLAN may specify its own direction.

There is a limit of 1024 source VLANs in each direction of a given mirror session.

Same VLANs can be configured as a mirror source for multiple sessions.

> **NOTE**
> When changing a source VLAN in an enabled mirror session (that is, adding, changing direction, or removing), mirrored packets being transmitted out the mirror destination port from other mirror sources may be briefly interrupted during the reconfiguration.

Direction of an existing source VLAN can be updated in one of two ways:

Procedure

1. Reenter the `source vlan` command with the new preferred direction.
2. Use the `no` form of the command with a direction (rx or tx) to selectively remove the specified direction. Specifying the last remaining direction for that VLAN will remove the VLAN from the configuration entirely.

Results

For packets bridged through the switch:

If the mirror is configured in 'both' direction, two copies of packets are mirrored, otherwise one copy of the packet will be mirrored.

For routed packets:

- If the mirror is configured in the 'rx' direction, packets are mirrored in the pre-routed form with the destination MAC address as the switch address.
- If the mirror is configured in the 'tx' direction, packets are mirrored in the post-routed form with the source MAC as the switch address. Destination MAC is the nexthop gateway or station.
- If the mirror is configured in the 'both' direction, one copy of the packet will be mirrored.

Control plane packets generated by the switch's CPU are processed both in the ingress and the egress packet processing pipeline. The following are the behaviors for mirroring with VLAN as source:

- If the mirror is configured in the 'rx' or 'tx' direction, the packets are mirrored to the mirror destination.
- If the mirror is configured in the 'both' direction, two copies of the packets are mirrored to the mirror destination.

Mirroring commands

Subtopics

- [source]
- [source interface]
- [source vlan]

`clear mirror`

Syntax

```
clear mirror [all | <SESSION-ID>]
```

Description

Clears the mirror statistics for all configured mirror sessions or a specified session

Parameter	Description
`all`	Specifies all configured sessions.
`<SESSION-ID>`	Specifies a numeric identifier for the session. Range: 1 to 4

Examples

Clearing mirror statistics for all configured mirror sessions:

```
switch# clear mirror all
```

Clearing mirror statistics for mirror session 1:

```
switch# clear mirror 1
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (`#`)	Administrators or local user group members with execution righ ts for this command.

`clear mirror endpoint`

Syntax

```
clear mirror endpoint [<NAME>]
```

Description

Clears mirror endpoint statistics for all configured mirror endpoints. The optional parameter can be added to clear a specific mirror endpoint.

Parameter	Description
`<NAME>`	Specifies name of the mirror endpoint instance to be cleared.

Examples

Clearing statistics for all configured mirror endpoints:

```
switch# clear mirror endpoint
```

Clearing mirror statistics for mirror endpoint test:

```
switch# clear mirror endpoint test
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manager (`#`)	Administrators or local user group members with execution rights for this command.

`comment`

Syntax

```
comment <COMMENT>
no comment
```

Description

Specifies a comment for the mirroring session.

When used in mirror endpoint command context, specifies a comment for the mirror endpoint.

The **no** form of this command removes the comment.

Parameter	Description
`<COMMENT>`	A comment string of up to 64 characters composed of letters, numbers, underscores, dashes, spaces, and periods.

Usage

Comments are optional and can be added or removed at any time without affecting the state of the mirroring session.

Adding a comment to a session that already has a comment replaces the existing comment.

Examples

Adding a comment to a mirror session:

```
switch(config-mirror-3)# comment This Mirror will be removed during next
maintenance window
```

Removing the comment from mirror session 3:

```
switch(config-mirror-3)# no comment
```

Adding a comment to a mirror endpoint:

```
switch(config-mirror-endpoint-test)# comment Monitor endpoint traffic
```

Replacing the existing comment for mirror endpoint:

```
switch(config-mirror-endpoint-test)# comment Monitor statistics on each
endpoint interfaces
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config-mirror-<SESSION-ID>`	Administrators or local user group members with execution righ ts for this command.

`copy tcpdump-pcap`

Syntax

```
copy tcpdump-pcap <FILE-NAME> <REMOTE-URL>
```

Description

Saves packet capture files to external storage.

Parameter	Description
	Specifies the packet capture file to save.

Parameter	Description
<FILE-NAME>	
<REMOTE-URL>	Specifies the external storage to which the packet capture file will be saved.

Usage

Only four files can be saved at any point on the switch. Packet capture files are not saved after a failover or reboot. View a list of saved files using **diag utilities list-files**.

Examples

Saving my_capture_file.pcap to sftp://root@10.0.0.2/file.pcap:

```
switch# copy tcpdump-pcap my_capture_file.pcap sftp://root@10.0.0.2/
file.pcap
root@10.0.0.2's passowrd:
Connected to 10.0.0.2.
sftp > put my_capture_file.pcap file.pcap
Uploading my_capture_file.pcap to /root/file.pcap
my_capture_file.pcap                              100%   156   219.8KB/s
00:00
Copied successfuly.
```

Command History

Release	Modification
10.08	Command introduced

Command Information

Platforms	Command context	Authority
5420 6200	Manager (#)	Administrators or local user group members with execution rights for this command.

`copy tshark-pcap`

Syntax

```
copy tshark-pcap <REMOTE-URL> [vrf <VRF-NAME>]
```

Description

Copies the tshark capture data to a file on a TFTP or SFTP server.

Parameter	Description
<REMOTE-URL>	Specifies the capture file on a remote TFTP or SFTP server. The URL syntax is: {tftp:// | sftp://**<USER>**@} {**<IP>**|**<HOST>**} [:**<PORT>**] [;blocksize=**<SIZE>**]/<FILE>
`vrf` *<VRF-NAME>*	Specifies the name of a VRF. Default: default.

Example

Copying the capture data to a file on SFTP server 10.0.0.2:

```
switch# copy tshark-pcap sftp://root@10.0.0.2/file.pcap
root@10.0.0.2's password:
Connected to 10.0.0.2.
sftp> put packets.pcap file.pcap
Uploading packets.pcap to /root/file.pcap
packets.pcap                                  100%  156   219.8KB/s   00:00
Copied successfully.
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420	Manager (`#`)	Administrators or local user group members with execution rights for this command.

Platforms	Command context	Authority
6200		

`destination cpu`

Syntax

```
destination cpu
no destination cpu
```

Description

The command causes the mirror session to transmit mirrored packets to the switch CPU. This destination may be configured for multiple sessions, however only one such configured session may be active at a given time.

The diagnostic utility Tshark may be used to view and capture packets transmitted to the CPU through this route. Ctrl+C must be entered to terminate a Tshark capture session. More details can be found in the **Supportability Guide**.

The **no** form of this command will immediately stops mirroring traffic to the CPU, but will not remove any sources from the mirror configuration.

Examples

Configuring a mirror session with CPU as the destination.

```
switch# config
switch(config)# mirror session 1
switch(config-mirror-1)# destination cpu
```

Removing the destination entirely.

```
switch(config-mirror-1)# no destination cpu
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-mirror-<SESSION-ID>`	Administrators or local user group members with execution righ ts for this command.

`destination interface`

Syntax

```
destination interface {<INTERFACE-ID>|<LAG-NAME>}
no destination interface {<INTERFACE-ID>|<LAG-NAME>}
```

Description

Configures the specified interface as the destination of the mirrored traffic.

The **no** form of this command immediately disables the mirroring session and removes the specified destination interface from the configuration.

Parameter	Description
<INTERFACE-ID>	Specifies a interface. Format: `member/slot/port`.
<LAG-NAME>	Specifies a LAG (link aggregation group) identifier.

Usage

Configuring a different destination interface in an enabled mirroring session causes all mirrored traffic to use the new destination interface. This action might cause a temporary suspension of mirrored source traffic during the reconfiguration. Mirroring traffic to a LAG destination with a device running LACP attached to the destination will cause the LAG link to flap if LACP packets are part of the mirrored traffic source.

Examples

Configuring a mirroring session and adding an interface as a destination:

```
switch(config)# mirror session 1
switch(config-mirror-1)# destination interface 1/1/1
```

Replacing the existing destination with different interface:

```
switch(config-mirror-1)# destination interface 1/1/12
```

Removing a destination:

```
switch(config-mirror-1)# no destination interface 1/1/12
```

Switch	Destination interface limit per mirror session (4 possible sess ions)
5420	64
6200	64
10040	1

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config-mirror-<SESSION-ID>`	Administrators or local user group members with execution righ ts for this command.

`destination tunnel`

Syntax

```
destination tunnel <TUNNEL-IPV4-ADDR> source <SOURCE-IPv4-ADDR>
   dscp <DSCP-VALUE> vrf <VRF-NAME> id <SPAN-ID>
no destination tunnel
```

```
destination tunnel <TUNNEL-IPV4> source <SOURCE-IPv4-ADDR>
   dscp <DSCP-VALUE> vrf <VRF-NAME>
```

```
no destination tunnel
```

Description

Specifies the tunnel where all mirrored traffic for the session is transmitted. Only one tunnel destination is allowed per session.

You may configure multiple mirror sessions with the same source/destination IP address pair, however, only one of those sessions sharing the same source/destination IP address pair can be enabled at a given time.

ERSPAN is not supported leaving the switch by the OOB port. If VRF management is configured for an ERSPAN session, the session will be in "mirror_err_tunnel_oob_port_not_supported" operation status.

ERSPAN is not supported leaving the switch encapsulated within another tunnel (e.g. GRE IPv4). When the path to the destination IP address will leave via a tunnel, the session will be in "tunnel_route_resolution_not_populated" operation status.

> **NOTE**
>
> The interface/LAG used to transmit ERSPAN packets should not be a source in the same mirror session.

The **no** form of this command will cease the use of the tunnel and disable the session.

Parameter	Description
<TUNNEL-IPV4-ADDR>	Specifies the tunnel address in IPv4 format (**x.x.x.x**), where **x** is a decimal number from 0 to 255.
<SOURCE-IPv4-ADDR>	Specifies the source address in IPv4 format (**x.x.x.x**), where **x** is a decimal number from 0 to 255.
<DSCP-VALUE>	Specifies the DSCP value to be carried within the DS field of E RSPAN packet header. Range: 0 to 63. Default: 0.

Parameter	Description
<VRF-NAME>	Specifies a VRF name. Default: default.

Examples

Creating a Mirror Session and adding tunnel destination, source, dscp, and VRF:

```
switch# config
switch(config)# mirror session 1
switch(config-mirror-1)# destination tunnel 1.1.1.1 source 2.2.2.2 dscp 10
vrf default
```

Replacing the existing tunnel destination:

```
switch(config-mirror-1)# destination tunnel 11.12.13.14 source 2.2.2.2 dscp
10 vrf default
```

Replacing the existing destination with a different DSCP value:

```
switch(config-mirror-1)# destination tunnel 11.12.13.14 source 2.2.2.2 dscp
2 vrf default
```

Removing the destination:

```
switch(config-mirror-1)# no destination tunnel
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	config-mirror-*<SESSION-ID>*	Administrators or local user group members with execution righ ts for this command.

diagnostic

Syntax

```
diagnostic
diag utilities tshark [file]
diag utilities tshark [delete-file]
```

Description

Captures packets from a mirror-to-cpu session, and save the most recent 32MB to pcap file which can then be copied and analyzed. When capturing a mirror-to-cpu session to a file, packets will not be dumped to the console.

> **NOTE**
>
> The `diagnostic` command must be entered prior to the `diag utilities tshark` command.

Use the **delete-file** form of this command to delete the most recent capture file.

Since **file** and **delete-file** are optional, the behavior of the base command **diag utilities tshark** does **not** save anything to a file, and instead dumps the tshark session to the console until **CTRL + c** is entered.

Parameter	Description
`file`	Saves captured packets to a temporary file.
`delete-file`	Deletes the most recent captured file.

Example

Performing diagnostic:

```
switch# diagnostic
switch# diagnostic utilities tshark file
Inspecting traffic mirrored to the CPU until Ctrl-C is entered
^CEnding traffic inspection.
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (`#`)	Administrators or local user group members with execution righ ts for this command.

`diag utilities tcpdump`

Syntax

```
diag utilities tcpdump [command <TEXT> | delete file <FILE-NAME> | list-
files |
  vrf <VRF-NAME> | count <COUNT-NUM> | proto <PROTO-NUM> | host-ip <IP-
ADDR> | source-ip
  <IP-ADDR> | destination-ip <IP-ADDR> | host-port <PORT> | source-port
<PORT> |
  destination-port <PORT> | verbosity <LEVEL> | print <DATA> | ethernet-
type <ETH-NUM>]
```

Description

Captures traffic received or transmitted over a network.

Parameter	Description
`command <TEXT>`	Captures packets based on a specified tcpdump command strin g.
`delete file <FILE-NAME>`	Deletes specified tcpdump list files.
`list-files`	Lists all the tcpdump capture files saved on the device.

Parameter	Description
`vrf <VRF-NAME>`	Captures packets on the specified VRF. If no VRF is named, the default is used.
`count <COUNT-NUM>`	Runs the tcpdump command until the specified number of pac kets are captured. Range: 1 - 2147483647.
`proto <PROTO-NUM>`	Captures packets of a particular type based on IP protocol num ber. Range: 0 - 255.
`host-ip <IP-ADDR>`	Captures packets matching with the source or destination IP ad dress.
`source-ip <IP-ADDR>`	Captures packets from the specified IP address.
`destination-ip <IP-ADDR>`	Captures packets sent to the specified IP address.
`host-port <PORT>`	Captures packets matching with the source or destination port.
`source-port <PORT>`	Captures packets from the specified IP port.
`destination-port <PORT>`	Captures packets sent to the specified IP port.
`verbosity <LEVEL>`	Captures packets of the specified verbosity. Range: level1 - l evel4. If no verbosity is specified, the default is level1.
`print <DATA>`	Captures the data of each packet. The maximum is 262144 by tes
`ethernet-type <ETH-NUM>`	Captures packets based on the particular ethernet type. Range: 0 - 65535.

Usage

- When using the **command** option, the only traffic captured will be packets that have been mirrored to the CPU.
- When using the **command** option, command line sanitization is performed to prevent options that may cause harm or security issues. The following options are blocked:
  - -i/--interface
  - -Z

- -B/--buffer-size
- -C
- -W
- -Z/--relinquish privileges

- Non-word operators such as "&" or "|" are not allowed. Use boolean keywords such as "and," "or," and "not."
- When using **command -r** to read a file, do not provide any directory path characters. Use list-files command to get the list of file names currently saved on the device, and then use those file names.
- A total of four files can be saved at any given point on the device. Packet capture files are not saved after a failover or reboot, but can be saved to external storage using the **copy tcpdump-pcap** command.

Examples

Inspecting traffic mirrored to the CPU via tcpdump and saving the output to my_capture_file.pcap:

```
switch# diag utilities tcpdump command -c 2 -x -w my_capture_file.pcap
Inspecting traffic mirrored to the CPU via tcpdump until Ctrl-C is entered.
2 packets captured
2 packets received by filter
0 packets dropped by kernel
Ending traffic capture.
```

Listing saved capture files:

```
switch# diag utilities tcpdump list-files
my_capture_file.pcap
```

Reading my_capture_file.pcap:

```
switch# diag utilities tcpdump command -r my_capture_file.pcap
reading from file /tmp/tcpdump/my_capture_file1.pcap, link-type EN10MB
(Ethernet)
  1  11:59:34.047867 IP6 localhost.40318 > localhost.ntp: NTPv2, Reserved,
length 12
        0x0000:  0000 0304 0006 0000 0000 0000 0000 86dd  ................
        0x0010:  600a 7e47 0014 1140 0000 0000 0000 0000  `.~G...@........
        0x0020:  0000 0000 0000 0001 0000 0000 0000 0000  ................
        0x0030:  0000 0000 0000 0001 9d7e 007b 0014 0027  .........~.{...'
        0x0040:  1601 0001 0000 0000 0000 0000            ............
  2  11:59:34.047915 IP6 localhost.ntp > localhost.40318: NTPv2, Reserved,
length 12
        0x0000:  0000 0304 0006 0000 0000 0000 0000 86dd  ................
        0x0010:  6b8d 23c5 0014 1140 0000 0000 0000 0000  k.#....@........
        0x0020:  0000 0000 0000 0001 0000 0000 0000 0000  ................
```

```
        0x0030:  0000 0000 0000 0001 007b 9d7e 0014 0027  .........{.~...'
        0x0040:  d681 0001 c016 0000 0000 0000
```

Removing my_capture_file.pcap:

```
switch# diag utilities tcpdump delete-file my_capture_file.pcap
Successfully removed file
```

Command History

Release	Modification
10.08	Command introduced

Command Information

Platforms	Command context	Authority
5420 6200	Manager (`#`)	Administrators or local user group members with execution righ ts for this command.

`disable`

Syntax

```
disable
```

Description

Disables the mirroring session specified by the current command context.

Usage

By default, mirroring sessions are disabled.

When a mirroring session is disabled, the **show mirror** command for that session ID shows an **Admin Status** of **disable** and an **Operation Status** of **disabled**.

Example

Disabling a mirroring session:

```
switch(config)# mirror session 3
switch(config-mirror-3)# disable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config-mirror-<SESSION-ID>`	Administrators or local user group members with execution righ ts for this command.

`enable`

Syntax

```
enable
```

Description

Enables the mirroring session for the current command context.

Usage

By default, mirroring sessions are disabled.

When a mirroring session is enabled, the **show mirror** command for that session ID shows an **Admin Status** of **enable** and an **Operation Status** of **enabled**.

If sFlow is enabled on an interface and a mirroring session specifies the same interface as the source of received traffic (the source is configured with a direction of

rx
or

both
):

- The attempt to enable the mirroring session fails and an error is returned.

> **NOTE**
> When adding, removing, or changing the configuration of a source interface in an enabled mirroring session, packets from other mirror sources using the same destination interface might be interrupted.

Example

Configuring and enabling a mirroring session:

```
switch(config)# mirror session 3
switch(config-mirror-3)# source interface 1/1/2 rx
switch(config-mirror-3)# destination interface 1/1/3
switch(config-mirror-3)# comment Monitor router port ingress-only traffic
switch(config-mirror-3)# enable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config-mirror-<SESSION-ID>`	Administrators or local user group members with execution righ ts for this command.

`mirror endpoint`

Syntax

```
mirror endpoint <NAME>
no mirror endpoint <NAME>
```

Description

Creates the specified mirror endpoint or enters its context if it already exists. The specifics of a mirror endpoint are created or altered while in the mirror endpoint context and the mirror endpoint is enabled or disabled from this context. It may be possible to support different encapsulations by different ASICs. For example, UDP for PVOS compatibility. Termination of GRE encapsulation is also supported.

The **no** form of this command removes an existing mirror endpoint. An enabled mirror endpoint is automatically disabled first before removal.

Parameter	Description
<NAME>	Specifies mirror endpoint name.

Examples

Creating a mirror endpoint named test :

```
switch(config)# mirror endpoint test
```

Deleting mirror endpoint named test:

```
switch(config)# no mirror endpoint test
```

Configuring a mirror endpoint named test :

```
6100(config)# mirror endpoint test
6100(config-mirror-endpoint-test)#
6100(config-mirror-endpoint-test)# destination
  interface  Specify interfaces to send traffic
6100(config-mirror-endpoint-test)# destination interface
  IFNAMELIST  An interface, a range or a comma seperated list of interfaces
6100(config-mirror-endpoint-test)# destination interface 1/1/3
  <cr>
6100(config-mirror-endpoint-test)# destination interface 1/1/3
6100(config-mirror-endpoint-test)#
6100(config-mirror-endpoint-test)# source 1.1.1.1 destination 1.1.1.2 id 1
```

```
vrf default
6100(config-mirror-endpoint-test)#
```

> **NOTE**
>
> Only physical ports can be configured as interface for mirror-endpoint destination. LAG port is not supported as interface for mirror-endpoint destination.

> **NOTE**
>
> The maximum allowed number of destination interfaces for both mirror-session and mirror-endpoint is 1.

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`mirror session`

Syntax

```
mirror session <SESSION-ID>
no mirror session <SESSION-ID>
```

Description

Creates a mirroring session configuration context or enters an existing mirroring session configuration context.

From this context, you can enter commands to configure and enable or disable the mirroring session.

The **no** form of this command removes an existing mirroring session from the configuration.

Parameter	Description
`<SESSION-ID>`	Specifies the session identifier. Range: 1 to 4

Examples

```
switch(config)# mirror session 1
switch(config-mirror-1)#
switch(config)# mirror session 3
switch(config-mirror-3)#
switch(config)# no mirror session 1
switch(config)#
```

> **NOTE**
> When configuring mirroring via the command-line interfacde, not all configuration errors will result in an immediate error message. After making configuration changes, always check the operation status of your mirror sessions using the **show mirror** command.

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`show mirror`

Syntax

```
show mirror [<SESSION-ID>]
```

Description

Shows information about mirroring sessions. If `<SESSION-ID>` is not specified, then the command shows a summary of all configured mirroring sessions. If `<SESSION-ID>` is specified, then the command shows detailed information about the specified mirroring session.

Parameter	Description
`<SESSION-ID>`	Specifies the session identifier. Range: 1 to 4

Usage

Information in the **Admin Status** column of the command output indicates the configured status. The admin status is one of the following values:

- **enable**: The mirroring session is enabled.
- **disable**: The mirroring session has been configured but not yet enabled, or has been disabled.

Information in the **Operation Statu**s colum indicates the status of the mirroring session. Operation status is one of the following values:

- **dest_doesnt_exis**t:The configured destination interface is not found in the system. The mirroring session cannot be enabled.
- destination_shutdown**: The mirroring session is enabled, but the destination interface is shut down. No traffic** can be monitored.
- **disabled**: The mirroring session is disabled and is not in an error condition.
- **enabled**: The mirroring session is enabled.
- **external/driver_error**: An internal ASIC hardware error occurred.
- **hit_active_sessions_capacity**: The mirroring session could not be enabled because the maximum number of supported mirroring sessions are already enabled.
- **internal_error**: An invalid parameter was passed to the ASIC software layer.
- **no_dest_configured**: The mirroring session does not have a destination interface configured.

- **no_name_configured**: A software error occurred. The mirroring session does not have a session ID in its configuration.
- **null_mirror**: A software error occurred. The session object reference is invalid.
- **out_of_memory**: The system is out of memory, reboot recommended.
- **tunnel_route_resolution_not_populated**: If the destination tunnel IP address is not reachable.
- **unknown_error**: An unexpected error occurred.

Examples

Showing summary information about all configured mirroring sessions:

```
switch# show mirror
ID  Admin Status  Operation Status
--- ------------- -----------------------------------------------------
1   enable        enabled
2   disable       disabled
3   disable       disabled
4   enable        internal_error
```

Showing detailed information about a single mirroring session:

```
switch# show mirror 3
 Mirror Session: 3
 Admin Status: disable
 Operation Status: disabled
 Comment: Monitor router port ingress-only traffic
 Source: interface 1/1/2 rx
 Destination: interface 1/1/3
switch#
```

Show the details of mirror session 1 with an empty LAG as destination:

```
switch: show mirror 1
Admin Status: enable
Operation Status: dest_doesnt_exist
Source: interface 1/1/1 rx
Source: interface tx none
Destination: interface lag1
Output Packets: 0
Output Bytes: 0
```

Show the details of mirror session 1 with a LAG and a LAG member of same LAG as sources. In this scenario, traffic received on all LAG member interfaces is mirrored, is it is not necessary to define 1/1/1 as a source since it is already part of LAG1; this does not impact the mirror output packet or byte counters.

```
switch: show mirror 1
Admin Status: enable
```

```
Operation Status: enabled
Source: interface 1/1/1 rx
Source: interface lag1 rx
Destination: interface 1/1/2
Output Packets: 0
Output Bytes: 0
```

Showing the details of mirror **session** 1 for a mirror configuration enabled with invalid interface which is not part of a lag:

```
switch# show mirror 1
Mirror Session: 1
Admin Status: enable
Operation Status: disabled
Source: interface 1/1/1 (unknown lag) both
Destination: cpu
Output Packets: 0
Output Bytes: 0
```

Certain configurations can cause this condition while the mirror source configuration remains unchanged:

1. Removing of a port from a LAG (Link Aggregation Group).
2. Port **split** or **unsplit** operations that result in LAG membership removal.

To recover from this issue, unconfigure the affected interface (previously part of a LAG), or restore the LAG membership configuration.

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (`>`) or Manage r (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show mirror endpoint`

Syntax

```
show mirror endpoint [<NAME>]
```

Description

Shows a list of all configured mirror endpoints, their Admin Status and their Operation Status.

The optional parameter will display the details of the specified mirror endpoint if it exists.

Parameter	Description
<NAME>	Specifies name of the mirror endpoint instance to be displayed.

Examples

Showing a summary of all configured mirror endpoints on the switch:

```
switch# show mirror endpoint
Name    Admin Status   Operation Status
-----  -------------- ----------------------------------------------------
test    enable         enabled
monitor disable        disabled
```

Showing the details of enabled mirror endpoint test:

```
switch# show mirror endpoint test
Mirror Endpoint: audit
Admin Status: enable
Operation Status: enabled
Comment: Mirror Endpoint Audit
Type: gre
Tunnel: source 1.1.1.1 destination 1.1.1.2 id 1 vrf default
Interface: 1/1/3
Output Packets: 123456789
Output Bytes: 0
```

NOTE

"Output Packets" in "show mirror endpoint [name]" is only supported for statistics.

"Output Bytes" in "show mirror endpoint [name]" is not supported due to ASIC limitation.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manager (`#`)	Administrators or local user group members with execution rights for this command.

`shutdown`

Syntax

```
shutdown
no shutdown
```

Description

Enables mirror endpoint from its default disabled state. To verify the mirror endpoint was successfully activated, run the **show mirror endpoint NAME** command and verify that the **Admin Status** and **Operational Status** has changed from disabled to enabled. If the status value remains disabled, consult the system logs to determine the reason for activation failure. To disable the mirror endpoint, first disable the remote mirror session on the switch that's originating the data. Next, use the `shutdown` command to disable the mirror endpoint.

Examples

Enabling a mirror endpoint:

```
switch(config)# mirror endpoint test
switch(config-mirror-endpoint-test)# no shutdown
```

Disabling a mirror endpoint:

```
switch(config)# mirror endpoint test
switch(config-mirror-endpoint-test)# shutdown
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`source`

Syntax

```
source <SOURCE-IP> destination <DESTINATION-IP> id <1-4294967295> [vrf
<VRF_NAME>] [type {gre}]
no source
```

Description

Configures tunnel parameters of the mirror endpoint. Configuring a tunnel parameter to a mirror endpoint will replace the existing configuration. By default the VRF is **default**, users can also explicitly provide a custom VRF. The default tunnel type is considered to be GRE and users also have the option to explicitly give type as GRE.

The **no** form removes the tunnel parameters of the mirror endpoint.

Parameter	Description
`<SOURCE-IP>`	Specifies L3 encapsulated IPv4 source in the form A.B.C.D.

Parameter	Description
<DESTINATION-IP>	Specifies L3 encapsulated IPv4 destination in the form A.B.C.D.
`id`	Specifies tunnel identifier from the encapsulated packet.
<VRF_NAME>	Specifies the name of VRF for which the tunnel belongs to.

Examples

Configuring a tunnel parameter to a mirror endpoint:

```
switch(config-mirror-endpoint-test)# source 1.1.1.1 destination 7.7.7.7 id
1 vrf default type gre
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`source interface`

Syntax

```
source interface {<PORT-NUM> | <LAG-NAME>} [<DIRECTION>]
no source interface {<PORT-NUM> | <LAG-NAME>} [<DIRECTION>]
```

Description

Configures the specified interface (either an Ethernet port or a LAG) as a source of traffic to be mirrored.

The **no** form of this command ceases mirroring traffic from the specified source interface and removes the source interface from the mirroring session configuration.

Parameter	Description
<PORT-NUM>	Specifies a physical port on the switch. Use the format **member/slot/port** (for example, **1/3/1**).
<LAG-NAME>	Specifies the identifier for the LAG (link aggregation group).
<DIRECTION>	Selects the direction of traffic to be mirrored from this source interface. There is no default for this parameter. Valid values are the following: • both: Mirror both transmitted and received packets. • rx: Mirror only received packets. • tx: Mirror only transmitted packets.

Usage

There is a limit of source interfaces in each direction of a given mirror session:

Switch	Source interface limit per mirror session (4 possible sessions)
5420	64
6200	64
10040	128

However, there is a practical limit to the amount of traffic that a mirror destination can transmit. For example, mirroring session with multiple 10G sources can overwhelm a single 10G destination.

> **NOTE**
>
> When adding, removing, or changing the configuration of a source port in an enabled mirroring session, packets from other mirror sources using the same destination port might be interrupted.

Examples

Configuring a mirrored traffic source interface:

```
switch(config-mirror-1)# source interface
  LAG-NAME       Enter a LAG name. For example, lag10
  PORT-NUM       Enter a port number
```

Creating a mirroring session and configuring a source interface to mirror both transmitted and received packets:

```
switch(config)# mirror session 1
switch(config-mirror-1)# source interface 1/1/1 both
```

Creating a second mirroring session and configuring two source interfaces. One port mirroring only transmitted packets and the other mirroring both transmitted and received packets:

```
switch(config)# mirror session 2
switch(config-mirror-2)# source interface 1/1/3 tx
switch(config-mirror-2)# source interface 1/2/1 both
```

Removing the first source interface:

```
switch(config-mirror-2)# no source interface 1/2/3
```

Configuring a source interface to mirror received packets only:

```
switch(config-mirror-3)# source interface 1/1/2 rx
```

Configuring a source interface to mirror both transmitted and received packets:

```
switch(config-mirror-1)# source interface 1/1/1 both
```

Configuring a LAG as source interface to mirror both transmitted and received packets:

```
switch(config-mirror-4)# source interface lag1 both
```

Stopping the mirroring of received packets from a configured source interface:

```
switch(config-mirror-4)# no source interface lag1 rx
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config-mirror-<SESSION-ID>`	Administrators or local user group members with execution righ ts for this command.

`source vlan`

Syntax

```
source vlan <VLAN-NUM> {rx | tx | both}
no source vlan <VLAN-NUM> {rx | tx | both}
```

Description

Mirroring with VLAN as a source is supported in the following traffic directions:

- **both** - traffic received and transmitted
- **rx** - only received traffic
- **tx** - only transmitted traffic

More than one source VLAN can be configured in a mirror session. Each such VLAN may specify its own direction.

There is a limit of 1024 source VLANs for a given mirror session. There is also a limit of 4096 source VLANs across all mirror sessions.

> **NOTE**
>
> When changing a source VLAN in an enabled mirror session (i.e. adding, changing direction, or removing) mirrored packets being transmitted out of the mirror destination port from other mirror sources may be briefly interrupted during the reconfiguration.

Direction of an existing source VLAN can be updated in one of two ways.

- Reenter the **source vlan <VLAN-NUM> <direction>** command with the new preferred direction.
- Use the **no source vlan <VLAN-NUM> <direction>** form of the command with a direction (**rx** or **tx**) to selectively remove the specified direction.

Specifying the last remaining direction for that VLAN will remove the VLAN from the configuration entirely.

Mirroring allows configuration of VLAN as a source. When VLAN source is configured in the **rx** direction, all packets are mirrored as they are received in the switch. When VLAN source is configured in **tx** direction, all packets are mirrored as they are transmitted out of the switch.

For packets bridged through the switch:

- If the mirror is configured in 'both' direction, two copies of packets are mirrored, otherwise one copy of the packet will be mirrored.

For routed packets:

- If the mirror is configured in **rx** direction, packets are mirrored in the pre-routed form with the Destination MAC address as the switch address.
- If the mirror is configured in **tx** direction, packets are mirrored in post-routed form with the source MAC as the switch address. Destination MAC is the nexthop gateway or station.
- If the mirror is configured in **both** direction, one copy of the packet will be mirrored.

Control plane packets generated by the switch's CPU are processed both in theingress and the egress packet processing pipeline. The following are the behavior for mirroring with VLAN as source:

- If the mirror is configured in the **rx** or **tx** direction, the packets are mirrored to the mirror destination.
- If the mirror is configured in the **both** direction, two copies of the packets are mirrored to the mirror destination.

The **no** form command will cease mirroring traffic from the specified source VLAN and remove the source from the mirror configuration.

Parameter	Description
`VLAN-NUM`	Selects the VLAN number.
`direction`	Specifies the direction of mirroring. **tx** (transmit), **rx** (receive), o r **both**.

Examples

Creating a mirror session and adding a VLAN as a source of traffic in both directions on that port:

```
switch# configure terminal
switch(config)# mirror session 1
switch(config-mirror-1)# source vlan 10 both
```

Creating a mirror session and adding two VLANs as sources of traffic:

directions:

```
switch# configure terminal
switch(config)# mirror session 2
switch(config-mirror-2)# source vlan 10 tx
switch(config-mirror-2)# source vlan 20 both
```

Configuring the source in session 2 to receive by specifying the source interface configuration:

```
switch(config-mirror-2)# source vlan 10 rx
```

Removing the first source interface in session 2 entirely, and removing the transmit direction from the other so that mirroring only occurs in the receive direction:

```
switch(config-mirror-2)# source vlan 10 rx
switch(config-mirror-2)# source vlan 20 tx
```

```
switch(config-mirror-2)# source vlan 2000 rx
The maximum number of source VLANs per mirror session is 1024 in each
direction
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution rights for this command.

Monitoring a device using SNMP

Configuring SNMP: Refer to the SNMP/MIB Guide for information on how to add SNMP so a device can be monitored from a network management system (NMS).

Configuring an SNMP trap receiver: Refer to the SNMP/MIB Guide and specific information about the `show snmp trap` command to enable SNMP traps.

Power-over-Ethernet

- The Power-over-Ethernet (PoE) subsystem manages power supplied to devices using standard Ethernet data cables. A Power Sourcing Equipment (PSE) supplies DC power as well as Ethernet connectivity to a Powered Device (PD) using a standard Ethernet cable. The maximum current depends on the PD Requested Class.
- A PoE subsystem contains two parts : a PSE and PD. A Power Sourcing Equipment (PSE) is a device that provides power through a standard Ethernet cable. A PoE capable switch functions as PSE. All Aruba PoE switches are considered as PSEs. A PD is a device powered by a PSE. Examples of PD are VoIP phones, Wireless APs, and IP cameras.
- When a PD or any network cable is connected to a PSE port, the PSE applies a detection voltage and measures the resistance value of the PD. If resistance is within IEEE 802.3 standard values (23 - 26k ohm), the connected device is treated as PD and classification begins. For legacy devices to be detected, you must enable prestandard detection on the switch.
- PDs are divided into different types and classes based on PD power requirements. The power supplied by the PSE is higher than the power PD draws to accommodate for the line losses that can result with the use of the standard maximum length cable(100m).
    - Type 1: PSE can supply maximum of 15.4W, and PD can draw a maximum of 13W.
    - Type 2: PSE can supply maximum of 30W, and PD can draw a maximum of 25.5W.

- Type 3: PSE can supply maximum of 60W, and PD can draw a maximum of 51W.
- Type 4: PSE can supply maximum of 90W, and PD can draw a maximum of 71W.

- Classes of PD:
  - Class 0: Type1 PD, it can draw a maximum of 13W.
  - Class 1: Type1 PD, it can draw a maximum of 3.84W.
  - Class 2: Type1 PD, it can draw a maximum of 6.49W.
  - Class 3: Type1 PD, it can draw a maximum of 13W.
  - Class 4: Type2 PD, it can draw a maximum of 25.5W.
  - Class 5: Type3 PD, it can draw a maximum of 40W.
  - Class 6: Type3 PD, it can draw a maximum of 51W.
  - Class 7: Type4 PD, it can draw a maximum of 62W.
  - Class 8: Type4 PD, it can draw a maximum of 71.3W.
- IEEE 802.3bt introduced 4-Pair PoE as a means of supplying higher power to PDs that need more than the current 25.5W supplied by IEEE 802.3at. To increase the available power without damaging the Ethernet cable, the standard introduced the ability to use all four pairs within the Ethernet cable instead of the two pairs used by previous standards (802.3at, 802.3af).
- Supported protocols:
  - Compatibility with IEEE 802.3af, 802.3at, 802.3bt and prestandard.
  - Long first class event supported on Type 3-4 PSE.
  - Support for Single Signature (SS) Type 0-6 and Dual Signature (DS) Type 0-4 PDs.
  - Multi-Event classification permits mutual ID of SS Class 0-6 and DS Class 0-4.
  - Support LLDP Data Link Layer (DLL) Type 1-2 extension 12-octet TLV and Type 3-4 extension 29-octet TLV.
  - Default PSE assigned class delivers the maximum PSE capable power at initial power up based on PD requested class.

- Always-on PoE is a feature that provides the ability for a switch to continue to provide power across user initiated reboots through software. Always-on PoE is enabled by default and no additional configuration is needed.

> **NOTE**
>
> PDs only remain powered, no data transfer or PoE power negotiation can occur until the switch has completely booted up and in normal operation. PD faults occurring prior to full switch boot up will result in PoE power removal and restart the detection process only after switch returns to normal operation.

Subtopics

PoE commands

All PoE configuration commands except **quick-poe**, **threshold configuration** and **always-on poe configuration** are entered at the **config-if** context. The PoE threshold command is used at the system level whereas the **always-on poe** and **power-over-ethernet quick-poe** commands are set at the slot level. These commands can only be configured in the global configuration context.

Subtopics

`lldp dot3 poe`

Syntax

```
lldp dot3 poe
no lldp dot3 poe
```

Description

Enables 802.3 TLV list in LLDP to advertise for Power over Ethernet Data Link Layer Classification. LLDP dot3 TLV is by default enabled for PoE.

The **no** form of this command disables 802.3 TLV list in LLDP.

Examples

Enabling 802.3 TLV list in LLDP:

```
switch(config)# interface 1/1/1
switch(config-if)# lldp dot3 poe
```

Disabling 802.3 TLV list in LLDP:

```
switch(config-if)# no lldp dot3 poe
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

`lldp med poe`

Syntax

```
lldp med poe [priority-override]
no lldp med poe [priority-override]
```

Description

Enables MED TLV list in LLDP to advertise for Power over Ethernet Data Link Layer Classification. Also enables the lldp-MED TLV priority to override user configured port priority for Power over Ethernet. When both dot3 and MED are enabled, dot 3 will take precedence. MED TLV is by default enabled for PoE. Priority over-ride is by default disabled.

The **no** form of this command disables MED TLV list in LLDP.

Parameter	Description
`[priority-override]`	System defined name of the interface.

Examples

Enabling and disabling LLDP MED PoE:

```
switch(config)# interface 1/1/1
switch(config-if)# lldp med poe
switch(config-if)# no lldp med poe
```

Enabling and disabling LLDP MED PoE priority override:

```
switch(config-if)# lldp med poe priority-override
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

power-over-ethernet

Syntax

```
power-over-ethernet
no power-over-ethernet
```

Description

Enables per-interface power distribution. Per-port power is enabled by default with priority low. PoE cannot be disabled for individual ports when Quick PoE is enabled for the entire switch or line module.

The **no** form of this command disables per-interface power distribution.

Examples

Enabling per-interface power distribution:

```
switch(config)# interface 1/1/1
switch(config-if)# power-over-ethernet
```

Disabling per-interface power distribution:

```
switch(config-if)# no power-over-ethernet
```

Showing Quick PoE enabled:

```
switch(config)# power-over-ethernet quick-poe 1/1
switch(config)# no power-over-ethernet
Interface PoE cannot be disabled when Quick PoE is enabled.
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

power-over-ethernet allocate-by

Syntax

```
power-over-ethernet allocate-by {usage | class}
no power-over-ethernet allocate-by {usage | class}
```

Description

Configures the power allocation method. Power allocation method is initially based on usage. PSE Allocated power value will change to LLDP negotiated power if and when LLDP exchange takes place between PSE and PD. When there is no LLDP negotiation, PSE Allocated Power Value will be the actual instantaneous power draw and reserve power based on actual consumption. In allocate-by class, power allocation is based on PD requested class and PSE allocated power value will be the LLDP negotiated power when LLDP exchange takes place between PSE and PD. When there is no LLDP negotiation, PSE Allocate Power will be based on PD class. Reserve power is based on PD Class. By default, power allocation is by usage.

The power allocation method can be changed on an interface through port-access (User roles or RADIUS). An allocation method when configured through port-access will replace the user configured method.

The **no** form of this command resets the action to default.

Parameter	Description
`usage`	Configures the usage - based allocation method.
`class`	Configures the class - based allocation method.

Usage

If you enable **pd-class-override** for an interface, the **allocate-by** configuration of that interface will be automatically changed to **class**. However, if you change the allocation method to **usage** when **pd-class-override** is still enabled, you will receive an error message stating that "The power allocation method cannot be changed when pd-class-override is enabled."

To remove **pd-class-override**, you can use the **no power-over-ethernet pd-class-override** command . It is important to note that **pd-class-override** requires the allocation method to be set to **class** and is enforced when configured through CLI. However, if you override the allocation method to **usage** via port-access, **pd-class-override** will not be in effect. Therefore, it is recommended that you do not override the allocation method to **usage** through port-access on interfaces configured with **pd-class-override**.

Examples

Configuring the power allocation method:

```
switch(config)# interface 1/1/1
switch(config-if)# power-over-ethernet allocate-by usage
switch(config-if)# power-over-ethernet allocate-by class
```

Resetting power allocation method:

```
switch(config-if)# no power-over-ethernet allocate-by class
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

`power-over-ethernet always-on`

Syntax

```
power-over-ethernet always-on <MODULE-ID>
no power-over-ethernet always-on <MODULE-ID>
```

Description

Always-on PoE is a feature that provides the ability to the switch to continue to provide power across a soft reboot. It is applicable only to the interfaces which were connected and delivering before the soft reboot. Also, power will not be delivered if power to the switch is interrupted. This command enables or disables the always-on PoE feature at the switch or the slot level. By default, always-on PoE is enabled at the switch or the slot level.

The **no** form of this command disables power distribution on soft reboot.

Parameter	Description
	Module number to apply always - on PoE configuration.

Parameter	Description
<MODULE-ID>	

Examples

Enabling per-interface power distribution:

```
switch(config)# power-over-ethernet always-on 1/1
```

Disabling per-interface power distribution:

```
switch(config)# no power-over-ethernet always-on 1/1
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`power-over-ethernet assigned-class`

Syntax

```
power-over-ethernet assigned-class {3 | 4 | 6}
no power-over-ethernet assigned-class
```

Description

Limit PoE power based on the assigned class. When an user assigns a maximum class to an interface, the PSE will limit the maximum power delivered to the PD up to a total power draw not exceeding the PSE assigned-class power. Power demotion occurs when a PD requested class is higher than the PSE assigned class, permitting the PD to receive power and operate in a reduced power mode. PoE ports cannot set an assigned class when Quick PoE is enabled on the sybsystem. The default assigned class is 4 for 2-pair capable PSE and 6 for 4-pair capable PSE.

The **no** form of this command resets the action to default.

Examples

Setting PoE assigned class:

```
switch(config)# interface 1/1/1
switch(config-if)# power-over-ethernet assigned-class 4
```

Resetting PoE assigned class to default:

```
switch(config-if)# no power-over-ethernet assigned-class 4
```

Showing Quick PoE enabled:

```
switch(config)# power-over-ethernet quick-poe 1/1
switch(config)# interface 1/1/1
switch(config)# power-over-ethernet assigned-class 4
Interface assigned class cannot be configured when Quick PoE is enabled.
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

`power-over-ethernet pre-std-detect`

Syntax

```
power-over-ethernet pre-std-detect
no power-over-ethernet pre-std-detect
```

Description

Before IEEE 802.3 released the first Power over Ethernet standard (802.3af), vendors had shipped PoE capable switches and PD's. HPE Aruba Networking switches are backward compatible and will support both IEEE standard and pre-standard 802.3af Power over Ethernet PD's concurrently. This CLI allows the user to enable or disable pre-802.3af-standard device detection and powering on the specific port. When pre-std-detect is enabled, power will be delivered on PairA only. Default is disabled.

The **no** form of this command resets the action to default.

Examples

Enabling pre-standard device detection:

```
switch(config)# interface 1/1/1
switch(config-if)# power-over-ethernet pre-std-detect
```

Disabling pre-standard device detection:

```
switch(config-if)# no power-over-ethernet pre-std-detect
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

`power-over-ethernet priority`

Syntax

```
power-over-ethernet priority {critical | high | low}
no power-over-ethernet priority {critical | high | low}
```

Description

Sets PoE priority for an interface Specifying critical, high, or low indicates the priority of the interface in the event of power over-subscription. Within the same priority level, higher power-priority line-module ports have higher precedence. With same PoE priority and same line-module priority, lower numbered line-module ports have higher precedence. Per-interface PoE priority is low by default.

The **no** form of this command resets the priority to default PoE priority "low".

Examples

Configuring PoE priority:

```
switch(config)# interface 1/1/1
switch(config-if)# power-over-ethernet priority critical
switch(config-if)# power-over-ethernet priority high
```

Resetting the PoE priority to default:

```
switch(config-if)# no power-over-ethernet priority high
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

`power-over-ethernet quick-poe`

Syntax

```
power-over-ethernet quick-poe <MODULE-ID>
no power-over-ethernet quick-poe <MODULE-ID
```

Description

Quick PoE is a feature that provides the ability for the switch to provide power to the connected powered device as soon as switch goes through cold reboot. When quick PoE is enabled on the subsystem PoE port disablement and PD demotion is not allowed. also quick PoE enablement is not allowed if any of the port is disabled on the subsystem. User should not over-subscribe the PoE power when quick PoE is enabled. Quick PoE saved configuration will work irrespective of the configuration change at reboot.

Enables quick PoE feature on the switch or the subsystem level. By default, quick-PoE is disabled for the subsystem.

The **no** form of this command disables quick PoE.

Parameter	Description
`<MODULE-ID>`	Specifies module number for quick PoE configuration .

Examples

Enabling and disabling quick PoE:

```
switch(config)# power-over-ethernet quick-poe 1/2
switch(config)# no power-over-ethernet quick-poe 1/2
switch(config-if)# power-over-ethernet quick-poe 1/1
PoE must be enabled on all interfaces before enabling Quick PoE
switch(config-if)# power-over-ethernet quick-poe 1/3
All interfaces must use the default assigned class before enabling Quick PoE
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

`power-over-ethernet threshold`

Syntax

```
power-over-ethernet threshold <PERCENTAGE>
no power-over-ethernet threshold <PERCENTAGE>
```

Description

Sets the threshold at which the system will send an excess power consumption notification trap. Default value is 80 percentage.

The **no** form of this command resets the action to default.

Parameter	Description
`<PERCENTAGE>`	Excess power consumption trap threshold. Range 1 - 99.

Examples

Setting the power-over-ethernet threshold:

```
switch(config)# power-over-ethernet threshold 75
```

Resetting the power-over-ethernet threshold to default:

```
switch(config-if)# no power-over-ethernet threshold 75
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420	`config`	Administrators or local user group members with execution righ ts for this command.

Platforms	Command context	Authority
6200		

power-over-ethernet trap

Syntax

```
power-over-ethernet trap
no power-over-ethernet trap
```

Description

This command enables/disables the SNMP trap generation for PoE related events at system level. PoE trap generation is enabled by default.

The **no** form of this command resets the priority to default PoE priority "low".

Examples

Enabling SNMP trap generation for PoE:

```
switch(config)# power-over-ethernet trap
```

Disabling SNMP trap generation for PoE:

```
switch(config-if)# no power-over-ethernet trap
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

`show lldp local`

Syntax

```
show lldp local-device [<INTERFACE-ID>]
```

Description

Displays information advertised by the switch if the LLDP feature is enabled by user.

Parameter	Description
`<INTERFACE-ID>`	Specifies an interface. Format: **member/slot/port**

Examples

Showing LLDP local device:

```
switch# show lldp local-device 1/1/10
Local Port Data
===============
Port-ID             : 1/1/10
Port-Desc           : "1/1/10"
Port VLAN ID        : 0
PoE Plus Information
PoE Device Type      : Type 2 PSE
Power Source         : Primary
Power Priority       : low
PSE Allocated Power: 25.0 W
PD Requested Power : 25.0 W
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`show lldp neighbor`

Syntax

```
show lldp neighbor [<INTERFACE-ID>]
```

Description

Displays detailed information about a particular neighbor connected to a particular interface.

Parameter	Description
`<INTERFACE-ID>`	Specifies an interface. Format: **member/slot/port**

Examples

Showing LLDP neighbor information when there is only one neighbor:

```
switch# show lldp neighbor-info 1/1/10
Port                            : 1/1/10
Neighbor Entries                : 1
Neighbor Entries Deleted        : 0
Neighbor Entries Dropped        : 0
Neighbor Entries Aged-Out       : 0
Neighbor Chassis-Name           : 84:d4:7e:ce:5d:68
Neighbor Chassis-Description    : ArubaOS (MODEL: 325), Version HPE_ANW IAP
Neighbor Chassis-ID             : 84:d4:7e:ce:5d:68
Neighbor Management-Address     : 169.254.41.250
Chassis Capabilities Available  : Bridge, WLAN
Chassis Capabilities Enabled    :
Neighbor Port-ID                : 84:d4:7e:ce:5d:68
```

```
Neighbor Port-Desc                 : eth0
TTL                                : 120
Neighbor Port VLAN ID              :
Neighbor PoEplus information       : DOT3
Neighbor Device Type               : TYPE2 PD
Neighbor Power Priority            : Unkown
Neighbor Power Source              : Primary
Neighbor Power Requested           : 25.0 W
Neighbor Power Allocated           : 0.0 W
Neighbor Power Supported           : No
Neighbor Power Enabled             : No
Neighbor Power Class               : 5
Neighbor Power Paircontrol         : No
Neighbor Power Pairs               : SIGNAL
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`show power-over-ethernet`

Syntax

```
show power-over-ethernet [member <MEMBER-ID>] [brief]
```

Description

Displays the status information of the full system.

Parameter	Description
<MEMBER-ID>	Displays the detailed status of given member.
<IFNAME>	Display the detailed status of given port.
`brief`	Display the brief status of all ports or the given port.

Examples

Showing sample output for show power-over-ethernet on standalone box with VSF capabiity:

```
switch# show power-over-ethernet
System Power Status for member 1
  Configured  Power Status      : No redundancy
  Operational Power Status      : No redundancy
  Total Available Power         : 740 W
  Total Failover Pwr Avl        :   0 W
  Total Redundancy Power        :   0 W
  Total Power Drawn             :   0 W +/- 6W
  Total Power Reserved          :   0 W
  Total Remaining Power         : 740 W
  Trap Threshold                : 80 %
  Trap Enabled                  : Yes
  Always-on PoE Enabled         : 1/1
  Quick PoE Enabled             : None
Internal Power
        Total Power
  PS    (Watts)        Status
  ----- -------------  ---------------------
  1     0              Absent
  2     740            Ok
System Power Status for member 2
  Configured  Power Status      : No redundancy
  Operational Power Status      : No redundancy
  Total Available Power         : 600 W
  Total Failover Pwr Avl        :   0 W
  Total Redundancy Power        :   0 W
```

```
  Total Power Drawn             :   0 W +/- 6W
  Total Power Reserved          :   0 W
  Total Remaining Power         : 600 W
  Trap Threshold                : 80 %
  Trap Enabled                  : Yes
  Always-on PoE Enabled         : None
  Quick PoE Enabled             : None
Internal Power
        Total Power
  PS     (Watts)        Status
  ----- -------------  ---------------------
  1      0              Absent
  2      600            Ok
```

Showing sample output for power-over-ethernet member:

```
switch# show power-over-ethernet member 1
System Power Status for member 1
  Configured  Power Status      : No redundancy
  Operational Power Status      : No redundancy
  Total Available Power         : 740 W
  Total Failover Pwr Avl        :   0 W
  Total Redundancy Power        :   0 W
  Total Power Drawn             :   0 W +/- 6W
  Total Power Reserved          :   0 W
  Total Remaining Power         : 740 W
  Trap Threshold                : 80 %
  Trap Enabled                  : No
  Always-on PoE Enabled         : 1/1
  Quick PoE Enabled             : 1/1
Internal Power
        Total Power
  PS     (Watts)        Status
  ----- -------------  ---------------------
  1      0              Absent
  2      740            Ok
```

Showing sample output for power-over-ethernet brief in a VSF stack:

```
switch# show power-over-ethernet brief
Status and Configuration Information for PoE
  Member 1 Power Status
    Available: 370 W  Reserved: 55.60 W  Remaining: 314.40 W
    Always-on PoE Enabled: 1/1
    Quick PoE Enabled: None
PoE      Pwr Power    Pre-std Alloc PSE Pwr PD Pwr PoE Port     PD      Cls
Type
```

```
Port      En  Priority Detect  Act   Rsrvd   Draw   Status       Sign
------- --- ------   ------- ----- ------  ------ ---------    -----  ---
----
1/1/1     Yes Low      Off     Class  0.0 W   0.0 W Denied       None   4   2
1/1/2     Yes Critical Off     Usage  1.6 W   1.5 W Delivering*  Single 0   1
1/1/3     Yes High     Off     Class 54.0 W  25.5 W Delivering*^ Dual   1/3 3
1/1/4     No  Low      On      Usage  0.0 W   0.0 W Disabled     None   N/A
N/A
  Member 2 Power Status
    Available: 600 W  Reserved: 0.00 W  Remaining: 600 W
    Always-on PoE Enabled: None
    Quick PoE Enabled: None
PoE       Pwr Power    Pre-std Alloc PSE Pwr PD Pwr PoE Port     PD     Cls
Type
Port      En  Priority Detect  Act   Rsrvd   Draw   Status       Sign
------- --- ------   ------- ----- ------  ------ ---------    -----  ---
----
2/1/1     Yes Low      Off     Class  0.0 W   0.0 W Searching    None   N/A
N/A
2/1/2     Yes Critical Off     Usage  0.0 W   0.0 W Searching    None   N/A
N/A
2/1/3     Yes High     Off     Class  0.0 W   0.0 W Searching    None   N/A
N/A
2/1/4     No  Low      On      Usage  0.0 W   0.0 W Disabled     None   N/A
N/A
*This port may go down in the event of a PSU failure.
^This port is power demoted due to user config or power availabilty.
```

Showing sample output for power-over-ethernet brief per-port:

```
switch# show power-over-ethernet 1/1/1 brief
Status and Configuration Information for port 1/1/1
  Member 1Power Status
    Available: 370 W  Reserved: 55.60 W  Remaining: 314.40 W
    Always-on PoE Enabled: 1/1
PoE       Pwr Power    Pre-std Alloc PSE Pwr PD Pwr PoE Port     PD     Cls
Type
Port      En  Priority Detect  Act   Rsrvd   Draw   Status       Sign
------- --- ------   ------- ----- ------  ------ ---------    -----  ---
----
1/1/1     Yes Low      Off     Class  0.0 W   0.0 W Denied       None   4   2
```

Showing sample output for power-over-ethernet brief for interface range:

For 6300 Switch series:

```
switch# show power-over-ethernet 1/1/1-1/1/2 brief
Status and Configuration Information for port 1/1/1-1/1/2
  Member 1Power Status
    Available: 370 W  Reserved: 55.60 W  Remaining: 314.40 W
    Always-on PoE Enabled: 1/1
    Quick PoE Enabled: None
PoE      Pwr Power    Pre-std Alloc PSE Pwr PD Pwr PoE Port     PD     Cls
Type
Port     En  Priority Detect  Act    Rsrvd   Draw   Status       Sign
-------  --- ------   ------- ----- ------  ------ ---------    -----  ---
----
1/1/1    Yes Low      Off     Class  0.0 W   0.0 W Denied       None   4   2
1/1/2    Yes Critical Off     Usage  1.6 W   1.5 W Delivering*  Single 0   1
```

Showing sample output for power-over-ethernet for a missing line card:

```
switch# show power-over-ethernet 1/3 brief
Module 1/3 is not physically present.
```

Showing sample output for power-over-ethernet brief for a missing member:

```
switch# show power-over-ethernet member 3 brief
Member 3 is not physically present.
```

Showing sample output for power-over-ethernet port when physical interface is not present:

```
switch# show power-over-ethernet 2/1/1
Interface 2/1/1 is not present.
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Aruba AirWave

You can manage and monitor the AOS-CX switch through Aruba AirWave. The following benefits and functions include:

- Configuration (partial configuration)
- Device topology
- Immediate and historical trend reports
- Monitoring of the device and user connected to the network.
- Network discovery
- Syslogs and trap receiver

For information about which versions of Aruba AirWave support AOS-CX, see the AOS-CX Release Notes.

Subtopics

SNMP support and AirWave

For AirWave to discover and monitor the switch, you must:

- Enable the SNMP services on the switch.
- Configure the SNMP agent to use the SNMP version supported by the management station.

SNMP on the switch

The switch provides SNMP services through the management channel and the data interfaces. Functionality, such as device discovery from NMS, syslog and trap forwarding, can be any channel configured by you.

Although the SNMP server can be enabled on both VRFs (`mgmt` and `default`), only one instance of SNMP can be running. The highest priority is on the `default` VRF.

For example, assume that SNMP is first enabled on the `mgmt` VRF (`snmp-server vrf mgmt`). Then, SNMP is enabled on the `default` VRF (`snmp-server vrf default`) without disabling SNMP on the `mgmt` (using an equivalent `no` form of the command). The `show running-config` command displays both `snmp-server vrf` commands; however, the SNMP instance is running only on the `default` VRF (highest priority).

```
switch# config
switch(config)# snmp-server vrf mgmt
switch(config)# snmp-server vrf default
switch(config)# show running-config
Current configuration:
!
!Version AOS-CX Virtual.10.01.
led locator on
!
!
!
snmp-server vrf default
snmp-server vrf mgmt
!
...
```

Supported features with AirWave and the AOS-CX switch

AirWave supports the following features with the AOS-CX switch:

Device management	Device discovery using SNMPv2C and SNMPv3
	Device dashboards
Monitoring management	Device health attributes (device status/reachability)
	Interface and VLAN management
	Initiates an SSH connection from Aruba AirWave to AOS-CX so that the device outputs from the AOS-CX CLI can be displayed in the Aruba AirWave user inter face.
	Firmware versions
	Displays neighbor devices connected to AOS-CX swi tches
	Device topology
Configuration management	Partial configuration

Alarm management	Alarm triggers (device and interface up/down, new device discoveries, custom event triggers)
	Syslogs and traps
Report management	Device inventory, interface utilization, and device reachability reports
	Summary report of device model, firmware, and boot loader version

Configuring the AOS-CX switch to be monitored by AirWave

Prerequisites

Aruba AirWave is active on the network.

Procedure

1. Enable SNMP on the switch by entering the `snmp-server vrf` command.
```
switch(config)# snmp-server vrf mgmt
switch(config)# snmp-server vrf default
```
2. Configure the SNMPv2C community to public by entering the `snmp-server community public` command. In this instance, `public` is a read-only community string.
```
switch(config)# snmp-server community public
```
3. The community-string is used by SNMPv1 and SNMPv2C for unencrypted authentication. SNMPv3 lets you encrypt the authentication mechanism. To enable SNMPv3, enter the `snmpv3 user` and `snmpv3 context` commands.
```
switch(config)# snmpv3 user Admin auth sha auth-pass ciphertext
AQBapZHf2d20GYr/xcGUzYzm0zjNf/4VKHtSqbNImqtfYbJYCgAAALkGFJVcSp3nZ3o=
priv des priv-pass ciphertext
AQBapb0H2poBQKXPoVsC9L9qzZyfJQnzR7hmTr7LGsOsI7K3CgAAAKP98Rq2jfTrFwQ=
switch(config)# snmpv3 context Admin
```
For discovering devices in AirWave through the SNMPv3 community, the SNMPv3 context name is not mandatory. Devices can still be discovered in Aruba AirWave without the SNMPv3 context name.
4. Enter the `logging` command for enabling syslog forwarding to a remote syslog server, such as AirWave:

```
switch(config)# logging 10.0.10.2 severity debug
```

5. SNMP traps enable an agent to notify the management station of significant events by way of an unsolicited SNMP message. Enable SNMP traps by entering the `snmp-server host` command:

```
switch(config)# snmp-server host 10.10.10.10 trap version v2c vrf default
```

   SNMP traps cannot be forwarded from AOS-CX 10.00 switches that have the VRF configured as mgmt. Later versions of AOS-CX support SNMP trap forwarding even when the VRF is configured as default or mgmt.

6. For information on how to add a device for monitoring in the Aruba AirWave user interface, see the documentation for Aruba AirWave.

AirWave commands

Subtopics

`logging`

Syntax

```
logging {<IPV4-ADDR> | <IPV6-ADDR> | <FQDN | HOSTNAME>} {udp [<PORT-NUM>]}|
{tcp [<PORT-NUM>]}|{|tls [<PORT-NUM>]}
   auth-mode {certificate|subject-name}
   disable
   filter <FILTER-NAME>
   include-auditable-events
   legacy-tls-renegotiation]
   rate-limit-burst <BURST>
   rate-limit-interval <INTERVAL>] ]
   severity <LEVEL>]
```

```
   vrf <VRF-NAME>]
no logging {<IPV4-ADDR> | <IPV6-ADDR> | <FQDN | HOSTNAME> }
```

Description

Enables syslog forwarding to a remote syslog server.

The **no** form of this command disables syslog forwarding to a remote syslog server.

Starting with AOS-CX 10.11, payload information is present in accounting logs.

The maximum REST payload that can be sent to RADIUS/TACACS server is 1024 characters, and the maximum of REST payload that can be sent to syslog server is 3500 characters. If this limit is is reached, the log will display three dots (...) to indicate that the log an exceeded the character limit and is incomplete.

Parameter	Description
`{<IPV4-ADDR> | <IPV6-ADDR> | <HOSTNAME>}`	Selects the IPv4 address, IPv6 address, or host name of the rem ote syslog server. Required.
`[udp [<PORT-NUM>] | tcp [<PORT-NUM> | tls [<PORT-NUM>]]`	Specifies the UDP port, TCP port, or TLS port of the remote sysl og server to receive the forwarded syslog messages.
`udp [<PORT-NUM>]`	Range: 1 to 65535. Default: 514
`tcp [<PORT-NUM>]`	Range: 1 to 65535. Default: 1470
`tls [<PORT-NUM>]`	Range: 1 to 65535. Default: 6514
`auth-mode`	Specifies the TLS authentication mode used to validate the cert ificate. • **certificate**: Validates the peer using trust anchor certificate based authentication. Default. • **subject - name**: Validates the peer using trust anchor certif icates as well as subject - name based authentication.
`disable`	Disable remote syslog confguration. This does not delete the co nfiguration, just disables/pauses the forwarding of syslog mess agesto the remote server. The config/forwarding can be reenabl ed (un - paused) again using the **no logging <hostname> disa ble** command.
`filter <FILTER-NAME>`	Specifies the name of the filter to be applied on the syslog mes sages.
`include-auditable-events`	Specifies that auditable messages are also logged to the remote syslog server.

Parameter	Description
legacy-tls-renegotiation	Enables the TLS connection with a remote syslog server supporting legacy renegotiation.
rate-limit-burst *<BURST>*	Specifies the rate limit for the messages sent to the remote syslog server.
rate-limit-interval *<INTERVAL>*	Specifies the rate limit interval in seconds. Default: 30 Seconds
severity *<LEVEL>*	Specifies the severity of the syslog messages: • **alert**: Forwards syslog messages with the severity of **alert (6)** and **emergency (7)**. • **crit**: Forwards syslog messages with the severity of **critical (5)** and above. • **debug**: Forwards syslog messages with the severity of **debug (0)** and above. • **emerg**: Forwards syslog messages with the severity of **emergency (7)** only. • **err**: Forwards syslog messages with the severity of **err (4)** and above • **info**: Forwards syslog messages with the severity of **info (1)** and above. Default. • **notice**: Forwards syslog messages with the severity of **notice (2)** and above. • **warning**: Forwards syslog messages with the severity of **warning (3)** and above.
vrf *<VRF-NAME>*	Specifies the VRF used to connect to the syslog server. Optional. Default: default

Examples

Enabling the syslog forwarding to remote syslog server 10.0.10.2:

```
switch(config)# logging 10.0.10.2
```

Enabling the syslog forwarding of messages with a severity of **err (4)** and above to TCP port 4242 on remote syslog server 10.0.10.9 with VRF **lab_vrf**:

```
switch(config)# logging 10.0.10.9 tcp 4242 severity err vrf lab_vrf
```

Disabling syslog forwarding to a remote syslog server:

```
switch(config)# no logging
```

Enabling syslog forwarding over TLS to a remote syslog server using subject-name authentication mode:

```
switch(config)#logging example.com tls auth-mode subject-name
```

Applying log filtering for syslog server forwarding:

```
switch(config)# logging 10.0.10.6 severity info filter filter_lldp_logs vrf
mgmt
```

Applying log filtering and enabling the rate limit for syslog server forwarding over TCP port:

```
switch(config)# logging 10.0.10.2 tcp 3440 severity err vrf mgmt include-
auditable-events filter filter_lldp_logs rate-limit-burst 3 rate-limit-
interval 35
```

Command History

Release	Modification
10.12.1000	**The disable** parameter is introduced
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`snmp-server community`

Syntax

```
snmp-server community <STRING>
no snmp-server community <STRING>
```

Description

Adds an SNMPv1/SNMPv2c community string. A community string is like a password that controls read/write access to the SNMP agent. A network management program must supply this name when attempting to get SNMP information from the switch. A maximum of 10 community strings are supported. Once you create your own community string, the default community string (`public`) is deleted.

The **no** form of this command removes the specified SNMPv1/SNMPv2c community string. When no community string exists, a default community string with the value **public** is automatically defined.

Parameter	Description
`<STRING>`	Specifies the SNMPv1/SNMPv2c community string. Range: 1 to 32 printable ASCII characters, excluding space and question ma rk.

Subcommands

```
access-level {ro | rw}
no access-level {ro | rw}
```

This subcommand changes the access level of the SNMP community. The default access level is read-only (**ro**).

The **no** form of this subcommand changes the access level of the community to default.

Parameter	Description
`ro`	Specifies Read - Only access with the SNMP community.
`rw`	Specifies Read - Write access with the SNMP community.

```
access-list {ip | ipv6} <ACL-NAME>
no access-list {ip | ipv6} <ACL-NAME>
```

This subcommand associates an ACL with the SNMP community. If an ACL is not associated with the SNMP community, the default access is allowed for all the hosts.

The **no** form of this subcommand removes association of the ACL with the SNMP community.

Parameter	Description
`ip`	Specifies the IPv4 ACL type.
`ipv6`	Specifies the IPv6 ACL type.

Parameter	Description
`<ACL-NAME>`	Specifies the ACL name. It supports a maximum of 64 characte rs.

Examples

Setting the SNMPv1/SNMPv2c community string to **private**:

```
switch(config)# snmp-server community private
```

Removing SNMPv1/SNMPv2c community string **private**:

```
switch(config)# no snmp-server community private
```

Configuring the access level for the SMNP community to read-only:

```
switch(config-community)# access-level ro
```

Changing the access level of the SNMP community to default:

```
switch(config-community)# no access-level rw
```

Associating an IPv4 ACL named **my_acl** with the SMNP community:

```
switch(config-community)# access-list ip my_acl
```

Removing the associated IPv4 ACL named **my_acl** from the SNMP community:

```
switch(config-community)# no access-list ip my_acl
```

Configuration supported for SNMP ACL:

```
access-list ip ipv4_acl
    10 permit any 4.4.4.4 4.4.4.1
    20 permit any 3.3.3.3 3.3.3.1
access-list ipv6 ipv6_acl
    10 permit any 2001::2 2001::1
    20 permit any 3001::2 3001::1
snmp-server vrf default
snmp-server community my_comm_1
    access-list ip ipv4_acl
    access-list ipv6 ipv6_acl
```

Configuration not supported for SNMP ACL:

```
access-list ip ipv4_acl
    10 deny any 6.6.6.6 6.6.6.1
```

```
access-list ipv6 ipv6_acl
    10 deny any 6001::6 6000::1
snmp-server vrf default
snmp-server community my_comm_1
    access-list ip ipv4_acl
    access-list ipv6 ipv6_acl
```

NOTE

hitcounts for SNMP ACL will not be incremented.

Example:show access-list hitcounts ip all will not show the hit count of SNMP ACL.

Command History

Release	Modification
10.14	Replaced the **ipv4** parameter with the **ip** parameter. The **ipv4** parameter is dep recated.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config` `config-community`	Administrators or local user group members with execution righ ts for this command.

`snmp-server host`

Syntax

```
snmp-server host <IPv4-ADDR | IPv6-ADDR> trap version <VERSION> [community
<STRING>]
[port <UDP-PORT>] [<VRF-NAME>] [notification-type <NOTIFICATION-TYPE>]
no snmp-server host <IPv4-ADDR | IPv6-ADDR> trap version <VERSION>
[community <STRING>]
[port <UDP-PORT>] [<VRF-NAME>] [notification-type <NOTIFICATION-TYPE>]
snmp-server host <IPv4-ADDR | IPv6-ADDR> inform version v2c [community
```

```
<STRING>]
[port <UDP-PORT>] [<VRF-NAME>] [notification-type <NOTIFICATION-TYPE>]
no snmp-server host <IPv4-ADDR | IPv6-ADDR> inform version v2c [community
<STRING>]
[port <UDP-PORT>] [<VRF-NAME>] [notification-type <NOTIFICATION-TYPE>]
snmp-server host <IPv4-ADDR | IPv6-ADDR> [trap version v3 | inform
version v3] user <NAME> [port <UDP-PORT>] [<VRF-NAME>] [notification-type
<NOTIFICATION-TYPE>]
no snmp-server host <IPv4-ADDR | IPv6-ADDR> [trap version v3 | inform
version v3] user <NAME> [port <UDP-PORT>] [<VRF-NAME>] [notification-type
<NOTIFICATION-TYPE>]
```

Description

Configures a trap/informs receiver to which the SNMP agent can send SNMP v1/v2c/v3 traps or v2c informs. A maximum of 30 SNMP traps/informs receivers can be configured.

The **no** form of this command removes the specified trap/inform receiver.

> **NOTE**
> Avoid configuring the same receiver for both SNMP traps and informs on the same UDP port for v1, v2, and v3.

Parameter	Description
<IPv4-ADDR>	Specifies the IP address of a trap receiver in IPv4 format (**x.x.x.x**), where **x** is a decimal number from 0 to 255. You can remove leading zeros. For example, the address **192.169.005.100** becomes **192.168.5.100**.
<IPv6-ADDR>	Specifies the IP address of a trap receiver in IPv6 format (**x:x::x:x**).
`trap version <VERSION>`	Specifies the trap notification type for SNMPv1, v2c or v3. Available options are: **v1**, **v2c** or **v3**.
`inform version v2c`	Specifies the inform notification type for SNMPv2c.
`trap version v3`	Specifies the trap notification type for SNMPv3.
`user <NAME>`	Specifies the SNMPv3 user name to be used in the SNMP trap notifications.

Parameter	Description
community *<STRING>*	Specifies the name of the community string to use when sendin g trap notifications. Range: 1 - 32 printable ASCII characters, excluding space and question mark. Default: **public**.
<UDP-PORT>	Specifies the UDP port on which notifications are sent. Range: 1 - 65535. Default: 162.
<VRF-NAME>	Specifies the VRF on which the SNMP agent listens for incomin g requests.
<notification-type>	Specifies the type of notification to be sent to the trap receiver. If no type is specified, all notifications are sent. The supported notification types are: • aaa - server • alarm • bgp • card • config • entity • fan • interface • lldp • loop - protect • mac - notify • mstp • mvrp • ospf • ospfv3 • port - security • power • power - ethernet • rmon • rpvst • stp • temperature • vrrp • vsf

Parameter	Description
	• vsx

Examples

```
switch(config)# snmp-server host 10.10.10.10 trap version v1
switch(config)# no snmp-server host 10.10.10.10 trap version v1

switch(config)# snmp-server host a:b::c:d trap version v1
switch(config)# no snmp-server host a:b::c:d trap version v1

switch(config)# snmp-server host 10.10.10.10 trap version v2c community
public
switch(config)# no snmp-server host 10.10.10.10 trap version v2c community
public

switch(config)# snmp-server host a:b::c:d trap version v2c community public
switch(config)# no snmp-server host a:b::c:d trap version v2c community
public

switch(config)# snmp-server host 10.10.10.10 trap version v2c community
public port 5000
switch(config)# no snmp-server host 10.10.10.10 trap version v2c community
public port 5000

switch(config)# snmp-server host 10.10.10.10 trap version v2c community
public port 5000 vrf default

switch(config)# no snmp-server host 10.10.10.10 trap version v2c community
public port 5000 vrf default

switch(config)# snmp-server host a:b::c:d trap version v2c community public
port 5000
switch(config)# no snmp-server host a:b::c:d trap version v2c community
public port 5000

switch(config)# snmp-server host 10.10.10.10 inform version v2c community
public
switch(config)# no snmp-server host 10.10.10.10 inform version v2c
community public

switch(config)# snmp-server host a:b::c:d inform version v2c community
public
```

```
switch(config)# no snmp-server host a:b::c:d inform version v2c community
public

switch(config)# snmp-server host 10.10.10.10 inform version v2c community
public port 5000
switch(config)# no snmp-server host 10.10.10.10 inform version v2c
community public port 5000

switch(config)# snmp-server host 10.10.10.10 inform version v2c community
public port 5000 vrf default

switch(config)# no snmp-server host 10.10.10.10 inform version v2c
community public port 5000 vrf default

switch(config)# snmp-server host a:b::c:d inform version v2c community
public port 5000
switch(config)# no snmp-server host a:b::c:d inform version v2c community
public port 5000

switch(config)# snmp-server host 10.10.10.10 trap version v3 user Admin
switch(config)# no snmp-server host 10.10.10.10 trap version v3 user Admin

switch(config)# snmp-server host a:b::c:d trap version v3 user Admin
switch(config)# no snmp-server host a:b::c:d trap version v3 user Admin

switch(config)# snmp-server host 10.10.10.10 trap version v3 user Admin
port 2000
switch(config)# no snmp-server host 10.10.10.10 trap version v3 user Admin
port 2000

switch(config)# snmp-server host a:b::c:d trap version v3 user Admin port
2000
switch(config)# no snmp-server host a:b::c:d trap version v3 user Admin
port 2000
```

SNMP trap notification type examples:

```
switch(config)# snmp-server host 10.10.10.10 trap version v2c community
public notification-type bgp fan interface power entity
switch(config)# no snmp-server host 10.10.10.10 trap version v2c community
public notification-type bgp
switch(config)# snmp-server host a:b::c:d inform version v3 user Admin
```

```
notification-type bgp fan interface power-ethernet
switch(config)# no snmp-server host a:b::c:d inform version v3 user Admin
notification-type bgp interface
switch(config)# snmp-server host a:b::c:d inform version v3 user Admin
notification-type ?
  aaa-server      Sends AAA notifications.
  alarm           Sends Alarm notifications.
  bgp             Sends Border Gateway Protocol (BGP) state change
notifications.
  card            Sends Card notifications.
  config          Sends Configuration change notifications.
  entity          Sends Entity notifications.
  fan             Sends Fan notifications.
  interface       Sends Interface notifications.
  lldp            Sends Link Layer Discovery Protocol (LLDP) notifications.
  loop-protect    Sends Loop Protect notifications.
  mac-notify      Sends MAC Notify notifications.
  mstp            Sends Multiple Spanning Tree Protocol (MSTP)
notifications.
  mvrp            Sends Multiple VLAN Registration Protocol (MVRP)
notifications.
  ospf            Sends Open Shortest Path First (OSPFv2) notifications.
  ospfv3          Sends Open Shortest Path First version 3 (OSPFv3)
notifications.
  port-security   Sends Port Security notifications.
  power           Sends Power notifications.
  power-ethernet  Sends Power over Ethernet (PoE) notifications.
  rmon            Sends Remote Network Monitoring (RMON) notifications.
  rpvst           Sends Rapid Per VLAN Spanning Tree (RPVST) notifications.
  snmp            Sends Sends Simple Network Management Protocol (SNMP)
notifications.
  stp             Sends Spanning Tree Protocol (STP) notifications.
  temperature     Sends Temperature notifications.
  vrrp            Sends Virtual Router Redundancy Protocol (VRRP)
notifications.
  vsf             Sends Virtual Switching Framework (VSF) notifications.
  vsx             Sends Virtual System Extension (VSX) notifications.
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`snmp-server vrf`

Syntax

```
snmp-server vrf <VRF-NAME>
no snmp-server vrf <VRF-NAME>
```

Description

Configures a VRF on which the SNMP agent listens for incoming requests. By default, the SNMP agent does not listen on any VRF. 4100i, 6000, and 6100 only support default VRF. **The SNMP agent can listen on multiple VRFs.**

The **no** form of this command stops the SNMP agent from listening for incoming requests on the specified VRF.

Parameter	Description
<VRF-NAME>	Specifies the name of a VRF.

Examples

Configuring the SNMP agent to listen on VRF **default**.

```
switch(config)# snmp-server vrf default
```

Configuring the SNMP agent to listen on VRF **mgmt**.

```
switch(config)# snmp-server vrf mgmt
```

Stopping the SNMP agent from listening on VRF **default**.

```
switch(config)# no snmp-server vrf default
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`snmpv3 context`

Syntax

```
snmpv3 context <NAME> vrf <VRF-NAME> [community <STRING>]
no snmpv3 context <NAME> [vrf <VRF-NAME>] [community <STRING>]
```

Description

Creates an SNMPv3 context on the specified VRF.

The **no** form of this command removes the specified SNMP context.

Parameter	Description
`<NAME>`	Specifies the name of the context. Range: 1 to 32 printable ASC II characters, excluding space and question mark (?).
`vrf <VRF-NAME>`	Specifies the VRF associated with the context. Default: default.
`community <STRING>`	Specifies the SNMP community string associated with the con text. Range: 1 to 32 printable ASCII characters, excluding space and question mark. Default: public.

Examples

Creating an SNMPv3 context named **newContext**:

```
switch(config)# snmpv3 context newContext
```

Creating an SNMPv3 context named **newContext** on VRF **myVrf** and with community string **private**.

```
switch(config)# snmpv3 context newContext vrf myVrf community private
```

Removing the SNMPv3 context named **newContext** on VRF **myVrf**:

```
switch(config)# no snmpv3 context newContext vrf myVrf
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`snmpv3 user`

Syntax

```
snmpv3 user <NAME>
      [auth <AUTH-PROTO> auth-pass [{plaintext | ciphertext} <AUTH-PASS>]]
      [priv <PRIV-PROTO> priv-pass [{plaintext | ciphertext} <PRIV-PASS>]]
      [access-level ro|rw]
no snmpv3 user <NAME>
        [auth <AUTH-PROTO> auth-pass [{plaintext | ciphertext} <AUTH-PASS>]]
        [priv <PRIV-PROTO> priv-pass [{plaintext | ciphertext} <PRIV-PASS>]]
        [access-level ro|rw]
```

Description

Creates an SNMPv3 user and adds it to an SNMPv3 context. The SNMPv3 security level (set with command **snmpv3 security-level**) determines which users are allowed to authenticate.

The **no** form of this command removes the specified SNMPv3 user.

> **NOTE**
> When updating the authentication protocols and privacy protocols for the existing SNMPv3 users, you must also update the access level. Otherwise, the access level will be set to read-only.

Parameter	Description
`<NAME>`	Specifies the SNMPv3 username. Range 1 to 32 printable ASCII characters, excluding space and question mark (?).
access - level	Configures the access level for the SNMPv3 user: • **ro**: Allow read - only access for the SNMPv3 user • **rw**: Allow read - write access for the SNMPv3 user
`auth <AUTH-PROTO>`	Sets the authentication protocol used to validate user logins. Su pported protocols are **md5**, **sha**, **sha224**, **sha256**, **sha384**, and **sha512**.
`auth-pass [{plaintext | ciphertext} <AUTH-PASS>]`	Specifies the SNMPv3 user authentication password. Range for **plaintext** is 8 to 32 printable ASCII characters, excluding space and question mark (?). Range for **ciphertext** is 1 to 256 printab le ASCII characters. Ciphertext is used when copying user confi guration settings between switches. **NOTE** Authentication passwords that include special c haracters must be enclosed in single quotation marks ('). For example, **'auth - pwd20246!@#'**.
`priv <PRIV-PROTO>`	Sets the SNMPv3 privacy protocol (encryption method). Suppor ted privacy protocols are **aes**, **aes192**, **aes256**, and **des**.
`priv-pass [{plaintext | ciphertext} <PRIV-PASS>]`	Specifies the SNMPv3 user privacy encryption password. Range for **plaintext** is 8 to 32 printable ASCII characters, excluding sp ace and question mark (?). Range for **ciphertext** is 1 to 256 pri

Parameter	Description
	ntable ASCII characters. Ciphertext is used when copying user configuration settings between switches. **NOTE** Authentication passwords that include special characters must be enclosed in single quotation marks ('). For example, **'priv - pwd20246!@#'**.

NOTE

When the authentication password is not provided on the command line, plaintext authentication password prompting occurs upon pressing Enter, followed by privacy encryption protocol prompting, and finally plaintext encryption password prompting. The entered password characters are masked with asterisks.

NOTE

When the authentication type and password plus the privacy protocol (encryption method) are provided on the command line but the encryption password is not provided, plaintext encryption password prompting occurs upon pressing Enter. The entered password characters are masked with asterisks.

Examples

Defining SNMPv3 user **Admin1** using **sha** authentication and **des** privacy encryption with provided plaintext passwords:

```
switch(config)# snmpv3 user Admin1 auth sha auth-pass plaintext F82#450h
                priv des priv-pass plaintext F82#4eva
```

Defining SNMPv3 user **Admin2** using **MD5** authentication and **AES** privacy encryption with provided authentication password and privacy encryption type but prompted encryption password:

```
switch(config)# snmpv3 user Admin2 auth md5 auth-pass plaintext F82#450h
                            priv aes priv-pass
Enter the privacy encryption key: ********
Re-Enter the privacy encryption key: ********
```

Defining SNMPv3 user **Admin2** using **MD5** authentication and **AES** privacy encryption with plaintext password prompting and privacy encryption selection:

```
switch(config)# snmpv3 user Admin2 auth md5 auth-pass
Enter the authentication password: ********
Re-Enter the authentication password: ********
Configure the privacy protocol (y/n)? y
Enter the privacy protocol (aes/des)? aes
Enter the privacy encryption key: ********
Re-Enter the privacy encryption key: ********
```

Removing SNMPv3 user **Admin1**:

```
switch(config)# no snmpv3 user Admin1
```

Creating an SNMP user on switch 1 and then creating the same user on switch 2 by copying from the switch 1 configuration:

On switch 1, configure a user named **Admin3**, and then use the **show running-config** command to display switch configuration. Save a copy of the full **snmpv3 user** command (shown by **show running-config**). This saved command is used on switch 2.

```
switch1(config)# snmpv3 user Admin3 auth sha auth-pass plaintext F82#450h
                 priv des priv-pass plaintext F82#4eva
switch1(config)# exit
switch1# show running-config
Current configuration:
!
!Version AOS-CX xx.xx.xx.xxxxxx
!
snmpv3 user Admin3 auth sha auth-pass ciphertext AQBaf2d...FJVcZ3o=
priv des priv-pass ciphertext AQBaH2p...2jfTFwQ=
ssh server vrf mgmt
!
interface mgmt
no shutdown
ip dhcp
vlan 1
```

On switch 2, execute the **snmpv3 user** command that you saved from switch 1 (as shown by **show running-config**). This creates the user on switch 2 with the same configuration.

```
switch2(config)# snmpv3 user Admin3 auth sha auth-pass ciphertext
AQBaf2d...FJVcZ3o=
                 priv des priv-pass ciphertext AQBaH2p...2jfTFwQ=
```

The following command sets a read-write access level for an SNMPv3 user with the user name **user1**.

```
switch(config)# snmpv3 user user1 auth md5 auth-pass plaintext abc1234
access-level rw
```

Command History

Release	Modification
10.13	Following authentication protocols are supported: **sha224**, **sha256**, **sha384**, an d **sha512**. Following privacy protocols are supported: **aes192** and **aes256**.
10.09	The **access - level** parameter was introduced.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

Subtopics

[snmpv3 user view]

`snmpv3 user view`

Syntax

```
snmpv3 user <USER-NAME> view <VIEW-NAME>
no snmpv3 user <USER-NAME> view <VIEW-NAME>
```

Description

Associates a user with an existing SNMP MIB view.

The **no** form of this command removes the associated user from the specified SNMP MIB view.

Parameter	Description
	Specifies the user name for the SNMP MIB view. Accepts a maxi mum of 32 characters.

Parameter	Description
`<USER-NAME>`	
`<VIEWNAME>`	Specifies the view name for the SNMP MIB view. Accepts a maxi mum of 32 characters.

Examples

Adding a user in the existing SNMP MIB view:

```
switch(config)# snmpv3 user nw-admin view my-nw-view
```

Removing the user from the SNMP MIB view:

```
switch(config)# no snmpv3 user nw-admin view my-nw-view
```

Attaching unconfigured or unknown SNMP view to an SNMPv3 user:

```
switch(config)# snmpv3 user nw-admin view myView
View myView is not configured.
```

Command History

Release	Modification
10.10	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

Support and Other Resources

Subtopics

[Accessing HPE Aruba Networking Support]

[Accessing Updates](#)
[Warranty Information](#)
[Regulatory Information](#)
[Documentation Feedback](#)

Accessing HPE Aruba Networking Support

HPE Aruba Networking Support Services	**https://www.hpe.com/us/en/networking/hpe-aruba-networking-support-services.html**
AOS-CX Switch Software Documentation Portal	**https://arubanetworking.hpe.com/techdocs/ArubaDocPortal/content/new-portal/aoscx.html**
HPE Aruba Networking Support Portal	**https://networkingsupport.hpe.com/home**
North America telephone	1-800-943-4526 (US & Canada Toll-Free Number) +1-650-750-0350 (Backup—Toll Number)
International telephone	**https://www.hpe.com/psnow/doc/a50011948enw**

Be sure to collect the following information before contacting Support:

- Technical support registration number (if applicable)
- Product name, model or version, and serial number
- Operating system name and version
- Firmware version
- Error messages
- Product-specific reports and logs
- Add-on products or components
- Third-party products or components

Other useful sites

Other websites that can be used to find information:

HPE Aruba Networking Developer Hub	**https://developer.arubanetworks.com/hpe - aruba - networking - aoscx/docs/about**
Airheads social forums and Knowledge Base	**https://community.arubanetworks.com/**
AOS - CX Software Technical Update channel on You Tube.	Videos on new features introduced in this release : **https://www.youtube.com/playlist?list=PLsYGHuNuBZcbWPEjjHuVMqP - Q_UL3CskS**
HPE Aruba Networking Hardware Documentation and Translations Portal	
HPE Aruba Networking software	**https://networkingsupport.hpe.com/downloads https://networkingsupport.hpe.com/downloads**
Software licensing and Feature Packs	**https://licensemanagement.hpe.com/**
End - of - Life information	**https://networkingsupport.hpe.com/end - of - life**

Accessing Updates

You can access updates from the HPE Aruba Networking Support Portal at **https://networkingsupport.hpe.com**.

Some software products provide a mechanism for accessing software updates through the product interface. Review your product documentation to identify the recommended software update method.

To subscribe to eNewsletters and alerts:

https://networkingsupport.hpe./notifications/subscriptions (requires an active HPE Aruba Networking Support Portal account to manage subscriptions). Security notices are viewable without an HPE Aruba Networking Support Portal account.

Warranty Information

To view warranty information for your product, go to **https://www.arubanetworks.com/support-services/product-warranties/**.

Regulatory Information

To view the regulatory information for your product, view the Safety and Compliance Information for Server, Storage, Power, Networking, and Rack Products, available at **https://www.hpe.com/support/Safety-Compliance-EnterpriseProducts**

Additional regulatory information

HPE Aruba Networking is committed to providing our customers with information about the chemical substances in our products as needed to comply with legal requirements, environmental data (company programs, product recycling, energy efficiency), and safety information and compliance data, (RoHS and WEEE). For more information, see **https://www.arubanetworks.com/company/about-us/environmental-citizenship/**.

Documentation Feedback

HPE Aruba Networking is committed to providing documentation that meets your needs. To help us improve the documentation, send any errors, suggestions, or comments to Documentation Feedback (**docsfeedback-switching@hpe.com**). When submitting your feedback, include the document title, part number, edition, and publication date located on the front cover of the document. For online help content, include the product name, product version, help edition, and publication date located on the legal notices page.